

SpringerBriefs in Modern Perspectives on
Disability Research

Wenn B. Lawson



Autism and Being Monotropic

What Medical and Other
Practitioners Need
to Know



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SpringerBriefs in Modern Perspectives on Disability Research

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Wenn B. Lawson

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What Medical and Other Practitioners Need to
Know

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The content of this book would be poorer had it not been for the vision, commitment, and support from my colleagues Dinah Murray and Mike Lesser (no longer with us in the flesh). Back in 1998, when we first met one another, we discovered a shared awareness and understanding of autism, as 'a matter of attention born from an interest system of mind'. This developed over the years into a 'cognitive theory' of autism that, as autistic people, we recognized and shared freely with others. Autistic people from all walks of life identified with this understanding and the term 'monotropism' was coined by a neighbour up the street from Dinah's home.

For many years, monotropism was seen as a novelty and not taken seriously. But, between us, especially for Dinah and I (as Mike had passed away on 1 July 2015) we refused to give up and took every opportunity to explore how monotropism played out practically in autistic lives as well as how to conduct research into 'monotropism'. Now, several years later and with an amount of work to support the idea of monotropism as

core to our autistic identity, this book aims to share our ongoing vision with readers.

Mike, I hope you found peace, my friend, and out there in the 'ethernet of all minds' that you know our work continues and is making a difference.

Dinah, my beloved friend, I miss you every day, but I know you are rejoicing in the knowledge of what we have built and the difference our understanding of autism is giving to countless others.

This book is dedicated to you.

Foreword

For far too many years, autism research has been dominated by the search for accounts and explanations of autism that all but ignore the actual life experiences of autistic people. The result has been both poor science and a failure of research to make a meaningful difference to people's lives.

Wenn Lawson has always stood against this trend. Decades ago now, along with his colleagues and friends, Dinah Murray and Mike Lesser, he began to sketch an account of autism that was heavily dependent on a rich understanding of the life experiences of autistic people. How is it, Lawson and colleagues asked, that autistic people understand the world around them? How do they respond to new, changing and sometimes challenging social situations? Where do they find the most meaning and fulfilment? What sustains well-being or flourishing? And what might underpin those different experiences when they are compared and contrasted with those of neurotypical or what Lawson calls Allistic people?

It is from this careful and considered attention to the reality autistic lives, combined with equally extraordinary process and rigour, that the idea of Monotropism first emerged. Monotropism, as Lawson has it, is 'the tendency for interests to pull or capture individual attention more strongly than in most people'. It is not intended just as a description of individual elements of autistic experience, but rather as 'an autism characteristic that underwrites Autism' itself. It is, in other words, a core feature of autism, driving and helping to explain others.

Crucially, for Lawson, understanding Monotropism and the role it plays enables Allistic people to appreciate both what makes certain situations difficult or damaging for autistic people and what makes other situations deeply enjoyable or productive. It is a way for the world, that is, to understand what autistic people need not just to survive but to thrive. Understanding it is thus crucial not only for those of us who research autism professionally but for everyone who works with, cares for, supports and is friends with autistic people. As such, it is no surprise that this book offers fundamentally important guidance for clinicians, educators, work colleagues, friends and neighbours.

As Lawson notes, despite all of this, it has taken an unacceptably long period of time for many outside of the autistic community to take the insights outlined in this book seriously enough. For years, Monotropism helped autistic people themselves make sense of their own lives, but it was overlooked by scholars and practitioners more keenly aware of theories and ideas that had come from more conventional voices in academic and scientific life.

Thankfully, that has now begun to change. After years of persistent scholarship, advocacy and practical guidance, Lawson has helped broaden the perspectives and deepen the learning of those who should always have known better in the first place. That is one reason why it is such a joy to be able to offer these short words at the start of this inspiring book. Another reason is that I know that everyone who reads it will learn something afresh from each and every argument it makes and story it shares.

Wenn Lawson has done more than almost anyone I know to advance our understanding of autism in the last decade or so and I am sure that you will enjoy considering the fruits of all of those labours in the pages that follow here.

University College London, London, UK
November 2024

Liz Pellicano

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About the Author

Wenn B. Lawson was born in England in 1952. He emigrated along with his family to Australia in 1984 and now lives fulltime in Australia. Dr. Wenn is an autistic lecturer, British psychologist, researcher, advocate, writer, and poet who has passionately shared professional and personal knowledge of autism over the past three decades. He has written/ contributed to over 30 books and many papers. Wenn is an Adjunct, A/Prof., and Senior researcher with Curtin University in Western Australia. He is a member of the Autism Cooperative Research Centre (ACRC), Qld., the Autism in Adulthood Editorial Board, the Advisory Board for the US Autism Association, and is on the Board of Directors for the Autism Research Institute (ARI), US. Wenn is a family man with neurodivergent offspring and grandchildren. As well as his passion for autism research, he has a passion for science fiction books, *Science Fiction Films and Television*, as well as for birds and animals.

Chapter 1

Introduction



When a client or patient comes for a consultation, usually certain circumstances have caused them to be referred to us or to contact us directly. Initially, the only information we have is by way of either known history (we already know them) or what's written on their referral. We may know they are autistic, are seeking an autism assessment, or have some 'mental health' issues. But whatever is written about the individual will likely be read, by us, via the lens of our understanding of the written statements, filtered through our own expertise and associated perceptions. If our knowledge of Autism is 'up to date' and founded upon recent research coupled with the 'Autistic voice', we will have a better chance of accurately translating what we read. If it's based upon outdated, traditional understanding of autism, pre-DSM-5 (2013, p. 22) criteria, we may still have an image of autism based upon an outdated notion arising from popular media of the time: 'Rain man' (1988): <https://www.imdb.com/title/tt0095953/>. The reason for writing this book is to offer an explanation of client behaviour that may appear puzzling to us, usually because of the lens we are viewing it through.

When Autistic individuals attend our clinics (diagnosed or seeking an assessment), their behaviour may not 'fit' our expectation and we risk missing the real reasons for their presentation. Some will 'appear' totally 'normal' and it won't be easy to move beyond the mask they present as being 'them'. Others will be close to 'a breakdown' or burn out and separating mental health issues, autistic issues and those of physical health, will be a major task. However, it is vital we do so and are able to see 'the big picture' the 'whole' person and that we are not distracted by a variety of 'layers' built up over time.

Many individuals will present with a picture that is painted by a total lack of 'ability' as in inertia, others will appear as if they are 'on speed' and some will not be able to explain what they are living with at all. If the picture appears to be that of inertia, it's a picture of total overwhelm, and total shut down. It's all just 'too much'. In Autism, this is due to every experience *being filtered through one stream*, therefore ALL streams: emotive, physical, environmental, social, tasks to do,

unexpected events as well as those expected, mental health, sensory, daily needs, and so much more have caused a traffic jam. It's overloading one highway causing the edges to break down and contents to spill out and come to a halt.

If the person seems to be going 500 miles an hour and much of what they say doesn't seem to add up, it's likely their over loaded highway of life experience has become a 'fast lane' to get through it all and even their top speed isn't enabling so much as disabling them.

If the person appears calm, collected, and together but their life story suggests otherwise, then they have probably masked who they are. They have taken on a different persona, as a way of dis-identifying from the overloaded highway of 'too much'.

This book seeks to explain what lies below these client presentations and help clinicians uncover the reasons for the behaviour we are seeing. Autism is a wide spectrum sometimes appearing with more differing characteristics than those that are similar. This can be disconcerting and cause us to doubt our own thoughts and findings.

A Word on Language What terminology should we use? Is it, 'Autism Spectrum Disorder' (ASD), the Autism Spectrum, or Autism? Language is very important because words conjure up images that lead to feelings and even beliefs. In this book, I will use terms such as Autistic, on the Autism Spectrum, on the spectrum and Autistics. I will not use terms that suggest an individual is 'with autism', 'has autism', or has an autism spectrum disorder.

For me, as an autistic person, saying 'I am with autism', 'I'm a person with autism' or that I have an autism spectrum disorder, is very uncomfortable because it feels like a lie. If I'm 'with autism' it implies I can be 'without' it. It reminds me of some appendage or extra article I carry that could be removed. This is not the case. The other reason for my choice of terminology is that the Autistic population prefer the term 'Autistic person' rather than 'a person with autism' (Kenny et al., 2016). Then again, avoiding language that stigmatizes and pathologizes helps an individual feel more whole. So, I prefer it because it moves us away from 'pathologizing autism', away from a 'medical dis-ease' and towards humanizing autistic people (Pellicano & den Houting, 2022). Of course, being respectful when working and/or supporting us, means it's important to ask us what language we are most comfortable with.

Also, when we view 'autism' as a part of neurodiversity (all brains are different) we are more likely to use 'neuro-affirming practice' (Dwyer, 2022). The term 'neurodivergent' is also popular because it lies beneath the umbrella term of 'neurodiversity' but illustrates and accommodates those of us who 'diverge' from the usual majority. This way of thinking about autism also moves us away from seeing autism as only a medical issue, which misses the bigger picture. As an alternative to the 'medical model' I prefer the term, biopsychosocial model which is also becoming popular with the broader community. The biopsychosocial model sees beyond the medical domain and further than the social model as it connects more to the whole person (Bölte et al., 2021).

Allistic When I refer to non-autistic people the term used is ‘Allistic’. Autistic implies ‘by oneself’ and Allistic implies ‘altogether’. In the past, ‘neurotypical’ is the term that has been used for Allistic individuals, but again it suggests ‘typical neurology’, the reality is that all brains are different, so this means a ‘typical’ brain is typical for that individual only.

For Autistic individuals, being understood and accommodated (by the medical, allied health and other professionals who support any and every aspect of autistic well-being) is rare. This is probably due to autistic and non-autistic individuals not comprehending one another’s communication style (Milton, 2012). To help bridge the gap between diagnostic criteria (criteria that sets the bar for autism) and understanding autistic individuals, this book sets out to explain autism in ways that go beyond the criteria in the Diagnostic and Statistical Manual (DSM-5, 2013; TR, 2022). It explores the very foundations and underpinnings of being autistic; the basis for the DSM-5 criteria. This means explaining what the reason is: for ‘rigid interests’, difficulties with changing attention, social interaction discomforts and sensory challenges (outlined in the DSM-5). It also explains the strengths autistic people can demonstrate. In a nutshell, autism is a matter of single attention, which implies single focus, often leading to individual strengths, a heightened and/or an underwhelmed sensory system, object permanence challenges, and disconnection from our interoceptive selves (Goodall, 2020; Lawson, 2011; Lawson & Dombroski, 2017; Murray et al., 2005; Pellicano et al., 2022; Pellicano & Heyworth, 2023), see also: <https://www.autism.org.uk/advice-and-guidance/professional-practice/interoception-wellbeing>

This book highlights current autism research, but importantly offers feedback, stories, and comments from the autistic community. There is an emphasis on ‘nothing about me without me’, and on ‘co-production’ from much of the research quoted. So, autism is explained by first-hand experience and complemented by research, much of which has been co-produced with autistic individuals. The foundational lens for exploring autism in this book is monotropism (single-focused attention) (Dwyer et al., 2024; Lawson, 1998; Murray, 1992; Murray et al., 2005; See also: <http://monotropism.org>).

From this foundation the book will also help readers to understand aspects of object permanence (Chap. 4), external sensing and internal sensing (interoception) (Chaps. 5 and 6) in autism which flow on differently, from the monotropic cognitive style, as opposed to the polytropic (Allistic) cognitive style. It will also consider ‘flow states’, inertia and mental fitness associated with the autistic person’s strengths and challenges. All of these ‘components’ in autism are vital for professionals working with autistic people to comprehend, so they are better equipped to understand autistic people and offer the appropriate supports.

The Non-autistic (Allistic) Cognitive Style The polytropic cognitive style in Allistic individuals allows for the change of attentional focus quite quickly. It facilitates movement between various interests and accommodates the sharing of interest when interest may not be the common denominator (Dwyer et al., 2024; Lawson, 2011; Murray et al., 2005). For social connectivity, rapid shifting of

attention and general societal connection, even when not particularly connected or interested, the polytropic cognitive style is more suited than the monotropic cognitive style. But the times where monotropism comes into its own are evidenced in the use of passionate interests that lead to deeper connections, possible new inventions, flow states, and productivity (Pellicano et al., 2020, 2022; Pellicano & Heyworth, 2023). The downside of the monotropic cognitive style are times of inertia (Rapaport et al., 2024) rumination (Dwyer et al., 2024) and awkwardness in social situations outside of mutual interest (Brown et al., 2024).

Milton, an autistic academic, illustrates how Autistic and Allistic individuals live in mutually exclusive worlds, with neither ‘getting’ the other (Milton, 2012; Milton et al., 2022; Zamzow, 2021). This non-recognition of ‘difference’ impacts daily life in a multitude of ways for all concerned (Murray et al., 2023). The autistic experience of ‘connection’ to everyday life from what is happening internally to one’s body (interoception) externally to one’s body (sight, smell, touch, taste, sound, proprioception), our kinesthetic sense and cognitively with object permanence, mirroring, and recognition of self as different to other are all one dimensional. This means, as Autistic people, we live in each place with our attention fully occupied by parts of an experience, rather than having open access to the ‘whole’ (Rapaport et al., 2024). Examples of this echoed by Autistic people are in our being literal, monotropic, and only noting concepts within our attention tunnel (Dwyer et al., 2024, p. 24; Grandin, 2006; Lawson, 2011; Murray et al., 2023; Murray et al., 2005; Pellicano et al., 2020; Rapaport et al., 2024).

The Diagnostic and Statistical Manual for Mental Disorders (DSM)-5

The DSM-5 TR (APA, 2013, 2022) states the criteria for an autism diagnosis, but it doesn’t tell us what’s behind this or offer the rationale or reasoning for this. Yes, it’s based upon observation, as currently there is no other way to diagnose autism (no biological test), but no explanation is given for the behaviour we see which we call ‘Autism’. There have been many studies to explore aspects of the brain that might be contributing to Autistic behaviour, as well as much work exploring genetic and environmental factors that may influence the behaviour seen in autism, but little research asking actual Autistic people about their experience (Pellicano & Heyworth, 2023).

Neuroscience Researchers have dissected the brains of autistic people, run genetic testing and explored our DNA as well as looked at environmental factors, but they haven’t identified causes or reasons for autism, only some aspects that may contribute to autism (Bai et al., 2019; Cheroni et al., 2020; Hallmayer et al., 2011; Ha et al., 2015; Landrigan, 2010; Kennedy, 2014; Pinto et al., 2010; Luo et al., 2018). There are many factors that contribute to being on the autism spectrum, just like there are many factors that contribute to being human. But, because there is no consensus diagnosticians are left having to ‘decode’ criteria for an autism assessment, often based upon their views of what ‘typical’ (normal) development looks like.

The DSM is ‘interpreted’ mostly by Allistic individuals who may not understand or even question why this behaviour exists. Because it’s ‘in the DSM’ the belief is that autism is a mental, medical condition, when it’s actually a ‘developmental condition’ based in our biology and is part of neurodiversity (Bolte et al., 2022).

For decades men were thought to qualify for autism and females couldn’t be autistic or were only autistic if they had a profound intellectual disability (Chelley et al., 2024; Kanner, 1943). Then, eventually, it was thought men were more likely than women to be autistic by 1 in 10.000 men, then 4–1 (Chelley et al., 2024; Rutherford et al., 2016). Currently, some research suggests male to female ratio is 1–3 (Loomes et al., 2017) and Tang et al. (2021) 1–2.8. This almost puts men and women on an equal par, with autism being almost as common in both genders (all genders). However, women may be diagnosed later in life due to their ability to mask many of their autistic ‘traditional’ characteristics, so they are overlooked and often misdiagnosed with other conditions (Chelley et al., 2024; Hull et al., 2020; Pellicano et al., 2021; Wing, 1981a, 1981b).

Too often the Autistic cognitive style is *masked* by Autistic individuals having adopted ‘adaptive morphing’ (Lawson, 2020) to enable them to access employment, seek a relationship or further their education (Lawson, 2020; Pearson & Rose, 2021). But this has often gone unrecognized and has negatively impacted autistic diagnosis, meaning females especially have gone undiagnosed (Pellicano et al., 2020). Adaptive morphing is one reason autism diagnosis can go undetected until ‘demand exceeds capacity’ (Pellicano & Rhodes, 2003), for many individuals, often diagnosed later in life. Masking one’s autism means attempting to do and be what ‘society’ expects, rather than being one’s authentic self. Masking our autism takes an awful lot of energy and is exhausting, it can lead to poor mental health and eventual autistic ‘burn out’ (Miller et al., 2021).

Autism as a Different Operating System Functioning via a different operating system to that of Allistic people, the autistic brain diverges from the majority. All brains are different and diverse, this is usual. Each person’s experience of being autistic is individual and different too. Yet, the common tendency in autism is to be monotropic (Dwyer et al., 2024; Lawson, 1998; Lawson, 2011; Murray et al., 2005, 2023). Having a brain that diverges from the majority allows Autistic people to use autistic behaviour as strategies to understand and interact with the world we live in. These strategies are used by neurologically divergent minds to aid the coping of an intensely experienced world (Dwyer et al., 2024; Elder-Robison, 2008; Lawson, 2011; Murray et al., 2023).

Practically, the world is founded on secular social interactions requiring individuals to notice self and others; allowing us all to work out expected social norms and relate across a wide social structure. It’s a given that people will be able to separate public from private, yours from mine, acting from real life, and so on. In Autism though these skills may be lacking, poorly formed or won’t generalize between situations and changing circumstances. Successfully navigating social environments requires the ability to shift attention quickly, but this will be difficult to navigate for Autistic monotropic minds. For example, in social situations there is a need to attend

to the multiple contexts of a conversation (voice tone, voice language, body language, context of surroundings, and so on) but a monotropic person might not process the full content of a conversation quickly (it requires rapid shifting of attention) or at all. Instead, being called away in one's attention to noticing or speaking about something of interest to them/us whether it's connected to the social encounter or not (Elder-Robison, 2008; Grandin, 2006; Lawson, 1998; Lawson & Dombroski, 2017; Williams, 2006).

This tends towards outcomes that lead to increased vulnerability for Autistic people. We are often traumatized due to living in a world that is not made for us (round pegs forced into square holes). One major concern is the digital world of technology. In Autism, this is often a preferred world as it appears logical, straight-forward, and less subject to a need for rapid shifting of attention or of reading facial and body language and so on. But, recognizing 'scripted from reality', the differences between the 'online world' and everyday life can be confusing or totally missed.

Although technology can be a very practical tool that allows for connection to others and to a way of sharing 'interest', it may not fully accommodate the vulnerabilities of a monotropic mind. Technology enables communication that might not otherwise happen, especially if you are an alternative communicator (rather than using speech to communicate you use typing or software allowing for communication by pictures and so on). This is so important and must be offered to individuals who need this to enable them to communicate. However, technology also offers instant gratification and leaves us open to a variety of risks and harms. It will mean, for example, (especially in a world where technology leads, and artificial intelligence is the norm), autistic people are drawn in higher-than-average ways, to pornography (Crabbe & Lawson, 2024). This could mean Autistic individuals may be more drawn to offending behaviour as well as to being victims of sexual abuse (Margari et al., 2024). This occurs due to the single focus of Autistic people who can get caught up in scams, extortion, porn, and so on and are more easily led astray by propaganda. Of course this happens to many individuals who are not autistic, but the monotropic cognitive style in Autism enhances this likelihood. It is so important Autistic individuals receive accessible sex education, learn how to critique the 'online' world and, above all, are having conversations about sex, mutuality, public, private, yours, mine, consent, and so on.

Monotropism, the tendency for interests to pull or capture individual attention more strongly than in most people, is an autism characteristic that underwrites Autism (Lawson, 1998; Murray et al., 2005). It rests on a model of the mind as an 'interest system'. We are all interested in many things, and our interests help direct our attention. Different interests are salient at different times. In a monotropic mind, fewer interests tend to be aroused at any time, and they attract more processing resources, making it harder to deal with things outside of the current attention tunnel (Dwyer et al., 2024; Lawson, 2011; Murray et al., 2005). So, when attention is sparked by interest, whatever connects to noticing, will take over and there won't be any 'spare' attention for anything else. Of course, being 'mono' can contribute to

Autistic strengths (more on this later in the book). It's also responsible for disabling an individual in social situations where it detracts from being able to divide one's attention to accommodate the social nuances of shared attention between 'self' and 'other'. Of interest though is when Autistic individuals get together socially there is usually lots to share and relate too, especially if shared interests are involved (Anthony et al., 2013; Lawson, 2011).



Being Monotropic (single minded and single focused) leads an Autistic individual towards focused interests and single flow states (Rapaport et al., 2024). In a world that is dictated to by polytropic interests (arising from divided attention seen more often in the Allistic population) the Autistic state of flow is not usually appreciated or accommodated. In the DSM-5 (2013, p. 22), aspects of this state of flow are referred to as rigid and/or repetitive interests (I prefer passions) some use the term 'focused or special interest'. For more on being monotropic, please explore the links below.

See: <https://www.youtube.com/watch?v=MUDQD1p2zFE>

And: <http://www.autisticscholar.com/monotropism/>

And: <https://en.wikipedia.org/wiki/Monotropism>

When, however, an Autistic individual is allowed to monopolize upon their 'state of flow' and utilize the very essence of who they are, they often thrive (Pellicano et al., 2022; Rapaport et al., 2024). Of course, having said this, no one multi-tasks well. It's of interest that even Allistic people find multi-tasking taxing.

See: <https://blog.rescuetime.com/multitasking/>

And: <https://www.apa.org/research/action/multitask>

And: <https://www.youtube.com/watch?v=5eQyfirx2HA>

This is important because although multi-tasking can be difficult for all of us, listening to many signals our body emits is usually an easier operation for Allistic individuals than it is for Autistic individuals. The main difference appears to be that the focus of autistic attention is deep and narrow and for Allistic individuals it's shallow and diffused or spread across a wider span. This makes it seem like Allistic

individuals are multi-tasking, but it's more about connecting attention more widely and noticing more widely too. So, for the external sensory system in Allistic people seeing, hearing, touching, tasting, smelling, as well as having a sense of balance and an awareness of where you are in time and space is more accessible than it is for Autistic people. When an Autistic person is governed by a narrowed attention, perhaps even hyper-focus, they may experience their external senses as disconnected but heightened or underwhelmed, and/or will move between the two (Elder-Robison, 2008; Grandin, 2006; Hendrickx, 2015; Lawson, 1998; Lawson, 2011).

For example, when an Allistic person smells dinner, they may notice they are hungry or not, make a move towards the dining table or not, experience mouth-watering sensations or a feeling of fullness as well as notice the way to move towards or away from the kitchen or dining table and so on. For an Autistic person, their attention may be already occupied with what they are doing and either not allow them to notice (they are hungry, full or somewhere in between and then can make a decision to leave their task to move on to the one their external senses are drawing them towards) or the sense may overwhelm them as too strong a smell and rudely invade their occupied attention causing them to feel distressed (Lawson, 1998; Lawson, 2011; Williams, 1996).

Autism Assessment in Adults: Differentiating the Complexities of Autism from Co-occurring Mental Health Issues

Seeing autism (via the lens of monotropism) and then exploring the characteristics of autism's impact upon an individual's life will help us separate autism from 'mental ill-health'. Once we 'comprehend' what autism looks like for any one individual, the idea of mental fitness and of poor mental health in comparison to their Autism will also be highlighted. The impact of a mental health diagnosis in adulthood can lead to self-injury, employment issues, relationship problems, and further mental health issues including suicidality (Hedley et al., 2022). Having an appropriate assessment, if autism and/or other neurodivergence is suspected, can lead to concepts that explain the nature of a difficulty to an individual and illuminate their experience to help gain insight and access appropriate support. Of course, autism and mental illness can co-occur, but we need to separate out what belongs to which so we can give appropriate support.

Compared to the Allistic population, the autism spectrum (AS) is characterized by difficulties/impairments in social communication combined with single focused (restricted) interests and behaviours (APA: DSM-5, 2013). There may also be sensory sensitivities, such as sound being perceived as too loud, light too bright, light touch as painful, food textures as uncomfortable and fragrances that may cause nausea, plus a number of variations on this theme (e.g. senses can be over or underwhelmed). I write more on sensory differences in Chaps. 5 and 6.

The Centre for Disease Control and Prevention (CDC) states Autism occurs at a rate of 1 child in 44—8-year old's across the USA. For adults, the CDC states more than 5.4 million people in the USA— or 1 in 45— over age 18 are on the spectrum (Dietz et al., 2020). Traditionally, researchers have put figures at 4–1 for autistic males to females. However, it has become clear that women are often missed in this figure.

With the DSM-5 TR (March 2022), I'm concerned on so many fronts. If an adult has lived most of their life 'struggling' to fit in with the Allistic community (where they do not fit) and have adaptively morphed (Lawson, 2020) into social society to keep safe, find friendship, employment, and so on, then separating Autism from the mask adapted for that purpose can be hard to do. If we are not careful, we will miss such individuals and this only adds to their woes, unemployment numbers, poor mental health, suicidality, and relationship break-ups (Hedley et al., 2017).

The External Sensory System (I write more about this in Chap. 5) Sight, touch, smell, taste, hearing, knowing where we are in time and space (proprioception) and vestibular (ability to keep our balance) are all means by which the human body detects outward sensory information that allows a person to notice information about their external environment.

For example, we can 'see' the sun setting at the end of a day at the beach; as we listen to and hear the ocean waves crashing onto the shore; as we smell the salty air arising from the ocean spray; and we can taste our ice cream as we walk along the beach and stay upright as we walk making sure we don't let the waves splash too close to us, so we stay dry!

The external sensory system keeps the human body informed appropriately so we can avoid external dangers (not walk into things usually), enjoy our dinner (taste), appreciate the forest after rain (smell); listen to the birds singing at dawn and so on. But the external sensory system does not alert us to internal goings on within our body. This is the job of our eighth sense (our interoceptive sensory system). I write more on this in Chap. 6.

Sometimes, for Autistic people, it can seem like we are very sensitive to our external surroundings; to what we hear, what we see, and what we smell. At times our senses can seem over tuned to picking up on these and things can seem too loud, too bright, and too 'smelly' while at other times we experience being underwhelmed by these senses. Then again, if the Autistic person is 'in control' (can access the volume button), they will be able to have music louder, lights brighter, and so on. This is a direct result of being monotropic. You can read more here: <https://en.wikipedia.org/wiki/Sense>

Autism and Cousins If an individual is living with other competing issues, such as ADHD (attention distractions with or without hyperactivity) which co-occurs with autism around 30–75% of the time (see: Young et al. (2020); learning difficulties, epilepsy and others, these will change the autism experience for the individual too.

For example, being autistic and ADHD will mean living with a circular monotropic disposition that might make it difficult to finish off on any particular interest, possibly having ‘fingers in many pies’ but, still battle with social anxiety and tend towards singular interests or the need to do one thing at any one time, even if it means several things on the go at once but serial fashion! Being monotropic explains much of those experiences that are not ADHD, but that co-occur. I write more about this in Chap. 10. Being autistic with various co-occurring dispositions doesn’t make life in this shared world any easier but it doesn’t have to mean we can’t live a fulfilling life; it just means our lives will be fulfilled differently.

The DSM-5 clearly states that autism can ‘manifest’ when demand outweighs capacity, which has been the case over the years for many individuals, they just didn’t have a name for it. Those individuals who do not have an intellectual disability nor do they show evidence of having a learning difficulty or personality disorder, may appear to fulfil criteria for some mental health condition. It’s important to recognize some ‘autistic characteristics’ which might mirror the issues seen in GAD (e.g. generalized anxiety disorder), for example. To an outsider and individual who doesn’t ‘get’ autism, especially in females, anxiety may look like general anxiety, especially when in situations of demand, in noisy, crowded or sensory overwhelming or underwhelming (boring) situations. So, learning how to sift through these and separate autism from poor mental health issues is vital.

If an individual has issues with attention switching, communication, a tendency towards rigid thinking and an exaggerated attention to detail, they may acquire OCD as a diagnosis, due to the repetitive nature of their actions. But, repetition in OCD is usually not welcome and is driven by obsession. This is not the case in autism as repetition is desired and wanted. Autism might cause individuals issues with rigid behaviour, anxiety, and processing social information quickly enough to respond with confidence, in situations outside of feeling connected and useful, or when feeling sensitive to being judged or criticized. Those who have learnt to mask and manage for short periods of time, but who cannot sustain this, will become easily fatigued. Fatigue can lead to loss of motivation, and this can lead to depression and despair.

The concept of ‘camouflaging’ or ‘adaptive morphing’ that is often seen in Autistic individuals who are struggling to find ways that keep them socially safe acts as a cover for many. But masking is exhausting and can rob an individual of all connections to ‘self’ making it difficult to have good self-worth and value, unless ‘masking’ who they truly are, which is unsustainable. Eventually this becomes too much and either an individual will limit ‘social exposure’, develop mental health issues, and/or feel ashamed and disconnect from their ‘real’ self.

So, the characteristics of autism that are common are:

- Being monotropic (foundational).
- Taking life literally.

- Having a hard time with grey areas; preferring ‘back & white’, rigid rules to live by.
- Liking and needing structure and routine (might be different if you have a Persistent Drive for Autonomy (PDA) profile).
- Not liking change; feeling distressed by change.
- Becoming quickly overwhelmed by too much sensory input.
- Having difficulty recognizing ‘connections’ from our interoception system (Internal senses) so may not notice when we are tired, hungry, unwell, changing our mood, thirsty, in need of the bathroom and so on.

Also, we may have difficulties with ‘Object Permanence’ (OP) too. This usual cognitive skill is acquired by children as young as 12–18 months. But, in autism all too often ‘out of sight means out of mind’.

Unfortunately, derailed OP can be mistaken for a lack of Theory of Mind (ToM). These are not the same thing, but behaviour can mirror these making it difficult to sift through.

For example, a family chooses to eat out. They take their Autistic daughter to a restaurant. Once there, she chooses where to sit and what to eat. A particular waitress is assigned to their table, so she serves them and takes their order. For the Autistic daughter, this whole event may become an understanding of what happens each and every time they eat out (a script for eating out). Being monotropic makes it hard to forward think so we stay with the steps of an event and expect these same steps each time we visit that event. For this family upon visiting the same restaurant, the young girl became very disruptive as the ‘expected’ table, chair, waiter, and menu had all changed and nothing was ‘as it should be’ in her mind and experience. Without recourse to an alternative way to understand this (social story, pictures that depict a change or/and forewarning of what ‘change’ may look like), she was distressed and traumatized. Her behaviour (shouting, fighting with Mum) could look like a lack of ‘theory of mind’, but it’s poor OP (Lawson & Dombroski, 2017).

In **summary**, monotropism is a cognitive style central to autistic cognition (Dwyer, 2022; Lawson, 2011; Murray et al., 2005; Rapaport et al., 2024). A monotropic mind is one that focuses its attention on a small number of interests at any time. This implies difficulty shifting attention and processing multiple stimuli.

The following chapters define, illustrate, and build upon each other in ways to help the reader gain a comprehensive understanding of monotropism, what it is, what it means for Autistic individuals, and what ‘supports’ may be needed because of it. Gaining this understanding will make it easier for Autistic individuals to avoid poor mental health outcomes as their medical and allied health practitioners will be better equipped in supporting and working with their autistic clients/patients.

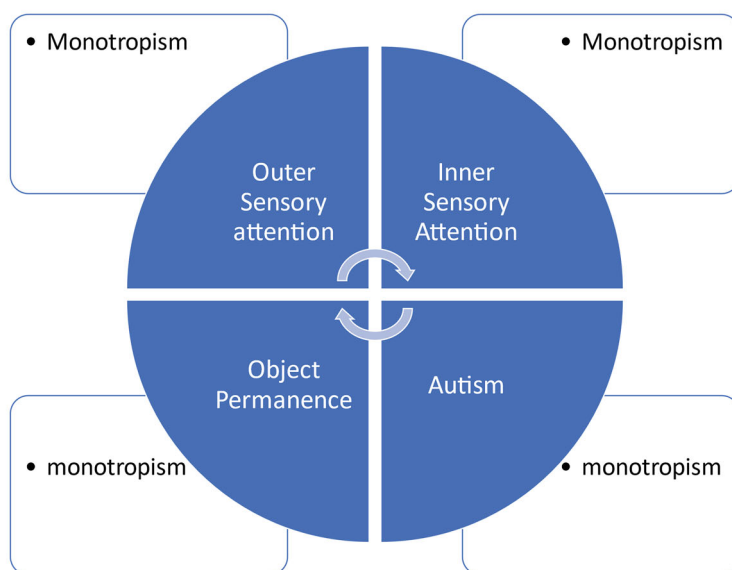


Diagram illustrating connections to the monotropic cognitive style in autism

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Chapter 2

Monotropism Theory



The monotropism hypothesis, also discussed under the name Single Attention and Associated Cognition in Autism (SAACA) (Lawson, 2009), is one of the modern, perception-based explanatory hypotheses of autism, alongside the model of Enhanced Perceptual Functioning (Motttron et al., 2006) and the Intense World Theory (Markram & Markram, 2010). As a proponent of the monotropism hypothesis, it is assumed that changed neurobiological structures, processes, and functions are at work in the brain compared to Allistic brains. However, a changed brain structure does not necessarily mean a deficit (Lawson, 2011, p. 27, 108). Rather, we see in monotropism a reflection of a strengths perspective, the cause of autism-specific learning behaviour and the reason for the often-cited special interests (Murray et al., 2005).

Murray (2018) states:

The central idea of monotropism (a word coined for Murray in 1992 by Jeanette Buirski) is that in autism, processing resource strongly tends to localize and concentrate to the exclusion of other input; an atypicality from which many other differences can be seen to follow. Understanding this concept fully requires a view of mind as a system of interests which inform cognitive, perceptual, and emotional processes (personal communication).

According to the current state of research, the exact connection between a changed genome, changed neurobiological structures, changed synapse activity of certain neurons, changed neuronal connectivity and functioning in the brain (e.g. reduced or increased connectivity between sensory and cognitive processes), changed cooperation between long-term and short-term memory, specific changes in brain areas such as in the limbic system (amygdala, angular gyrus) as well as in the neurotransmitter system, and the concrete effects on the behaviour and experience of an Autistic person are still unknown (Donovan & Basson, 2017; Hashem et al., 2020; Lawson, 2011).

Against this background, the starting point for the approach of the monotropism hypothesis is the assumption that a person's attention is quantitatively limited such that there is 'competition between mental processes for the scarce attention ... in

shaping the cognitive process' (Murray et al., 2005, p. 140). The strategies used to allocate attention are largely genetically determined and correspond to a normal distribution.

While Allistic individuals can focus and distribute their attention on many activities and areas of interest almost simultaneously, Autistic people may focus on few interests and usually do not divide their attention outside of a captured focus and/or a state of flow (Rapaport et al., 2024). This concentration on a few, but 'highly aroused' interests that demand a lot of attention and may not be connected to one another is much more intense. This monotropic form of perception, is the cause of changed thinking, experiencing and acting by Autistic people compared to Allistic people who, with their broad attention and many less emotionally charged, interconnected, and corresponding interests, tend towards polytropism (Lawson, 2011, p. 20), consequently also towards the so-called multi-tasking. It's worth noting here that no one multi-tasks well, and it's actually not as productive as focusing on one thing at any one time (See: <https://www.verywellmind.com/multitasking-2795003>). But switching attention as Autistic people can be a slower and more arduous process.

Figuratively speaking, one could also describe it in such a way that a neurotypical person's use of attention can be compared to the use of the broad beam of a torch to highlight more of the bigger picture, whereas an autistic person's attention is like the bright core, or focused beam of that torch. This is very effective in a small field and can bring more detailed structures to light (Mundy & Newell, 2007, p. 269).

Many of the typical autistic characteristics and experiences can be explained on the basis of this model of understanding (Lawson, 2011, p. 21).

For this purpose, there follows five cognitive components of the SAACA concept (monotropism):

1. **Literality** (e.g. literal meaning of words, sentences, idioms, situations, forms of expression) rather than reading the 'hidden curriculum' of human behaviour.
2. **Thinking in closed concepts** (e.g. staying with rules, habits; resisting change, not broadening or connecting concepts). For example, I have this pen from your desk in my hand, so it is mine.
3. **Context and scale** (e.g. difficulty generalizing things, situations, or learning, taking in the wider perception/proposal). For example, the scale of final, as in death, or the notion that sleep is temporary.
4. **Timing, sequencing and predicting** (e.g. difficulties in organizing timings, intentions or the completion of various tasks). For example, appreciating consequences which will negatively effect executive functioning.
5. **Non-social priorities** (e.g. preferring casual wear, that one has grown fond of, instead of expected formal wear on festive occasions; not engaging in 'small talk').

Following the monotropism hypothesis, it is concluded that we (Autistics) have a different 'wiring structure' in our brain as Autistic people compared to the general population, which also influences how a person's attention and interests are directed.

As a result, repetitive, similar, and focused interests and behaviours for us, as Autistic people are more likely to be attributed to altered attention processing and are less to be seen as a sign of a poorly developed ‘Theory of Mind’ (Lawson & Dombroski, 2017).

Of course, one could argue that the inability to put yourself in another person’s shoes is a clear sign of selfishness and complacency, (why anyone would want to wear another person’s shoes is beyond me). The term autism (autos: self) allows for such a short-sighted conclusion; but what if this perceived self-centredness resulted from an Autistic person’s total attention being consistently focused on a small spectrum of activities, rather than the person dividing their attention between self- and other-related activities (Gomot et al., 2008)? What if this problem was more due to attentional differences, and what if Autistic individuals could compensate for this by connecting through their interest systems? An explanation or answer to these questions unfolds as the reader further the chapters in this book.

For all of us, as human beings, many of our cognitive processes include the ability to access:

Executive functioning (EF) (a formal brain function) necessary to:

- plan ahead
- organize oneself
- take into consideration a number of events
- be mindful of others’ emotions

However, having good EF in autism is often an issue. We might become passionate about things that interest us and if this includes planning and ordering fair enough but, without appropriate access to the bigger picture we will find concepts associated with EF very difficult.

Central Coherence (CC) the activity of the brain to:

- draw all the bits of information that are coming our way through our senses, together, to create a fuller picture.

In autism, CC has been said to be lacking or poorly formed. However, it’s not the CC ability that’s lacking but rather it’s the difficulty ‘picking up on’ the signs or concepts associated with having good CC that we have issues with (Lawson, 2011). This might mean we are accessing an incomplete picture when compared to individuals with highly developed central cohesiveness. However, once these ‘signs’ are portrayed in accessible formats (placed within our ability to grasp) we are enabled to ‘get the picture’. When it comes to other disability fields, visual impairment for example, would we say that because an individual is visually impaired, they cannot ‘get the picture’? No, we would enable them to ‘see’ using different means (e.g. brail; touch; talking books and so on). This is what needs to happen in autism. I believe that using our focused attention and interest is the key. This is the connecting factor or common denominator.

In being Allistic if your attention is in many directions it will mean:

- The need for ‘executive functioning’ is high, but there is a low need to attend closely. Therefore, EF comes relatively easily giving rise to multiple connections (or to a global map allowing cognitive, emotional, behavioural, and thinking plans to take place).

In autism though:

- We tend to attend to one thing at any one time (unless we have shared things already within our attention tunnel (e.g. related interests).
- This means interests dictate.
- We are driven to organize and plan.
- So, thinking ahead may only occur in connection to interests.
- Our focused attention only applies to chosen selected details.
- Therefore, the big picture may be viewed as unimportant, or simply missed.
- However, coherence is not weak.
- But some signposts are missing.
- In autism generally the need to order, orchestrate, and achieve coherence is very high.
- But this will necessitate using up attention in large amounts, which is very extreme and very tiring.

Allistic

If you are not autistic, you are most likely part of a community with learning styles informed by being multi-channelled (focus enabled via ability to divide attention) and able to accommodate the non-literal sense of language (spoken and body language). This enables you to read between the lines and to attend to the hidden social curriculum (those social constructs understood without specifically being spelt out such as understanding facial expressions, body language, yours and/or mine, turn taking, public and private, and so on).

It means you cope relatively well with change and even invite it.

You can construct a richly connected information store and access memory (factual and non-factual) after storage. You learn initially by noting and modelling on other people’s interests (events that capture joint attention).

This means the general learning style for Allistics is a polytropic one.

- Polytropism (Murray et al., 2005; Lawson, 2009, 2011)—divided attention governs the interest system (able to have and move between different interests).
- Being non-literal enables you to cope well with metaphor and reading between the lines in all aspects of communication.
- This also enables you to think in open concepts—(e.g. the past informs the present and the future).

- Being governed by generalizability—(the transfer of understanding across different domains) is usual for you.
- Gives you the understanding of timing and sequencing needed for daily life, usually allowing you to plan and not panic.
- You are enabled to forward think outside of your interest—therefore, you may be particularly sensitive to others, but know how to let go of ‘stuff’ that is not yours, know about appropriateness and propriety.

In other words:

- Polytypism: (thoughts/feelings connected).
- Non-literality: (understands can’t (s), glance, and stance. Practices propriety).
- Thinking in open concepts: (space for accommodation of change, difference, and diversity).
- Generalizability: (puberty to adulthood).
- Understanding timing and sequencing (when, where, and how).
- Can forward think outside of one’s interest: (understand and accommodate needs of other).

Autism: Single Attention Governs Interest System (Difficult to Accommodate ‘Different’ Interests)

- Literality—rules and social understanding will be ‘fixed’. Literality: (instant gratification: Now).
- Thinking in closed pictures—the past does not inform the present so accommodation might not occur.
- Lack of generalizability—little transference of skills between situations, expectations, problem-solving opportunities, and so on. Also: Lack of generalizability in accommodating beyond the self (his, hers, ours, etc.).
- Timing and sequencing difficulties.
- Forward thinking tends to occur when connected with interest only.
- Might mean you only attend to self or your passion, sometimes one passion might involve another, but it will always mean we attend to feelings or thoughts, not both at once).
- Thinking in closed pictures: (feelings govern; difficult to be open to reason (which is thought based)).
- Timing/sequencing difficulties (timing off).

Polytypism for example: if the person one is relating to is also used to dividing their attention, then the way the two of you relate will probably be mutually understood. You will understand power dynamics, relational balance, hierarchies, and so on.

Child to adult (who is in charge?)

- Employee to employer (who is in charge?)
- Equal relationship; equal say (partners)
- Student/teacher
- Boy to girl (child to child)
- Policeman to pedestrian
- Patient to doctor

If, however, one's learning style and cognitive processing abilities are dominated by single attention and associated cognition in autism (Monotropism or SAACA) the story will be different:

- Attention is all in one place (difficulty switching). Therefore, difficult to work out 'intention, context and scale' (physical, emotional, or social intention; physical, emotional, or social context; physical, emotional, or social scale).
- World around us is unpredictable, fragmented, disconnected, and alarming.
- Information store confusing (unable to decipher it). We need a reservoir of strategy and learnt rules, which constantly need adding to, because some aspects of intuition may not be available as a result of not being great at dividing attention and reading between the lines.
- You could argue that we have differing perceptive qualities. Our perception is like that illuminated by the narrow beam of the torch (lighted area pinpointed) compared to the allistic population who operate by knowledge gained via the wide beam of a torch (light spread further afield highlighting a wider area).
- Therefore, what do I understand, what do you (I might not get the same picture) so finding ways that work for us both so that our communication is accessible and mutual are needed.
- Cognitive qualities may lead to difficulty processing quickly or/and accurately. As Autistics, we are more likely to be geared towards constants, sameness, familiarity, and routine. This will mean a need to be prepared so we can access any change that may occur.

For information on some of the issues, these differing neuro qualities infer, see Milton's double empathy problem (Milton, [2012](#)).

In autism what does it all mean/require of me?

- **Emotional consequences:**
 - What are my dominant emotions? I may be controlled by anxiety, stress, and fear with lots of discomfort and mistrust (if I am not connected to understanding) or I may feel comfortable, safe, and accommodated (when my needs are accounted for and accommodated).

An Example of Being Literal Within a Counselling Situation and How this Impacts the Bigger Picture or Client Perception

Being literal in one's thinking isn't only about taking words literally. It might mean reaching a conclusion about something, someone, some feeling, and so on, by association. The autistic thinking demonstrated below shows how an individual response can be taken literally leading to a core belief based in paranoia and poor self-esteem. It also shows the reasoning used by this older female (aged 55) as an attempt to integrate the experience (gestalt) to relieve their discomfort.

Scenario: Counselee has a session with their counsellor and 'pours their heart out'. The counsellor is encouraging, warm, and accepting. The session eventually ends, and they part company until the next time. The counselee is out shopping and passes the counsellor crossing the road. The counselee smiles towards the counsellor who appears to deliberately look away, without acknowledging the counselee. The counselee, an autistic female, is deeply wounded. What the counselee saw and perceived from this event and the 'beliefs' it led to are below:

The picture: A person I have shared deeply with doesn't acknowledge my presence.

The belief:

1. I am not important to that person.
2. When we were in the session, they only acted caring towards me because it's their job.
3. People will always reject me because I don't have the skills or personality that connects with others.
4. These people cannot be trusted.

The outcome for this individual adds fuel to her depression and poor self-esteem. She is reluctant to see this therapist again. But, in an attempt to put her discomfort to bed she has a go at rationalizing it. She tries to find a reason, other than the above, to explain the counsellor's behaviour.

Upon meeting up with a friend, our counselee mentions her experience and shares her discomfort. Her friend empathizes with the scenario and says a similar situation happened to her. They chatted together over this and conclude that may be there is a protocol for professionals and, the counsellor was following this protocol. If so, she might have chosen not to acknowledge the counselee outside the professional setting, so to keep things separate: the professional working relationship and the social one. If engaged in a professional relationship, one doesn't pursue a social one. The counselee accepts this was, therefore, the reason for the counsellor's 'rude' behaviour. No other possible reasoning is added. The above reasoning appears sound and the counselee is relieved of her discomfort.

Problem

The autistic individual needs to fill the gaps in their thinking. This leads to the need to find ways to explain and rationalize things that happen. In the typical world, Allistics use attention in ways that allow them access to scenarios based upon broader awareness of human imperfections, changing circumstances, personality, not taking things literally, and so on. In autism, ‘reading’ others based upon these characteristics is usually offline. We tend towards taking things literally, building gestalt from ‘perceived’ projections and then fitting a potential personal puzzle piece into the picture we are building.

So, the above ‘solution’ to the counselee’s dilemma seemed reasonable, rational, and all conclusive, allowing the counselee to feel better. However, it is too simplistic, fails to broaden awareness and could be misleading. This is not to say that keeping professional conduct separate from social conduct is not ‘protocol’, it is. But the nuances of how that can happen are being overlooked.

The counselee, in this situation, is prone to fitting puzzle bits of social information and situations, even fleeting ones, into a single attentional mode. By this I mean, one based upon ‘one core belief’. This belief, so common in girls on the spectrum, is: ‘I am not valuable’. ‘No one likes me, everyone rejects me, I will always get it wrong’ and so on. So, each social encounter is knitted into the garment of this belief and is worn on all occasions. Attempts to undo this established way of thinking are difficult to do.

Problem-Solving

1. Each belief needs to be clearly spelled out and each row of knitting needs to be unpicked.
2. To do this, the individual needs to be willing to explore different scenarios, in different settings, with different people. But this needs to be a safe environment with chosen trusted and committed others, who are in it for the long haul.
3. Role play can be employed to act out various scenarios that show individual human behaviour. Concepts need to be taught. Concepts such as human infallibility; areas of greyness rather than only black and white; yours and mine; me and others; kindness, rudeness, and indifference; good self-esteem; personality differences; intention and actual action, and the possible meaning behind these and so on.

Guidelines and protocol seem all inclusive, but even these have degrees and grey areas that cross over and between borders. It’s these concepts and many associated others that, as autistics we miss and, Allistics taken for granted.

Monotropism and the autistic profile of PDA (pathological demand avoidance) **or persistent drive for autonomy.**

PDA is important to understand because an Autistic person with this profile will not benefit from the structure of routine and scaffolding to support their day. Rather the expectation found in routine will be experienced as demand. When PDA individuals encounter demand, even of their own volition as well as that of others, their environment, their interoception system of sensing and of general daily needs, may cause them to ‘freeze’, fight, flop, fawn, or take flight, to get away from the demand. Demand can be experienced as ‘sheer dread’ and can totally immobilize an individual.

Persistent Drive for Autonomy

PDA is most often considered as a particular profile or specific set of characteristics within the autism spectrum (See: <https://www.autism.org.uk/advice-and-guidance/topics/behaviour/demand-avoidance>).

Although PDA is not in the DSM-5 (2013, p. 22), it is widely understood as a ‘profile’ for some autistic individuals (Kildahl et al., 2021; O’Nions et al., 2013). PDAers share similar qualities and characteristics as many others in the world of autism (e.g. dislike of uncertainty and difficulties with change, sensory over or underwhelm), but the obvious difference that clinicians need to understand when assessing for autism, or working with PDAers is the way an individual can use social expectation to mask their difficulties. This might mean seeming to have good eye contact, being able to ‘blend in’ with social situations and have an apparent understanding of friendship, mutuality, and accommodation of ‘other’. However, these traits are often superficial, ‘practiced’ social skills that are developed as secondary characteristics to ‘appear’ capable and confident as ways to hold onto autonomy.

At times an individual’s behaviour can look like that seen in oppositional defiance disorder (ODD) placed under the new category ‘Disruptive, impulse-control, and conduct disorder’ in DSM-5 (2013). But, in PDA the root cause for ‘manipulation’ to avoid any type of demand is more to do with stress, anxiety, and fear, rather than to be spiteful, revengeful, and vindictive (as seen with the ODD individual) See: Andrews & Istanffy (cited in Kimbel, 2012).

In PDA, this is not wilful defiance or simply wanting to get out of having to do something, it’s a product of fear and anxiety based around uncertainty. Even when a PDA individual chooses an activity they want to do, panic can be triggered, and the PDA brain moves them into avoidance so they can gain back some control and autonomy. PDA is the driver, and the central common denominator is the perception of demand in any action required of them (Green et al., 2018; Stuart et al., 2019).

In PDA, the usual tools employed in autism to help an individual predict and understand change and expectation will need to be different to those traditionally used in autism. Direct instruction can cause panic. For example, when I said to my granddaughter, ‘Mummy cooked dinner, we can wash up the dishes, so we are helping too. How about I wash, and you dry, or we could do it the other way around?’ Her response to me was:

Yes it would be good to be helpful, and usually I am. But today I'm a dog and I don't have hands, only paws, so I can't help with the dishes.

This was her way of avoiding the fear and anxiety the request stirred within her. Even though she has completed such tasks often, she still has difficulty envisaging what it might mean (poor object permanence). Her verbal ability is very good, and she said all this with a smile. It would be so easy to believe she is being silly and difficult, and I should just be firm with her. But, in PDA we need to work in relationship with a foundation of trust and remove demand from our conversation. For example, at a different time, when I said to my granddaughter, 'Mummy cooked our dinner, there's the dishes, but I can't work out what I should do'. She said: 'You wash I'll dry'. Now, this way of removing demand from our interaction won't always work. But, understanding PDA and aiming to reduce demands where possible, make actions fun and use distraction and games, build upon trust in our relationships, keep things light-hearted, and use negotiation, will mean we have a better chance to grow and work together towards a positive outcome. If we don't, we risk our kids becoming the product of PTSD as they constantly experience trauma in reliving anxiety and fear.

<https://www.youtube.com/watch?v=0gCXwBh2saQ>

<https://www.pdasociety.org.uk/>

<https://www.facebook.com/PDAAustraliaNewZealand/>

Conclusion

Although monotropism as a foundational 'theory' about autistic attention and its relationship to 'interest', research and autistic voices are in agreement. In the past, Autistic people have not been included in research about their lives and their experience, but this is changing, and the change is bringing an exciting window of understanding that has not previously been realized. Being monotropic impacts every aspect of our lives as Autistic people. This chapter has only considered our 'base' autism and also how the PDA profile in some Autistic individuals can be an extension of monotropism. The following chapter explores monotropism from the autistic perspective.

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Chapter 3

Autism from the Autistic Perspective



Introduction

Every autistic person (like any other person) is unique and individual. We are all very different from one another and this is because of age, personality, gender, and various other intersectionalities including culture, sexuality, social status, ethnicity, and every other human characteristic as well as our unique autistic profile. The common denominator for us all though, as autistic people, appears to be in how we process information or in our ‘cognitive style’ (Murray et al., 2005). Our cognitive disposition includes being monotropic, which is the lens through which we view everything else. Because we are singly focused and better at noticing and doing one thing at a time, rather than attempting several things practically simultaneously, this puts us in the minority. However, it’s common for many Allistic individuals to feel confident in conversation, while walking, thinking, processing, and noting things ahead (danger, hunger, boredom, personal needs, and so on), but this may not be the case for Autistic individuals. Usually, we focus upon one thing at any one time. This means we use up an enormous amount of concentration and energy but for smaller demands on our time (Elder-Robison, 2008; Grandin, 2006; Lawson, 1998; Lawson, 2011; Williams, 1996).

Being autistic means many different things to different people. Being monotropic has lots of benefits, at least when it comes to being able to focus upon areas of interest. But, in the world outside of autism, the one where Allistics live, being singly focused (in social situations especially) is not considered an asset. This is because Allistics have a learning style that makes it hard for them to stay focused upon one thing for long periods of time (OK, they might find it easier if they need to be focused, but they still hear their name and can respond, which is quite different from those of us who are Autistic when we are deeply focused and possibly in ‘a flow’ state). Instead, Allistics operate via many (poly) different angles/channels (tropic) making it easier to connect socially, and to cope better with change, interruptions, and so on.

For social engagement to happen well, usually individuals need to join one another's interests and share interest across varying domains. Allistics take no effort to look at someone, while listening, thinking, processing, and even moving, all at the same time. This results from their polytropic cognitive processing style. Allistics have a sensory system that is well connected too, so looking while listening, while walking, and so on, all comes naturally to them. But this is not so for us, as Autistics (Dwyer et al., 2024; Elder-Robison, 2008; Lawson, 2011; Perkis, 2006; Williams, 2006).

It's interesting because within our Autistic community this is a given. We gravitate towards others who might share the same interest or disposition. For example, I'm very much at home with other Autistics who love talking and sharing about autism (one of my interests) dogs and animals or birds. These are all passions of mine. If, however, I'm in a restaurant or pub and people are talking about Sport (the footy), I'm switched off and cannot join their conversation. Often, if I really want to connect to the group I'm with, I'll direct the conversation towards something I'm interested in, so I can engage. Outside of interest though I find it very difficult to maintain a conversation.

I recognize my Autism, like most of us, means a very spikey profile. It means I have passionate areas of interest where I can excel and others I just 'can't do'. For some of us, this may be the washing up!

For example: '...I love to draw and my art work gives me great joy, but housework, like doing the dishes, is just too hard...I mean, I can't work out where to start, it all looks too messy and too overwhelming' (Personal communication, 2022).

Autism rarely travels alone, it has cousins. So, aphasia, auditory processing disorder, ADHD, poor interoception, external sensory challenges, alexithymia, dyslexia, dysgraphia, dyscalculia, and others will impact upon our abilities with various responsibilities that could mean we just can't manage some things without support (I write more about these in Chap. 10). This could mean someone else doing these for us. It can look like we are being lazy, not caring, being difficult or procrastinating, but some things are just not in our arena, no matter what we try! It's one of the reasons I wrestle with 'Imposter syndrome'. How can I have a PhD but not be able to do my own shopping? Well, the answer is 'autism' and accepting my 'limitations' and requesting 'help' is not saying I'm not bright enough or I'm being difficult. It is saying I'm learning to have 'self-compassion' and recognize the need to take care of my 'mental health'. See: <https://researchers.mq.edu.au/en/publications/cultivating-self-compassion-to-improve-mental-health-in-autistic-#:~:text=Self-compassion%20may%20improve%20mental%20health%20in%20autistic%20adults,strategy%29%20or%20its%20interactions%20with%20emotion%20regulation%20processes>

Why Do We, as Autistics, Connect and Learn in the Way We Do?

Coupled with the understanding of how autism impacts upon individuals in various ways, there is research showing why. Although this is for a variety of reasons, there is consensus that Autism is a developmental disposition. This allows a number of different brain architecture differences from neuron-usual brain development (Allistic) that make it harder to multi-task, forward think, cope with ‘change’, access a generalized understanding of object permanence, and process general information (especially from one’s senses, felt, spoken, and/or written) (e.g. Gomot et al., 2006; Lawson, 2013a, 2013b; Lawson & Dombroski, 2017; McKavanagh et al., 2015) while enhancing local discrimination and allowing attention to things beyond one’s interest (Lawson, 2011, 2013a, 2013b; McKavanagh et al., 2015; Prato et al., 2018).

The above is why we need to pay attention to ways to increase motivation in Autistic learning, so we can discover more opportunities for connection. This means all medical personnel working with us need to understand not only ‘autism’ in general, but the autism that the person they are seeing, is living with. Autism as a spectrum, (AS) often comes with cousins, as mentioned above. For example, around 33% of autistic individuals also have an intellectual disability (IQ below 70) but 67% do not (Bourke et al., 2024; Shenouda et al., 2023). Other issues influencing learning relate to: sensory issues (hyper, hypo, and interoceptive) about three quarters of autistic individuals live with a disrupted sensory system (<https://www.spdstar.org/basic/co-morbidity>); epilepsy (5–46% of autistic people have epilepsy, compared to an estimated 1–2 percent of the general population (Spence & Schneider, 2009); attention deficit disorder with or without hyperactivity (ADHD) co-occurs with autism at an estimated level of between 35% and 75% (See: Young et al., 2020), then we may have learning difficulties such as specific learning difficulty (SLD), for example, (previously known as dyslexia, dyscalculia, dysgraphia, and so on) (<http://specificlearningdisabilities.weebly.com/what-causes-sld.html>) and other arbitrary dispositions that too often are unrecognized, such as gastro-digestive problems that parents report govern their children’s lives (<https://www.empowher.com/autism/content/study-looks-digestive-problems-autistic-children>), Pathological Demand Avoidance Syndrome (PDA) (I prefer Persistent Drive for Autonomy that Emily Wilding writes about) See: <https://emilywilding.com/neuropsychology/neurodiversity/pathological-demand-avoidance/> and others.

High and Low Functioning Autistics

I’m hopeful that such terminology as above is rapidly being dropped from popular use! These terms are ableist and not productive or accurate in how they describe Autistics. They imply that any person who differs from the mythical ‘norm’ is somehow ‘less than’. Because they presume that those who appear to manage life

tasks relatively well don't need any support, and those who have obviously higher support needs are less capable. As stated, Autistic abilities fluctuate and can be completely different on different days, at different times and during different expectations. Because these terms reduce the person to a 'functioning label' they rob individuals of their humanity and fail to see us as individuals with individual needs. See: <https://the-art-of-autism.com/understanding-the-spectrum-a-comic-strip-explanation/>

Rebecca's original comic strip story here: theoraah.tumblr.com

Sensory Differences

Many autistic individuals experience differences in the way connection to our body awareness occurs. This means our sensory systems have a tendency towards being hypo- or hyper-sensorially connected. This translates to individuals living with communication and social challenges in a world where sound, sights, tactile stimulus, gustatory sensations, and fragrances can over or overwhelm us. More on this in Chaps. 5 and 6.

Synaesthesia Synaesthesia is a rare blending of the senses (sight, hearing, taste) in which the stimulation of one sense triggers a c in another sense. For example, someone with synaesthesia may be able to hear colours, feel sounds, or taste shapes. At times this feeling is only subjective. Most people with synaesthesia are born with this condition, so we don't know anything different. However, once we tell people how we experience the world, we might be told we are hallucinating or going crazy. Being diagnosed with synaesthesia though is often a relief in these situations. (See: <https://www.wikihow.com/Tell-if-You-Have-Synesthesia>).

The team of scientists from Cambridge University found that whereas synaesthesia only occurred in 7.2% of Allistic individuals, it occurred in 18.9% of Autistic people: <https://www.cam.ac.uk/research/news/synaesthesia-is-more-common-in-autism#:~:text=The%20team%20of%20scientists%20from,they%20should%20not%20share%20anything>

Interoception

Interoception is thought of as the eighth sense and refers to the awareness of inner bodily workings, such as connection to hunger, thirst, pain, sexual desire, temperature, need for digestive elimination or urination, being able to notice heartbeat, breathing, and so on. If an individual isn't connected to this sense (due to not noticing because their attention is elsewhere focused), it makes it very difficult to self-regulate. Self-regulation is necessary for us to take charge of our emotions and behaviour and to delay gratification. The lack of connection to this sense has huge

implications for learning! I write more about this in Chap. 6. As Autistic people, we often have poor interoception due to our monotropic attention being already engaged, so no attention to notice other things, even pressing things like the need to use the bathroom.

See: <https://www.education.sa.gov.au/interoception>

Literality

Autistics are known for being very literal. This means we take language literally and can find translation of ordinary conversation very uncomfortable. For example, people talk in metaphors and don't 'say what they mean and mean what they say'. Some examples of typical conversations are:

...you could have knocked me over with a feather; it was such an eye opener to finally see through her; even though she dotted all her 'i's' and crossed all her 't's.

... it's raining cats and dogs out there. I couldn't see two feet in front of me. The weatherman got it wrong! He said it would be a toaster this weekend.

For many Autistics, the sentences above present images of feathers with muscles, eyes and can openers, an invisible woman with dots and 't's all over her; Dogs and cats falling from the sky, invisible feet in sets of two and toasters that act as 'ends' to a week.

When communication is presented verbally and contains lots of metaphors, some Autistics find it incredibly exhausting trying to decode the terminology!

Object Permanence (OP) is the understanding that objects have a life of their own, even when out of sight. This process of 'knowing' something can still exist, even if you can't see it, is a developmental one and typical children have grasped this often before their first birthday, and even as young as 8 or 9 months. However, in autism this process is likely to be delayed (Lawson & Dombroski, 2017). For example, an Autistic 5-year-old is invited to play a game of '... hide and seek'. But, when she cannot see or hear her sister (who is hiding) she becomes excessively distressed as she believes her sister is gone, forever. I write more about OP in Chap. 7.

Helpful Modifications for Some Sensory Experiences

These modifications may include decreasing reverberation by covering reflective surfaces (e.g. black/white boards not in use, linoleum or wood floors, untreated ceilings), using properly spaced acoustic dividers, using other absorption materials throughout open or empty spaces (e.g. unused coat areas), and/or changing the location of 'study' sites. External noise sources can be eliminated or moved away from learning spaces (e.g. aquariums, fluorescent lights that hum, an open door or a

wall) (p. 14 of 26) See: American Speech-Language Hearing Association (April, 2005) in reference list. From website: <http://www.asha.org/docs/html/TR2005-00043.html>

The Science of Motivation in Autism

Apart from numerous autobiographical accounts of how motivational interest in autism has opened doors to learning, to social understanding, and to both further education and ultimately to employment (e.g., Elder Robison, 2007; Grandin & Scariano, 1986; Holliday Wiley, 1999; Lawson, 1998, 2000; Murray 2006; Perkis, 2006; Sainsbury, 2009; Santamaura, 2012; Williams, 1998, to name a few), the role of autistic ‘passion’ and its place in learning is still queried.

A study conducted in 2013 by Ryan Kiser, Louisville University seemed to show that autistic participants had GAMMA in excess during a novel task concerned with auditory and visual sensory stimulation (Kiser, 2013; Lawson, 2013a, 2013b). Observations were made however, that showed when Autistic children in the study were interested and employed with a favourite game or topic it changed how GAMMA looked. Instead of being in excess and scattered, GAMMA appeared to be switched on and activated (Lawson, 2013a, 2013b).

When GAMMA is activated, it is doing its job of taking information from various brain waves (Alpha, Beta, Theta, Delta) which carry singular information enabling a person to notice, from alert states, to alert and connected to winding down and eventual sleep. GAMMA is known as the ‘binding’ band width enabling the brain to access the bigger picture. So, if GAMMA is offline the brain’s ability to process context, forward thinking or outcomes, issues related to object permanence, sensory dispositions, and other issues (as outlined above) are also offline. However, if autistic individuals, via motivation, are enabled to connect as GAMMA is switched on, this is the key to enabling connection in autism. I write more about this later in this chapter.

When individuals live with a sensory system which doesn’t build typical (polytropic-multiple) connections but is dictated to by a brain wired to process stimuli one at a time (monotropic-single), it’s very difficult to multi-task. Noise interferes as an added dimension because it calls away attention, often ‘highlighting’ the noise and exaggerating it. Yet, communication (between self and someone else) requires multi-tasking. Individuals need to notice the person, sign, and device ‘speaking’ to them; decode what is being heard, seen, or offered, then take any action, if one is required. These are hard things to do in autism, especially if your attention is called away by added extras, such as ‘noise’! Interestingly, as many families have noticed, Autistic children WILL multi-task when they are working, playing, or acting in an area of interest that motivates them. This is probably one reason in Wood’s (2020) study children communicated well due to motivation (e.g. when they wanted something different to that on offer).

When individuals have their sensory needs accommodated (they may need to wear sun glasses, a cap, a squeeze jacket or weighted garment, have chair/table legs padded, wear ear defenders, and so on), when specific learning difficulties are recognized and taken into account (possibly needing an adjusted curriculum, things colour coded and material ‘chunked’ rather than pushed all together) when students have time tables presented in ways that make sense to them (e.g. visually, on an iPad, and so on) and when their interest is central to everything, not simply used as a reward, but intertwined and interspersed throughout our learning, we will be happier, motivated individuals with less stress-related behaviours and more socially adept.

Autistic Attention

Dwyer et al. (2024) highlight the role ‘monotropism’ plays in Autistic attention pointing out the areas of challenge, as well as those of strength for Autistic individuals. Murray et al. (2005) note monotropism’s impact upon all areas of functioning; although these authors eventually all were recognized as being autistic, only one author on their 2005 paper had a formal diagnosis at that time. From the Autistic perspective when listened to and engaged with, current research where co-production is foundational, also supports the idea of monotropism being core to the Autistic experience (Rapaport et al., 2024; Dwyer et al., 2024).

The idea that Autistics lack empathy and are not able to join in social activities, is not supported, so much as the nature of the activity and the connecting interest for the Autistic person, is what counts. Murray et al. (2023) suggested that:

some characteristics long thought intrinsic to autism, such as social mis-perception and reduced empathy, may be alternatively understood as state-dependent outcomes contingent on specific contexts and interactions (p. 3).

Practical Example Reflecting back on my first labour (giving birth) experience delivering my first child, which was very painful and very long, (not unusual), was not connecting for me nor did it ‘typically’ occur. I was induced, again not unusual, but the lack of information given, let alone offered in ways that might make sense for me as an Autistic person, was sadly lacking. So, not being prepared for the hospital experience (sensory overload) what it meant to have an episiotomy, the pain and the attitude of the attending midwife (‘people have been having babies for thousands of years, it can’t be that bad, now stop making such a fuss...’) was extremely traumatic.

In the end, the sedation used for me caused me to be totally ‘out of it’. When I came to and asked about my baby, I was told that he was fine and, in the nursery, and that I would see him the next morning. I remember thinking that wasn’t what I expected and was very surprised and unhappy. I had read that ‘baby’ should be put to the breast after birth and there was no need to separate mother and child. My monotropic literal sense led me to believe certain things, which did not happen. I

was not supported to develop an appropriate connection to understanding and my mental wellbeing was threatened.

‘Being and Feeling’ these things didn’t mean I could easily convert that feeling into awareness and voice it. I worried very much about how my son was being fed and what was happening to him. My mental state meant I couldn’t sleep or rest and felt quite panicky. When I finally got to see him the next day, I was also surprised that I didn’t have any specific ‘feeling’ towards him. I mean, in the films on television mothers cried as they held their newborn. But I didn’t feel like crying. I only felt a kind of detachment and strangeness. This is the result of ‘false gestalt’ and a lack of generalizing of concepts as well as poor OP and literal thinking.

Looking back over that time and subsequent times for my following further three births over 8 years between 1974 and 1982, many changes to medical understanding had occurred. But the one thing that didn’t change then and is still an issue today is having the voice and needs of autistic individuals taken seriously. I echo here what I’ve written above: ‘But “being and feeling” these things didn’t mean I could easily convert that feeling into awareness and voice it’.

Another example of connecting the dots and being able to practically care for my baby, entailed the need to ‘do’ practical care for my baby, over and over, and over again. Watching how to change a nappy, bathe a newborn and hold them wasn’t something I could comprehend from being told, nor from doing it one time only. There is quite a bit of literature that suggests Autistics are visual learners (Grandin, 2006). But this isn’t the case for all of us. I’m an auditory and kinesthetic learner despite having auditory processing disorder. It’s important to remember Autistic individuals have varied learning styles as well as their monotropic cognitive style, personality, culture, and social disposition.

I remember my own birth. I remember the sensation of being pushed and squeezed along a tight space. I remember the blinding light as I gushed from my mother’s womb into the outside world. I also remember the harsh material I was wrapped in and feeling like I was being brushed up the wrong way.

Being Autistic begins before we are born, and our sensory needs do too. Having these understood and accommodated is the primary step towards helping us feel comfortable in any medical setting. We need medical and allied health teams to understand autism and to listen to the autistic individuals’ words of apprehension, fear, and even panic at times.

There will also be those among our ranks who will not connect to their experience due to our interoception disconnection (I write more about this in Chap. 6). There will be others who almost ‘over connect’ for similar reasons. All their monotropic attention will be in one place, possibly heightening certain experiences and being completely taken up with these. Rumination in Autism stems from this ability to focus on one thing, but an individual might not know how to let it go, so it continues to solely occupy one’s mind and thinking.

When clinicians are exploring possibilities of whether, or not, individuals are on the Autism spectrum they are guided by the description in the Diagnostic and Statistical Manual of Mental Disorders, edition 5 (DSM-5; APA, 2013, p. 22) and

in Europe, the ICD-11 (WHO, 2022). These manuals help describe criteria for many mental health disorders (DSM-5) and health and related disorders (ICD-11).

The DSM-5 houses criteria for many different ‘neurological’ dispositions from the autism spectrum to specific learning difficulties/disabilities (previously known as: dyslexia; dyscalculia; dyspraxia, Attention deficit disorder with or without hyperactivity (ADHD), and so on) to personality disorders, depressive disorders, and clinical mental health disorders (e.g. bipolar disorder, generalized anxiety disorder, schizophrenia, and anorexia nervosa) to other neurological issues such as Alzheimer’s and so on. Although Autism is not thought of as ‘an issue of mental health’ but rather as a pervasive developmental disorder, it qualifies for the DSM-5 due to the influence of autism on a variety of areas in a person’s life. For example, a mental disorder may refer to:

a health condition characterized by significant dysfunction in an individual’s cognitions, emotions, or behaviors that reflects a disturbance in the psychological, biological, or developmental processes underlying mental functioning (American Psychiatric Association, 2012, p. 465)

Mental ill health is very common in autism, and this can be mitigated when who we are, what we need and how we need things, are accommodated. The DSM-5 added a section referring to ‘sensory issues’ which wasn’t in previous versions. This will add to the profile of how autism may impact some individual’s communication. Sensory dispositions in Autism may mean an individual’s ‘senses’ (sight, sound, smell, touch, taste, movement, and various interoceptive senses) are either offline, reduced in connective capacity, mixed with other senses to distort or mis-configure a ‘pure’ sense (synaesthesia) are hyper-sensitive, hypo-sensitive, or a mixture of these (Ward et al., 2017).

The criterion for assessing autism suggests individuals must have difficulties with social understanding and social communication as well as difficulties owing to narrow, repetitive interests. So, if an Autistic person has a brain ‘wired’ to be single-minded (repetitive interests), with senses that might be seeking sensory stimuli or over reactive to sensory stimuli, this combination will lead to a cognitive style of thinking which often misses the bigger picture needed for broader connecting, social communication, and socializing in general (Happé, 1999; Happé & Frith, 2006). Such a cognitive style will have many strengths though and, provided Autism is accommodated for the person (e.g. environmental adjustments are made, and sensory accommodations are made, education and work schedules adjusted to individual needs and the person has appropriate motivation), autism can be a legitimate lens to see and engage with the world both within and around us (Lawson, 2011; Prato et al., 2018). In fact, some of the communication differences in autism have created walls to communication between Autistic and Allistic people, not because of autism but because we have ‘missed’ what is being communicated by Autistic individuals (Wood, 2019).

To explore why communication in autism can be so difficult, it’s important to understand how monotropism impacts us across the broad-brush strokes of all aspects of communication. Monotropism impacts how our sensory systems perceive

information and process it as well as how sensory information differences impact autistic communication and highlight the areas of good communication expressed by autistic individuals that are often overlooked.

Putting It All Together

Rethinking Autistic Communication

In Wood (2020) research study, it was found the autistic children communicated very well, just not in ways expected or desired by a teacher (e.g. they made it clear when they were not interested in a choice or topic on offer as they moved towards what they were interested in, for example, they chose an activity from a choice card that a teaching assistant didn't want them to choose; they enjoyed silence when they were focused and motivated to read or employed in a motivational activity they enjoyed, but might not have known how or when to be quiet otherwise). Wood noticed from her study children who were said to be 'non-verbal' would respond to another by using sounds, babble, or a jumbled mix of words that were an obvious attempt at communication, which was not understood by others.

For example, a 5-year-old, a lively and engaging talker, who enjoyed conversation, but struggled with pronunciation, language recall and narrative order, as this extract from one of Wood's semi-structured interviews demonstrates:

Boy: *The ra, the ra...another day (i.e. the other day) there is, there is a man sitting at the road, there's, there is a, there is a man sitting at the road and he got really rundover (...) An ambulance got in and he went to hospital. Boy (shaking head) He didn't, he really didn't, he, he, he, he looked at the road, he didn't dowan like that, he, he, he croshed the road, he was looking, and he went crosh the road to our house.*

Similarly, a 9-year-old found it difficult to relate anecdotes and pronounce words:

But I'm still. . .I'm still old. . .I'm still. . .I know how to. . .I know how to. . .like. . . use the. . .my sister's compers (i.e. computer). . .she didn't. . .she won't ask me. . .she always. . .she wants. . .her ones

Then a 7-year-old, the only female participant among the children in Wood's study, sometimes struggled to answer the questions during the interviews, of which the extract below is an example:

And what about Maths? Do you like Maths?

Yeah

Do you find Maths easy or difficult?

(pause)

Or sometimes easy sometimes difficult.

(pause)

You're not sure about that one?

Yeah

OK, that's all right.

These short extracts exemplify some of the difficulties in speech and communication (Wood, 2019, p. 5–7) as Autistic individuals we may have. It's of interest too that this doesn't just apply to children, nor to the 'level' of Autism given (currently, according to the DSM-5 (2013, p. 22)) Autism comes in as level one (needing some support; level 2, needing much more support and level 3 needing full time support). It's too often assumed that if you are an Autistic person operating mostly from 'level' 1, you will have good speech and general ability. But, in my experience when I'm overloaded/overwhelmed, either with demand or sensory information, speech is the first thing to be compromised! For more on rethinking Autism, see Jac den Houting at: <https://www.youtube.com/watch?v=A1AUdaH-EPM>

Wood (2019) suggests the noise in schools creates places for misunderstanding, confusion, and miscommunication (it can be at home or anywhere). For example, noise from children playing, from school bells, from noisy activities such as assembly, dancing, sports, musical instruments, children screaming, equipment such as air conditioners, hand driers, doors opening and closing, and so on. Yet, school often complained of the 'noise' made by Autistic children. Somehow school yard noises or screams of other children were not considered 'noise' but if an autistic child cried, screamed, or shouted, this was considered 'strange' or 'difficult', or abnormal. This not only applies to school, but the school nurse, counsellor, or any other professional needs to pay attention to this and understand the reasoning and connection to a monotropic nervous system.

When individuals live with a sensory system which doesn't build typical (polytropic-multiple) connections but is dictated to by a brain wired to process stimuli one at a time (monotropic-single), it's very difficult to multi-task. Noise, light, fragrances, changes in expectation and others, interfere as added dimensions because it calls away attention, often 'highlighting' those other things and exaggerating them.

To illuminate previous research and the monotropic brain re: connection via 'interest'

All communication processing happens via brain waves which are responsible for conveying information in electrical format:

- *Beta waves* (12 to 38 Hz) *Beta brainwaves* dominate our normal waking state of consciousness when attention is directed towards cognitive tasks and the outside world. *Beta* is a 'fast' activity, present when we *are* alert, attentive, engaged in problem-solving, judgement, decision-making, or focused mental activity.
- *Alpha waves* predominantly originate from the occipital lobe during wakeful relaxation with closed eyes. *Alpha waves* are reduced with open eyes, drowsiness, and sleep.
- *Theta waves* generate the *theta* rhythm, or signals, recorded either from inside the brain or from electrodes glued to the scalp. ... *Theta waves* with a lower frequency range, usually around 6–7 Hz, are sometimes observed when a rat is motionless but alert.

- *Delta waves* are usually associated with the deep stage 3 of *NREM sleep*, also known as *slow-wave sleep* (SWS), and aid in characterizing the depth of *sleep*.
- *Gamma*: A *gamma wave* is a pattern in humans with a frequency between 25 and 100 Hz, though 40 Hz is typical. According to a popular theory, *gamma waves* may be implicated in creating the unity of conscious perception (the binding problem) (see: <https://lucid.me/blog/5-brainwaves-delta-theta-alpha-beta-gamma/>) helping us draw together ‘the big picture’ from information associated with the other brain frequencies. If GAMMA is scattered, in excess or offline, the bigger picture information isn’t available to us.

In Kiser’s (2013) study, it’s of interest that seemingly scattered GAMMA changed when the children were motivated or interested, and this enabled GAMMA to be normalized. Interest (motivation) could be anything that captures attention, not just traditional ways of thinking about ‘interest’ as a hobby. For humans to make sense of the information being communicated to and with them, they need to notice the bigger picture it connects to. The brain wave band width ‘GAMMA’ is the one responsible for connecting information from the other bands and enabling an individual to put the information together so they can access the bigger picture, rather than individual details provided when differing brain waves act on their own (they usually act together but this is dependent upon the circumstances).

For example, if the doorbell rings it might mean someone is at the door requesting it be opened. If you are autistic and single-minded, you may not notice the doorbell being rung, especially if you are otherwise engaged. If you do notice, you might not realize something that needs to be done with this information. This is part of being monotropic (Murray et al., 2005). If as in Kiser’s (2013) study GAMMA tends to be scattered and in excess in autism, it would make it difficult to shift attention, especially when sensory issues are present. So, accessing the bigger picture, outside of motivational interest, is difficult to do. But if motivated, Autistic individuals will be ‘enabled’ and are often able to operate from a wider lens.

Translating Information into Communication

So, information is carted around the brain by means of electrical impulses or brain waves. These ‘pulses’ operate at differing band widths and are responsible for relaying sensory and motor (information from the muscles) in ways so we can make sense of the information. For communication to be effective, it must be taken on board, be understood, then be appropriately acted upon. Communication also depends upon good working memory, short- and long-term memory, and the motivation and retrieval from memory. In autism, sensory information can infiltrate memory and knock out context. For example, even when an individual knows what to do, when and how, if their senses are overwhelmed the retrieval of such information isn’t available to them. (see: <http://www.autism.org.uk/sensory>)

For example: ‘If I get overloaded then I just shut down; you get what’s known as fragmentation... It’s weird, like being tuned into 40 TV channels’... (National Autistic Society, 2020).

In autism, we have the same brain waves as in typical development but, the communication between them is different. At times we are not ‘as connected’ in general as Allistic people, but we can be deeply connected at other times. So much so we don’t hear when our name is called. But if we are motivated, we hear better. As autistics we operate, so to speak, from ‘the narrow beam’ of a torch with an intense narrow focus that takes in details of interest. Allistic people more readily access the wider ‘beam’ of the torch so are privy to more information with wider connectivity but often miss the detail. What does this have to do with sensory issues and communication? Everything!

In autism if one’s sensory system is in overdrive, this takes up all one’s attention. There won’t be any ‘spare’ attention to notice when someone speaks or even walks towards you. The way past this is to accommodate the sensory profile for autistic people. Taking time to work through a sensory check list, for example, (see appendices) will aid in understanding any sensory issues a person may have (see: <http://www.aetraininghubs.org.uk/wp-content/uploads/2012/05/37.2-Sensory-assessment-checklist.pdf>).

When schools, hospitals, shopping malls, and so on are too noisy, it might be useful for Autistic individuals to wear ear defenders, a baseball cap, sunglasses, and so on. Using a squeeze jacket (see: <https://www.squeasewear.com/>), could be helpful for some.

A Pressure Garment and/or Sensory Supports

A Squeeze Jacket is a garment that an individual can wear that allows them to ‘pump’ air into special channels which puts air pressure into the garment which ‘pushes firmly’ onto their upper torso. This acts like a ‘squeeze’ and can help calm someone. When I wear mine, I feel like there is a distance between myself and others. It’s like having a force field around me and not only do I feel protected and safer I also feel enabled to be more present around others. So, for me this is helpful in crowded places. Sometimes weighted blankets or/and soft toys can be useful. But, after about 20 min or so, the brain accommodates the weight, so they only work for a short time. With my squeeze jacket I can pump it up or let out the air... so, my brain always accommodates the pressure because it’s freshly pumped up when I need it, rather than being there all the time.

Self-regulation is vital for good communication. How can we know when it’s our turn to talk or when we need to be quiet? We depend upon our sensory and motor systems being in sync; (not over- or under-stimulated) and upon accessing the big picture. So, creating an autism friendly environment that allows the building of such connectivity is essential to the processes of inclusion and allows sensory modalities

to be monitored and better communication to take place, See: <https://neurodivergentinsights.com/blog/the-power-of-sensory-regulation>

For us, as Autistic people, sometimes communication happens best when we don't look at someone. Looking at someone may accost our Autistic senses. Sometimes communication works best via text, email, comic strip stories, video, and TV. Communication in autism must consider the sensory systems we live with, must give us time for processing and must be autism friendly.

Conclusion

Communicating with Autistic individuals makes the most sense when we are in a position to attend to the message being given. This will mean if we have sensory issues taking over our attention that they are accounted for. For example, when my Autistic granddaughter came to visit, we glued green snooker table felt to the 'bottoms' of the chair and table legs. This meant she didn't cover her ears in pain each time a chair was moved at the table because the chair no longer 'scraped' the floor. When we went out to the shopping mall, my granddaughter wore her ear defenders so the noises in the shopping centre were dulled, allowing her to help me with the shopping. At times my granddaughter needs to follow her lessons on her iPad with her headphones, rather than be attending to teacher in a noisy classroom. There are times she needs instructions texted to her or emailed because she can take her time over this type of communication and can re-read it as often as she likes. This gives her time to check in on her understanding and the content/context.

Technology is so often a very useful tool for Autistic individuals. It comes without body language (which can confuse) is a single focused piece of equipment (not talking, walking, and waving hands about all at the same time) and can be text or picture based (See: <http://www.spectronics.com.au/product/20032/popup>).

True communication needs all concerned parties to hear and be listened to. Are we trying to make Autistic individuals fit into a mould they are not designed for or are we adapting our communication styles to fit in with them so both our communication styles can be understood? True inclusion says: 'I accept you where you're at and will walk with you to our mutual destination at the pace you need me to'.

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Chapter 4

Monotropism and Cognitive Flow on Effects



Introduction

As Autistic individuals, being monotropic is our default setting. This means monotropism will impact on our other cognitive states (either negatively in a way that reduces productivity or positively in ways that increase productivity), such as memory, processing, accessing a wide range of connected and unrelated information, executive functioning, central coherence, forward-thinking (this includes Object Permanence (OP)) interoception, and others. If we envisage monotropic attention as a highway flowing in one direction, but one with side roads that come off at various intersections, we can appreciate all its traffic (attention) is flowing one way with small signposts to those side roads. But, the signpost itself won't explain the reasons for these so understanding the meanings may be missed or misunderstood (Lawson & Dombroski, 2017). If we label a couple of side roads that carry important traffic forward to a particular destination, we can better appreciate their importance. This chapter will seek to do this.

From the previous chapters we can understand autism's foundation is key to comprehending what we see in clients presenting for autism assessments, or who already have a diagnosis and are individuals we may work with in our profession. Recognizing the wider umbrella of neurodiversity and then neurodivergence, should aid in comprehending the differing cognitive styles and help avoid 'the double empathy problem' (Milton, 2012). Although the DSM-5 (2013, p. 22) stipulates criteria for autism (including passionate (rigid) interests and difficulties with change which impact upon communication and social understanding), it leaves it to the clinician to translate these criteria into how this presents in an individual. Autistic expression, based upon an individual's monotropic disposition (e.g. single use of attention with discomfort at having to shift attention rapidly or a monotropic disposition impacting flow states, inertia, and hyper-focus), personality, cultural and linguistic lifestyle (CAL), and the flow on implications for interoception and OP how these are experienced in autism is different to those seen in Allistic people

(Ashinoff & Abu-Akel, 2021; Dwyer et al., 2024; Grotewiel et al., 2022; Lawson, 2011; Lawson & Dombroski, 2017; Rapaport et al., 2024; See also: <https://differentbrains.org/object-permanence-chronic-pain-conditions-introspection-on-the-autism-spectrum/>). This is important to comprehend as it's all too easy to miss this cognitive foundation in autism when it's delayed or 'offline' and to see 'behaviour' which we may judge as a lack of empathy, bad manners, being inconsiderate, being rude, and so on.

Although traditional tools showing degrees of monotropism have not existed, currently there is a monotropism questionnaire that is rapidly gaining favour within the research and autistic community (<https://monotropism.org>). This tool is under investigation and is showing good validity and reliability in certain research projects (See: <https://osf.io/wpx5g/>). There is also a hyper-focus assessment tool (Hupfeld et al., 2019). These tools can help us gauge a client's 'connection' to OP so we can tell what has been understood and what we can expect from our client with regard to how 'monotropic' they might be.

An Example of Non-functioning OP in a High School Student

John is a 13-year-old non-binary autistic student who is noted for too many absent days from school. As a 'school refuser', he is referred to the school psychologist who is trying to support John and his family.

After several 'sessions' with the school psychologist, it became obvious that although John who was very bright and was very able in a number of school subjects, was not happy at school. John often talked about 'missing their dog' who was a particular support to John. John also missed his mum, who worked 3 days a week. During school hours, John seemed inattentive and wasn't focusing on his schoolwork.

After a staff meeting with John's parents and teachers, it was concluded that John was happy at home but didn't like to be parted from his dog or his Mother. Even though John was a teenager, it was decided that he might benefit from a 'social story' (See: Garol Gray, 2015; Timmins, 2017). The story was written in 'comic strip style' and explained that mum went to work, so did Dad, but John came to school. It showed John getting ready for school, getting on the bus for school, arriving at school, enjoying school, and happy about what he did at school in the classroom and at recess. It also showed John who they could talk to if they felt unhappy or sad. The story echoed to John that the school counsellor/psychologist and John's home group teacher were safe people to talk to at any time. John was given the social story and encouraged to read it at home before coming to school and again at school during recess. For more on Social Stories, see: <https://www.autism.org.uk/advice-and-guidance/topics/communication/communication-tools/social-stories-and-comic-strip-conversations>

It was hoped that this would help John to understand the differing role he and his family played and that this might support John to feel included and happy to come to

school, knowing there was a safe place for him if he felt troubled. However, this made no difference to John's school attendance.

Eventually, the school psychologist asked John to share some photographs with her; these were to be of things that were important to John and to add some text beneath the pictures of John's thoughts about them. John brought six photos to school, three were of his dog and three were of his mum. Beneath each photo John wrote the name of his dog, where the picture was taken, the date it was taken on, and what he was doing at the time the photo was taken. John wrote the same statements beneath the photos of his mum. Then, beneath each photo of his dog, after the other statements, John wrote the statement: 'How do I know my dog still exists if I am at school'? It feels like she is gone. Beneath each photo of his mum, after the other statements, John wrote: How do I know my mum still exists if I am at school'? It feels like she is gone. The photographs brought home to the school psychologist that John worried his dog and his Mum did not continue to exist once they were out of sight. John felt bereft, their dog and their mum ceased to exist for John, outside of their physical presence with them. The experience John is having is one of 'poor OP'... John hasn't any 'sense' of his dog or his mum unless John is with them. This was the root of John's school refusal. The answer for John was in the school allowing John to bring their dog (as a service animal, supporting John's wellbeing) to school and John was free to text their mum 4 times during the school day to check she still existed. John also carried photos of his dog and mum in his wallet. John might eventually develop OP and not need prompts to enable a connection to him mum and to his dog, but if John continues to find this difficult, they are able to access 'tools' (photos in wallet; mobile phone check-ins) to help them.

In addition to understanding monotropism, the degree of OP is also relevant to both an autism assessment and supporting autistic clients. Autism presents behaviours clearly denoting if an individual's cognitive OP ability is fully connecting to them. However, traditionally, when exploring OP the existing tools have not considered 'adult' presentation in Autism so are not sensitive enough to this population. Most research suggests OP in adults only changes if the brain is subject to extreme damage or personality disorder: <https://psychcentral.com/disorders/borderline-personality-disorder/object-constancy-understanding-the-fear-of-abandonment-and-borderline-personality-disorder>. But, in Autism where monotropism is foundational, forward-thinking (an outcome of good OP) will be difficult due to attention being taken over by existing interest. To notice that an object still exists (even when out of sight) requires attention to be flexible and move from 'now' and notice 'then, when, how and what if'. These emotive states require an individual to shift between attention states, rather than only be 'aware' of the 'now'. This shifting of monotropic attention towards other things, situations, emotions, and so on is very uncomfortable in autism and is not an Autistic person's 'natural state'.

Although the research and understanding of monotropism is in its infancy, the autistic community endorses and recognizes it in their everyday lives: (Dwyer et al., 2024; Lawson, 2011; Murray et al., 2005; Rapaport et al., 2024). The consensus is Autistic people are monotropic in how their attention is utilized (single focused, single minded). Whereas Allistic people are understood as being polytropic (divide

attention as a natural part of who they are) and can move through interests or shift their attention more easily, hence comprehending OP in its usual way, finishing incomplete statements and context, as well as appreciating consequences and being able to forward think.

For example: When I was a child, I remember my mother saying:

‘if you do that again I will take away your book’, no more reading at the dinner table, but you can have it again after dinner is finished.

For me the message (script) I understood was: ‘I must bring my book to the dining table and read, then Mum will take it away from me, then Mum will give it back to me when dinner is over’. I had not understood the intention behind mum’s statement. This was a typical scenario for me, so I was often ‘in trouble’ but failed to join the dots in the way my Allistic siblings did.

Considering the monotropic cognitive style is essential to understanding object permanence difficulties in autism. The Allistic ability to shift attention and marry/align ‘interests’ (whatever is held within the attentional focus) with that of the community or person involved in the communication (e.g. family, class, team, colleagues, medical staff) enables the broader social picture to be understood. In turn this allows for relating across a wide section of interest or/and connections to things being attended to in ways that are socially interchangeable, to occur.

For Autistic people, however, noticing the difference between attended to ‘now’ or ‘then’ or ‘next’ or ‘when’ requires a shifting of attention that may not occur easily or at all. So, when a person is out of sight they may also be ‘out of mind’. For autistic children experiencing ‘school refusal’, it might be a matter of disconnection between the appreciation of ‘now’ but not of concepts outside of now. For example: If I’m at school, is my dog, my bedroom, my mother, and so on, still at home or do they cease to be?

Autistic individuals may find it difficult to identify emotions of self or others and have difficulties to generalize concepts or interpret context. This is not about empathy or feelings; it’s about noticing and naming them. Concepts and context require divided attention and connection from self to other. Difficulties in these areas can be misinterpreted as poor theory of mind. However, object permanence requires a person to attend to multiple stimuli in order to appreciate concepts such as ‘here, gone, still here but unseen, temporarily out of sight, back soon, and so on’.

A Conversation with an Autistic Adult

Wenn: One ‘themed’ experience regards a sense of incomplete self, when ‘Named Other’ is not available...

Christina: ...as an adult, I miss the presence of a particular person and realise that it is the interaction with that person that enables specific aspects of my own personality to surface...

Wenn: This is so true Christina... I notice around certain 'Others' I thrive and am connected to life in ways that don't exist if they are not there.

Christina: ... so in a way the presence of particular others enable the existential existence of my inner self and without the interaction with these specific persons that part of me ceases to be.... that part of me is existentially dead... or in a coma state ... I am not wholly myself.

Wenn: I understand what you are saying and, I also experience this. On the other hand though, as I've come to experience my 'fuller self' on a more complete level, I don't feel so 'disconnected' and left searching for 'Other' to complete me.

Christina: This I feel is more important to autistics who are unable to or find it difficult to bring their inner personality/self into being existentially as readily as Allistic people seem able to do. They seem to have more stylized, stereotypic 'inner selves' more readily available and can find others who facilitate their being via a wider medium. Or, they can connect more widely to 'themselves' in a variety of concepts without needing to 'check in' with others.

Wenn: I understand this statement but I'm not sure Allistic people are as connected to their 'inner' selves as we might think. They have more access, this I believe, but this is a different scenario.

Christina: I'm not sure if I'm explaining what I mean adequately but I know some parts of the deeper real me are unable to exist experientially, for months, years, decades even. Yet when the particular other is present that inner (sleeping) part of myself is resurrected ... and engages in that deep inner mode of being.

Christina: SO, the absence of particular others may, particularly in childhood, be experienced as a fear of both the loss of self and loss of that other that facilitates the engagement of that particular inner self... If the other is not present a panic state can exist in that it is not known if and when the other will reappear and also when and if the inner self that engages with that other will be able to take life again.

Wenn: Yes, I'm sure this is so, especially for young children who may not have developed autonomy. Even though I now experience autonomy in ways I never have before, and even though there are particular individuals who spark life and connections in me that are not there when they aren't... I still have the taste of that life, even in their absence. This enables me to feel 'complete', even outside of their physical input. I have the memory of their presence and the mutuality of our togetherness, which can still feed me. But, without an established connection to OP, I could easily be cast aside in an ocean of 'nothingness' feeling lost and alone these are indeed important understandings to have.

This situation can be frightening in that there isn't the opportunity to reinforce and evolve particular inner aspects of self without the other with whom relationship facilitates the existence of this Self.....so for an individual it may be experienced as an unidentifiable fear associated and embodied in that absent other... who at that time and space doesn't exist experientially relationally and also that hole, the void in 'self', becomes apparent with the absence of that particular other.

Some Other Examples

- The doctor's appointment is booked for 9 am. Autistic person, being very literal and possibly having difficulty in forward-thinking, leaves when no one calls them at 9 am.

Tell us in advance the expected wait time, or how you might let us know when it's our turn. We will not naturally appreciate concepts around 'running late', 'delayed', the need to wait, and so on.

- The Autistic person receives a letter telling them they need to come to the hospital for a review.

They may expect the review to 'happen now'. Even if they read the letter fully, they may not grasp the concepts beyond 'come' in its current tense. Say what you mean, mean what you say.

- Autistic person says 'yes' to a request but then something they are attending to takes over and they are not able to follow through with their commitment.

Please send text reminders or programme an alarm on my smart watch to help capture my attention and facilitate my ability to follow through with actions.

- **OP issues:** When a partner, family member, or a support worker is out of sight, do they still exist?
- If I go shopping, are my home, my dog, still there?
- What about paying bills, insurance, keeping my commitments to communicate and let folk know how I am?
- How can I know something when it ceases to be physically present?

Some More Examples of Underdeveloped OP

The carer tells the individual they are leaving the room; they leave to go outside; the individual seems fine. The carer returns after a short time away to find the individual upset/angry and saying 'Where were you I couldn't see you; I thought you left me'.

- Tom aged 50 years is invited to join a game. Tom is autistic. He is playing hide-and-seek. But, as soon as others are out of sight, Tom begins to shout and rock. He is utterly convinced they are gone. Tom repeated this experience over and over again. It was very upsetting for others and awful for Tom. This is not a 'problem behaviour' or an IQ issue. It is due to Tom's learning style and cognitive adaptation to change. Tom lacks OP.

In autism being monotropic with all of our attention flowing towards the 'thing' attended to, the side road to object permanence might be under construction, incomplete or totally lacking, there are all sorts of implications for health & wellbeing in autistics of any age or gender.

When it comes to 'executive functioning' (knowing how to orchestrate one's plans, knowing what comes first and next, knowing how to separate what is urgent

from what can wait, and so on), monotropism and OP also play a big role. Being monotropic will impact the ability to forward think so planning ahead will be difficult, uncomfortable, or not available. How can an individual understand consequences and outcomes once the initial event ceases to be? How can I tell when something is over or needs to finish if I have no connection to when it started or to what 'finished' looks like? Medical personnel need to be mindful of these factors when conducting medical appointments, medical examinations, and medical procedures. It's so important to communicate clearly what you are going to do, are doing, and when you will stop doing it.

Another example is: As with Tom above, when we were playing a game of 'hide go seek' with our granddaughter, she was terrified when we moved out of sight as she could no longer see us and believed we were gone. But, having a Walkie-Talkie and speaking to us was very helpful in helping her understand we were not gone, just out of sight.

Ian's Story

Ian's story highlights his literal thinking and his gaps in object permanence.

'My father died a few years ago. My mother thought it was too upsetting for me to attend his funeral, so I was kept at home. My mother said father had gone to heaven. She gave me some pictures of him, which I placed around my room. I had a photograph of dad and I fishing, which was taken when we went camping the year before he died; I kept this picture in my wallet. As the days went by, I told my mother I was sad that dad died and how much I missed him. In fact, I echoed this sentiment every day and every day mum said the same thing: '...We all miss your Father but he is happy in Heaven'. I felt very hurt by this and wondered why dad chose to go to heaven rather than stay with us. Each day when I looked at dad's picture I told him to come home, but each day passed and he stayed away. Eventually mum met Michael and she even let Michael kiss her the way dad did. I was very upset with her and I started to fight with her. All the things I felt and thought were locked inside me because I didn't have the connections to language and necessary concepts to explain them'.

Problem

Ian's father died but Ian hasn't understood the concept that his father isn't coming home again. This is compounded by his mother telling Ian his father is in heaven, a place he prefers over being home with Ian. Maybe Ian could board a bus or train and go to visit his dad? The problem is Ian isn't aware that heaven is a place of mystery not accessible here on earth to the living. Entry to heaven is by way of death...

sometimes we use metaphor stating when we are very happy, we are in heaven. But this is a replacement term meaning ‘extremely happy’. So, each day when Ian looks at the photographs of his father, he is reminded that his father isn’t there with him.

Possible Solution

Ian needs to be told that when his father died it was an event outside of his dad’s choice and one that has taken him away, in person, for all time. The pictures Ian has of his father are reflections of when he lived; these pictures continue, but Ian’s father does not. It might be helpful if a farewell ceremony could happen. A time and place where Ian can put some things that were his father’s into a box and put them into a place where they cannot be retrieved. Ian hasn’t had an opportunity to say goodbye and he needs this. It’s important in autism to remember that we are very literal thinkers and it’s good to say things in an authentic manner: ‘...say what you mean and mean what you say’. Ian’s father isn’t ‘asleep’ nor has he ‘gone to a better place’ and so on; Ian’s father is dead, and he no longer lives.

Where there is the death of a loved one for a Allistic individual, a therapist might use photographs and have them talk about their lives together to capture the ‘essence’ of the person who died. This helps that person to connect and grieve appropriately. This works because the Allistic person isn’t literal in their thinking, has a concept of object permanence, and can access the bigger picture. Concepts such as then, now, tomorrow, here, gone, and so on are understood. Each one can exist without taking away from the other. But, in autism, this may not be the case.

Conclusion

The information above illustrates the connection OP has with being monotropic. It offers explanations and examples to help clinicians build good rapport with clients and enable mutual communication that aids both mental and physical health. Believing an Autistic individual doesn’t have empathy because their behaviour appears egocentric or not considerate may be more about delayed or underdeveloped OP and not about lacking in empathy. Once bridges to connection are built these can offer ways to cross over from a state of disconnection and confusion to a place of connection and understanding. Knowing this and ways to work with Autistic individuals to make this happen are paramount as useful tools in any clinician’s tool kit.

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Chapter 5

External Senses and Their Relationship to Monotropism



One's external senses aid us in knowing where we are in time and space as well as where others might be. They inform us about being touched, or touching something, they allow us to notice and see, hear, taste, and smell. This information is uploaded to our brain and in turn our brain informs us of what, if anything, we need to do about it. For example, if we see food, our brain may tell us we need to eat. Our mouths may start to 'water' which helps motivate our digestive juices and the desire to eat. This in turn is aided by senses of hunger, smell, and taste. If we are not hungry, the sight of food may cause us to want to move away and not have our senses accosted by food smells.

Monotropic individuals may not synchronize sensory information but rather experience it sequentially as opposed to experiencing it as a collective. This will mean for some individuals the information is heightened, brighter, louder, and possibly overwhelming, while for others the information appears 'diluted' muted and not well noticed, so underwhelming. With such a disposition to external sensory stimuli, autistic individuals may behave in ways that are confusing and challenging, both to themselves and to others. For example, we may have our fingers in our ears when noise is too loud (whispering may be too loud for us); we may prefer not to look at you when you talk to us so we can reduce the 'noise' coming from your eyes and facial expressions. On the other hand, if our senses are dulled, we may be sensory seeking (as a way to connect to a feeling) so we might get too close to you (not aware of personal space) or stare into your eyes too long and so on.

It's all too common that the sensory environment we all share can be over stimulating to many autistic people. Some of the things common to autism is how often we will put our hands over our ears, prefer to avoid 'eye contact', appear not to be listening, not to hear you, get very uncomfortable if we are made to wear clothing that we experience as scratchy, or need to 'sniff' our food and drink before consuming it. This may happen during what is experienced by others as normal everyday things that others may not notice or be bothered by (Bogdashina & Casanova, 2016).

The sensory world in autism often dictates ‘behaviour’ so seemingly small things that Allistic people hardly notice can be experienced as ‘pain’ in autism (Bogdashina, 2003) and very painful things might not be noticed at all.

Waiting in a doctor’s, dentist’s, or clinician’s waiting area at a clinic can be really unbearable for some of us as Autistic people. There is all the noise of chatter, possibly a child(ren) or baby crying, phones ringing, music playing, clicking of computer keys on a computer keyboard, noises and sights coming off the environment, peoples clothing and hair as well as the irritating background noises from headphones. Some of these can be irritating to Allistics too, but having a polytropic cognitive style usually aids sensory integration rather than highlighting sensory discomforts as in autism.

I have always said that as an autistic individual I am disabled due to the non-accommodation of my difficulties.

(Lawson, 2008, p. 62)

For many of us as Autistic individuals, sensory sensitivities can have a huge impact on our ability to cope with seemingly ordinary, everyday things. But we may not have developed the skills to communicate our discomfort to others or have found strategies to help lower, dilute, or heighten (if underwhelmed) the sensory experiences that cause us discomfort and even pain. As medical professionals, we need to use our powers of observation (looking at behaviour via a ‘sensory’ monotropic lens) so we can identify the challenges someone may be experiencing. This is echoed in Wood (2019, p. 46) where she describes the sensory sensitivities that some children encounter.

the school (or other people filled environments) ...can present autistic children with a multi-sensory onslaught in terms of sounds, smells, textures and visual impacts that constitutes both a distraction and a source of discomfort.

Honeybourne (2018, p. 83) explains the sort of impact these sensory sensitivities can have on individuals as they try to go about their daily lives:

The environment around us can have a considerable impact on our wellbeing, as well as on our ability to learn and concentrate.

There is a check list at the back of this book that will aid the gathering of sensory information to help us create sensory friendly spaces for monotropic individuals. Sensory needs are different for everyone, but what we implement for autistic individuals will work for everyone.

The implications sensory issues may have in relation to behaviour are significant, and understanding an individual’s sensory profile is an important element when planning an appropriate environment...

(Miller, 2018, p. 31)

Ideas to Support an Autistic Individual with Hyper-Sensitive Senses and Those with a Hypo-Sensitive System of Sensing

Touch

- Sometimes there are surfaces, materials, people, textures, and so on that as Autistic individuals we may avoid. Usually, it's because we are uncomfortable, not sure what these might mean, or such textures and patterns 'hurt' us. So, finding ways to mitigate this would be helpful. It might mean keeping walls and furniture one colour and no patterning. It might be helpful to place photographs in a leaflet and using this to 'prime' us before we see the environment we are coming into. It might be beneficial to visit the clinic and become familiar with it and its people before an actual appointment. Also, we would benefit from words that are used to describe environments and people that portray what to expect. For example, if there are 'lumpy, bumpy, uneven, shiny, damp surfaces' or if certain people are in attendance on certain days, but not other days and if they are away on the day, it's good to let us know so we can appreciate it might be another person taking their place.
- If it involves us taking part in an activity, find ways to adapt it so that we can still complete the activity without having to experience something we are uncomfortable with (e.g. maybe wearing gloves while exploring a medical instrument such as a stethoscope, rather than use bare fingers).
- Remember we may be very sensitive to light touch, so please give us a warning if you are going to touch us; let us know what part of us you will touch and how firm that touch will be.
- Certain textures will be uncomfortable. If we need to change into a gown, for example, check in with us about it.
- Some of us are particularly sensitive to labels, seams, buttons, and certain materials. Knowing this is important and it should inform how we relate to our patients and clients (e.g. ask us, 'would you prefer a gown without a label?')

For Underwhelm or Hypo-Sensitive to Touch

- Some of us benefit from holding a 'fidget aid' or having something in our hands. This may help the need to 'touch' by reducing the drive, as we already are touching an object. Sometimes we need to touch an item of furniture or whatever might be sitting in front of us, on a desk for example. If you know this about us in advance, you can make sure only appropriate things are on your desk or in your room that you are happy for us to touch.

Sight

- Creating a low arousal environment can be helpful. Consider using calming colours such as cream or pastel colours for walls, floors, and upholstery. Avoid brightly coloured patterns as these can be confusing.
- Use natural lighting or soft lighting if possible. Avoid florescent lighting as many people say they flicker. Also, florescent lighting can be a trigger for epilepsy and this condition is not uncommon for Autistics.
- Consider the window covering. Having net curtains can help to block out some of the light or fit semi-transparent window film which can also defuse the bright sunlight. If possible, avoid blinds with slats as they can cause distracting light patterns in bright sunlight.
- Create an organized and uncluttered environment with most things out of view. We can be distracted by ‘clutter’.
- Encourage us to wear sunglasses or glasses with tinted lenses. Wearing a hat with a large brim or a cap can also limit the amount of light reaching our eyes.

Wearing a baseball cap can be a helpful strategy:

As a younger person (still do this) I “lived” in my baseball cap. My cap served to slightly limit my field of vision to help prevent sensory overwhelm, gave pressure to my head so I knew where I ended and helped me navigate open space. If I was made to remove my cap, my attention went with it and I became inattentive...

Wenn Lawson (2019, p. 11, Foreword in Wood).

Vision Hyposensitivity

Although it appears to be less likely you will come across this, it’s not out of the question. Some of us may be driven by a sensory need for visual stimulation that seems ‘over the top’. There is a drive to stare for ages at someone’s face or body, to ‘stim’ on the shape, feel or look of an object or piece of furniture. It seems difficult to look away and we will constantly be drawn to do this. One way to limit this need is to give an opportunity to do this safely and appropriately beforehand and have the experience on offer afterwards too. Another way to work with this need might be to join us in the activity, of course check in with us first that the person is happy to have us share with them. It must always be ‘nothing about us without us’.

Hearing

- Background noises such as ticking clocks, humming electrical devices, projectors, and lawn mowers can be very irritating and prevent us from concentrating. Consider how these sounds can be reduced. It might mean turning off a device, closing a window, giving us ear plugs/headphones, ear defenders, and the like.

- Acknowledge our fears and discomforts. Always warn us when to expect a sensory stimulus we may not like. Sometimes, if we are in control of when the noise is going to happen it can reduce anxiety. For example, one child hated it when his mum did the hoovering but could cope if he turned it on for her and then he would go to another room.
- As Autistic individuals, we often use the strategy of covering our ears with our hands to block out sudden noises such as emergency vehicle sirens. But wearing ear defenders or earphones to help block out sounds in lots of noisy environments can be helpful. Earphones can be worn without having music playing, but many of us find that playing music is calming and can help to block out sounds we find uncomfortable.
- Consider the surfaces used in buildings, for example, carpets can reduce the impact of sound rather than using hard surfaces such as tiles and wooden flooring. Even a scatter rug can help.
- Consider where you can provide a quiet area for us to go to if we find the noise level in the building too loud.
- Avoid noisy environments if possible. If we need to visit a noisy place think about when it will be at its least noisy. Always warn us that it is likely to be noisy and offer us the opportunity to take calming sensory objects with us which can help to calm and distract us from the sensory experience we find unpleasant.

Hypo-Sensitive Hearing

At times you will have clients/patients who enjoy loud noises that they choose to make, or who will be drawn towards noise (such as emergency vehicles sirens) and seeking this sensory input might put them in danger. It can be quite a balancing act to accommodate noise seekers. It requires creative thinking that keeps everyone safe. One young Autistic man I knew loved banging drums, cymbals, a tambourine and anything 'bang able'. He loved music class where he was able to be part of a percussion band. It was helpful for him to record the band 'music' and listen to it over his headphones. He was more likely to work with us when he had his headphones on and/or after playing percussion in the music room.

Taste

- Do not pressure us to eat foods we find uncomfortable. Rather check in with us to see if there is a way we can adapt a 'food' that we may need to eat for various medical procedures. For example, dissolving a powder in a favoured drink, placing a crushed medication inside a chocolate treat, and so on.
- Some of us do not like different textures of food mixed, so separate them by placing them on different plates, but please check in with us first.

Hypo-Sensitive to Taste

Others of us seek out textures and will put all manner of ‘non-edible’ things into our mouths. It’s a very strong urge to connect with an oral sensation, but this can be dangerous. Finding ways to safely accommodate this need is very important. This might range from ‘edible crayons’ to edible homemade play dough and so on.

Smell

- Please respect us if we say a smell that you find pleasant is unpleasant to us and try to find a way to avoid it. The smell of certain things can be overpowering, and we may need to leave the room. Putting a scent of something, we like on to a wristband or tissue may help to disguise the uncomfortable one.
- Please avoid using deodorants and perfumes, heavily fragrant makeup and colognes, also please use unperfumed soaps, shampoos, washing powder, and cleaning products.

Hypo-Sensitive to Fragrance

Some Autistic people are fragrance seekers and will sniff everyone, including other people. Again giving ‘appropriate smelly objects’ can be helpful and enable a monotropic person to stay grounded and more likely to appreciate what you might need them to do.

Vestibular

- Some activities (having an X-ray, climbing onto a dentist’s chair and so on) can be difficult to navigate if we have balance issues. We may need support to negotiate medical equipment.
- Always check in with us though; if touch is an issue, we might not feel comfortable holding a hand, but we might be OK to hold onto a strap.

Hypo-Sensitive to Movement

Loving to swing, twirl, run, hang upside down, and so on may come from a need to ‘connect’ to our vestibular selves. An Autistic person may twirl round and round very fast but not get dizzy when they stop. They are seeking a sense of connection through movement that they don’t get any other way.

It can be helpful for them to sit on a wobble ball rather than a chair, for example.

Proprioception

- Some of us have proprioception issues and can appear clumsy. Please be considerate when placing furniture and other items that might ‘hurt’ us if we bump into them.
- Some of us need to move often or all the time, it might even help us to focus. If our movement interferes with attention though, offer regular movement breaks to allow us to take a few minutes to move and then refocus.
- At times, doing some physical exercise before we need to concentrate on an activity can help us to focus our attention on what is needed from us.
- Deep pressure provided by weighted jackets, or a weighted soft ‘toy’ can have a calming effect on the proprioceptive system, which may reduce our anxiety.

Hypo-Sensitive Proprioception

Knowing where we are in time and space is essential to being able to stand up straight, sit upright, walk in a straight line, and appreciate a sense of time. For monotropic individuals who are hypo-sensitive in this sense, the need to explore things upside down, from various angles or just to constantly jump on a trampoline, can be very high. It’s important not to stop us if this is what we need but to safely facilitate the activity. When my son was young reading to him while he jumped, swung, or hung from monkey bars was his best way of attending to the story or instruction.

Interoception (or Our Eighth Sense)

In Chap. 6, more information on interoception and its connection to monotropism is explained. Below is a brief section that could be thought of as connecting external and internal sensing systems.

- For some of us, use of a visual (photo, picture, or flow chart) can support us to know what is expected now and next. In autism, forward-thinking can be difficult. We may not notice if we are in pain, hungry, or need the bathroom. So, having a visual of these, or receiving a text message or programmed alarm on a smart watch, for example, can help.
- Breathing techniques and exercises are helpful for calming as well as paying attention to what is going on inside our bodies.
- Mindfulness can be helpful too. But it is important to know when to use this support and when to avoid it. Mindfulness can mean we focus upon a thought and this can heighten the thought in a negative way.

Sensory Overload

Chris Packham (2019) describes how stimuli from different sensory systems can cause real difficulty:

I rarely go into supermarkets. I find that environment really challenging, all of the bright lights, the confusion of the enormous complexity of goods in there, plus all the smells and the sounds. It's a difficult environment.

<https://www.youtube.com/watch?v=-1aTVfKstrs>

Many environments or events are just too much for us and we aim to avoid them. I've always found it easier to enter some environments either with my sunglasses on and a cap, or if I know I'm only going to be in this environment for a short time I might cope OK without these.

Vestibular

- Providing movement activities such as swinging, jumping, and spinning will stimulate the vestibular system.
- Turning upside down or doing cartwheels and handstands can give intense vestibular input that might be very exciting and feel good.
- Provide a 'wobble cushion' for an individual to sit on. These are air-filled cushions that create a 'moving surface' which accommodates the need to be on the move; while sitting still.

Proprioceptive

- Activities that involve input to the muscles and joints as well as being helpful for calming the hyperreactive individual can also help those who are hyporeactive. This can also increase attention and level of arousal. The activities that will do this are unique to each individual and it is important to observe and identify which ones will work best. Movement gives proprioceptive input but some activities that involve the whole body such as pushing or pulling, oral actions such as chewing or sucking and activities involving the hands such as squeezing or pinching actions will all give increased proprioceptive input.
- Some hyporeactive individuals present as lethargic, but encouraging movement breaks and opportunities for them to get up and stretch will help.

Be Prepared

Having things explained to us beforehand is very important. I need to know what I'm in for. Attwood (2015, p. 57) suggests using items that offer 'alternative sensory experiences that neutralises specific sensory sensitivity' and are 'mesmerising enough to block out sensory experiences...' may be helpful. Other sensory objects, such as a squishy ball to squeeze, glitter tubes to watch the rainbow glitter move, tactile fidget toys, blue tac, or a drink with a straw, might also be useful.

Discovering what helps us to stay calm and self-regulate is trial and error. But, checking in with us gives us a chance to let you know what we may need. Completing a sensory checklist ahead of any medical encounter is also helpful (see Appendix).

The Setting

The medical centre can be a busy, noisy, and overwhelming place for the Autistic person. It is vital that practitioners have received training on autism and are able to recognize and be sensitive to the person's individual needs as without an understanding of sensory differences they may misinterpret the person's behaviour. As Lawrence (2019) emphasizes

It is essential that as teachers we are sensitive to these sensory issues, rather than being distracted by responding to the behaviours-avoidant, anxious, self injurious- which the sensory overload may prompt.

Lawrence (2019, p. 19).

The SEND Code of Practice (2015, 6.9, p. 93) requires schools to make 'reasonable adjustments' to prevent children being at a disadvantage. Often small changes can make an enormous difference.

As Mahler says:

I believe one of the best solutions that we can use as professionals is to proactively provide a sensory friendly ... environment. Learn about each ... person and his or her unique sensory needs. Try to consider these sensory needs throughout the day and implement modifications that will maximise sensory comfort.

(Mahler, 2016, p. 6)

Beany and Kershaw (2014) explain that the room environment can have a dramatic effect on the individual's behaviour and the ability to focus and suggest that the Autistic person (and others) will benefit from:

- The day will be a lot easier for the Autistic person if regular routines can be established. This will help to reduce anxiety often associated with social demands. Autistic individuals usually like routine and things to be predictable and changes can make them anxious. Warn them about any alterations to the routine of the day or changes in staffing to reduce their anxiety.

- An organized and uncluttered office can help with creating a calmer presence. Put displays within borders, books on a shelf and possible distracting material out of sight.
- Well labelled and organized equipment—equipment that is needed should be easily accessible and items that would distract, out of sight.
- A designated space that individuals can go to if they become sensory overloaded and need a place that is quiet with no demands.

Conclusion

We are all accosted by various sensations and sensory experiences living in this shared world exposes us to. But if you are Autistic with a monotropic disposition you will find sensory information from the external world is often either overwhelming or underwhelming. The above chapter has given us some ideas about the Autistic sensory world. It has also laid out some examples and ways to ‘dull down’ the intake of sensory information or make it louder if needed. Please remember than one Autistic person can be either, hyper- or hypo-(sensitive) but can also move between these several times in 1 day. It’s so important to check in with us directly so the external sensory world can be outfitted in the way we need it to be. I know I can’t often ‘change’ the external environment to how I need it to be so I carry with me certain pieces of ammunition that can help me. I have my tinted glasses, my cap, my ear defenders and ear plugs as well as certain sensory ‘toys’ that stimulate my senses if need be.

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Chapter 6

Interoception and Relationship to Monotropism



Introduction

Neurodivergent individuals may find it difficult to identify emotions of self or others and have difficulties to generalize concepts or interpret context. Concepts and context require divided attention and connection from self to other. Difficulties in these areas can be misinterpreted as poor Theory of Mind. However, object permanence, (the knowing that objects, people, emotions, seasons and so on, continue to exist, even when out of sight), and *interoception*, one's inner sensing system require a person to attend to multiple stimuli to appreciate concepts such as 'here, gone, still here but unseen, hungry, and so on', may not be accessible in Autism as they are in the Allistic population. This chapter focuses our attention on the connections between monotropism and interoception.

Interoception

Sometimes thought of as our eighth sense is a lesser-known sense that helps us understand and feel what's going on inside our body. Individuals who struggle with the *interoceptive* sense may have trouble knowing when they feel hungry, full, hot, cold, thirsty, well or unwell, in pain, in love or tired! Having trouble with this sense can also make self-regulation a challenge. For a fuller description of interoception, the way it works, and of its likely connection with autism, see: <https://autismawarenesscentre.com/what-is-interoception-and-how-does-it-impact-autism/>

And

<http://web.seru.sa.edu.au/pdfs/Introception.pdf>

See also: <https://www.autism.org.uk/advice-and-guidance/professional-practice/interoception-wellbeing>

Our interoception ‘sense’ is vital to how we understand our basic needs and our relationship to these as well as how we understand other people (body language, mood changes, stress and anxiety, hunger, desire, and more). As Autistic individuals our body is giving us the messages to make sense of our internal sensing system, but sometimes our brain is not noticing these, due to being single minded and having all our attention elsewhere focused. This is especially true for monotropic individuals who may have their attention already occupied and don’t have any spare attention to notice their interoceptive signals (Murray et al., 2005). This might mean not knowing we are sick, in pain, in need of the bathroom, hungry or tired, to name a few.

Neurodiversity means all brains are different; it also means that neurological differences are to be recognized and respected as in any other human variation. Some of these differences (e.g. Dyspraxia, Dyslexia, Attention Deficit Hyperactivity Disorder, Dyscalculia, Autistic Spectrum, Tourette Syndrome, and others) may prefer to be thought of as neurodivergent (because they diverge from the majority). In individuals who are both Autistic and ADHD (AuDHDers), we know that monotropism reigns even more so than in Autistic individuals who are not ADHD (Dwyer et al., 2024). Many characteristics of Autism and of ADHD overlap so this will mean interoception awareness will be lower in both populations and possibly more so in AuDHDers.

Interoception helps develop our connection to self, as when we understand how we feel, we learn to connect to ourselves, and we are more likely to connect to others. So, neurodiversity is the diversity of human brains and minds—the infinite variation in neurocognitive functioning within our species. The greater the diversity of the pool of available minds, the greater the diversity of perspectives, talents, and ways of thinking—and thus the greater the probability of generating an original insight, solution, or creative contribution (Goodall, 2020).

Valuing minds of difference leads to empowering divergent minds too.

Supporting individuals with logistical challenges is so important. Learning as much as we can about how our minds work best and what our interoception needs are can also help us to understand others. Maybe you have had times when you have been so engrossed in your work, or some other activity, you missed a meal or failed to recognize you were hungry, tired, or needed the bathroom. For monotropic individuals, this is our natural state of being and we will often benefit from outside prompts such as those from our smart phone, smart watch, a text or email, and so on.

Goodall (2020) points out ‘Interoception, the conscious perception of internal body signals, tends to be atypical in people with developmental disabilities, mental health disorders, trauma, intergenerational trauma, and chronic stress...’ (p. 1).

The human brain with its neuroplasticity allows us to learn and develop various skills throughout our lifetime (Aikens et al., 2013). Various interventions with young people have enabled them to learn self-regulation skills which have been found to improve self-esteem and internalizing behaviour (van Genugten et al., 2016), but emotional dysregulation was found to be linked to suicidal ideation (Davey et al., 2018).

How Interoception Exercise Can Support Interoceptive Connection

Physical exercise is known to help in alertness and in brain connection (Ruiz-Ariza et al., 2017). It's not uncommon to see school or college young people involved in some form of physical exercise before going into class.

We know too that physical exercise in older individuals is also a precursor to not only keeping physically fit but also intellectually fit. When physical activity becomes too hard due to illness or physical disability, this can negatively affect blood pressure, cardiac fitness and mental fitness too.

Some brief interoception exercises I have done with my own children to help them recognize when their bladder was full, when they were in pain (earache), or when they were hungry, included the following.

The Need to Urinate

When I asked my youngest son if he needed to use the toilet, he just looked at me. It was as if he were asking me 'How can I tell if I need to pee?' So, I said to him, each time I asked him this question, 'When you press on your tummy low down, is it comfortable or uncomfortable?' When I asked this question, I gave him about 10 s to experience and interpret the feeling from the pressure of his hands on his tummy. 'If it's uncomfortable' I would say 'It might feel better if you pee into the toilet. Go to the toilet and pee and see if it does'.

Pain

There were times when my son seemed 'out of sorts' or possibly unwell. I used to say to him: 'Do you have pain; does it hurt somewhere on your body. Do you have some discomfort?' *At this question he would look at his hands to see what he had... what he was holding...* it was obvious he didn't understand so I rephrased my question. 'Does it hurt here if I press?' I did this for various places on his body, such as his arm, his head, his tummy ... I gave him time to connect to any 'feeling' he was having. If he was too apprehensive about me touching him, I asked him to use his own hand. It helped too if he closed his eyes. This was helpful because he could tell what was comfortable and what wasn't, then he could point to where his discomfort was.

I remember thinking 'he must be hungry' but he didn't respond when I said: '... are you hungry?' So, instead I made him his favourite, a snack I knew he liked, made sure he wasn't absorbed in a favourite activity, then asked him to close his eyes. I waved the fruit or meal under his nose, and I made sounds of enjoyment and

pleasure. I asked him to breathe in and smell the lovely fragrances from his food and as he took his time to connect, I'd say: '... you are hungry, that's why your mouth is making saliva (watering) and your body is telling you to eat this'. His smile told me he now knew he was hungry.

If an individual's interoceptive sense is disconnected, it's important to build interoceptive exercises into their day. In one South Australian School, they quoted their 'school exclusions' were down by 64% (personal communication, 2018) due to practicing interoception during school hours. I know (we all know) excluding a student from school achieves nothing, except possibly to confirm to that student they are not valued, and no one is listening to them (Parker et al., 2016)!

Some Ideas to Help with Interoception

- For the individual who doesn't recognize their internal body sensations such as the need to go to the bathroom, the need to have a drink or to eat, using a visual prompt photo/card as a reminder can be useful. At one of my workplaces, we had 'urine charts' on the wall. These consisted of a poster with three pictures showing one that was almost colourless, one that had more colour and the last one that was the deepest colour. These were all shades of 'yellow' depicting what the urine will look like from a well-hydrated person to one who was lesser hydrated and one that needed hydration.
- Some individuals may not be able to feel their heart is beating faster when they have completed strenuous exercise. So, sometimes specifically focusing on an activity to raise (or lower heartbeat/pulse) can be useful. Writing these into a graph or chart, can give a great visual to make these more visible. This may also help us to recognize the feeling and be aware of changes to our heart rate when we need to notice. But, just building awareness of heart rate isn't enough we need to also know what it means and if there might be anything we need to do about it (e.g. my heart rate is fast, showing I am excited, have done strenuous exercise, might be very anxious or unwell). As we consider these and which one is applicable, we then need to determine what, if anything, we should do? For example, 'Do I need to practice some mindful breathing to reduce my anxiety?', 'Do I need to see a medical person if I'm sick?', and so on).
- Some of us may not recognize when we are becoming too hot or too cold. Using a thermometer to check the room or outdoor temperature could be a useful way to let us know we may need to remove an item of clothing or put one on (take off a jumper or put on a coat over a jumper). Just knowing it's too warm without the added understanding of what this means for me, isn't enough. We must make sure we understand what is needed, what it means, and how to meet that need.
- To help us recognize our emotions is a difficult task. I will know I'm uncomfortable or I'm comfortable but naming an emotion in more detail may escape me. Using photographs of body language showing different feelings and talking about these may be helpful.

- When feelings are labelled, it can be a useful exercise. If an individual is finding it difficult to put words to their feelings, we might need an alternative way to describe what we feel. For example, as a synesthete I experience feeling states as coloured auras. The colour red means 'excitement', brown is angry, anxious is blue, and frustration is grey. You wouldn't know this about me though unless you asked me about my experience of 'feeling'. The first time an individual asked me how I was feeling I answered, 'With my fingers, how do you feel?' But you might state that you are feeling well, as in, you are not unwell. Then you might ask me if I'm well. Please avoid saying or using words such as 'good' or 'bad'. When I think of 'bad', I see an apple that is very bruised and has become discoloured and soft. Sometimes a person says they had 'a bad day' and this is hard to get a picture of!
- If an individual is completing a sensory checklist, it might be a useful time to chat about sensations they have and how these are experienced. This is one way which might help to raise awareness of body sensations, both external and internal, then we must add in the next step, what do we need to do about it.

Self-regulation is vital for good communication. How can we know if it's our turn to talk or when we need to be quiet? We depend upon our sensory and motor systems being in sync; (not over- or under-stimulated) and upon accessing the big picture. So, creating an autism friendly environment that allows the building of such connectivity is essential to the processes of inclusion and allows sensory modalities to be monitored and good communication to take place (hopefully).

As Autistic people, sometimes communication happens best when we don't look at someone. Looking at someone may accost the senses, and be too uncomfortable to process a response. Sometimes communication works best via text, email, comic strip stories, video, and TV. Communication in autism must consider the sensory systems we live with, must give us time for processing, and must be autism friendly.

Conclusion

Communicating with Autistic individuals makes the most sense when we can attend to the message being given and being accepted. This will mean if we have sensory issues taking over our attention that they are accounted for. For example: If my attention is taken over by anxiety it's not a good time to ask about pain or other discomforts I might be experiencing. Rather, first let's name our anxiety, see it is there, then practice taking in a breath and letting it out again. We may be able to 'imagine' the 'anxiety' as being carried away from us on our exhaled breath. If this is difficult to 'see', we could let the person know we understand they are anxious and we will look at the anxiety further, but first we need them to take some deep breaths and let them out again. We might go through a process of checking in on their body relaxation, starting with their head and ending with their toes.

The point is, we need to help Autistic people connect to their interoception sense, understand what, if anything they need to do about it, then move on in their daily lives. Interoception communication is vital to understanding self, understanding others, and to being able to regulate one's emotions in order to live a happy and well-regulated life.

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Chapter 7

Looking at Monotropism as a Strength-Based Cognitive Theory



Introduction

When we understand monotropism is foundational to understanding autism, it's all too easy to lose sight of what this may mean for a strength-based understanding of autism. After all, our insurance companies often only support someone for their 'medical condition'. Medical condition implies there is something wrong with you. Even in Australia, our National Disability Insurance Scheme (NDIS) which is meant to take note of an individual's 'functional capacity needs' not necessarily their 'medical condition' still might not recognize the impact one's 'autism' has upon daily function. This means the NDIS only offers financial support for an individual who can show formal evidence of 'disability' as in Autism level 2 and 3, and not level 1. For some individuals, however, an assessment of autism might lead to a formal report saying they operate at level 1... which does not account for those days when their autism leads them to sensory overwhelm and shut down, eventually to 'burn out'.

Pellicano et al. (2022) point out that when we consider what a person can do, rather than what they cannot, we are coming from a 'strength based' model. Also, capability is not an innate characteristic of the individual but more about how they are facilitated to be who they might be with appropriate environmental and other supportive factors. Both the physical and mental health institutions we are all engaged with need to consider these factors:

First, according to the capabilities approach, a human "capability" is not an intrinsic ability that a person has or does not have solely by virtue of who they are. Instead, "capability" refers to the actual opportunity to be or do something that is facilitated or constrained by features of the person and by the broader contexts in which a person is embedded. The relevant contexts can include close family and household influences; everyday community interactions; educational institutions; economic factors, including the cost of living; services and supports, including accessibility and performance of healthcare institutions; and the broader social and political context, including social attitudes towards autism. Second, flourishing human lives are characterized by a set of these capabilities which enable a person

to achieve any number of a range of outcomes, rather than by the attainment of a small number of pre-specified outcomes. These capabilities are considered foundations for a range of doings and beings; they shape what a person can do and, critically, who and how they can be in the world

(Pellicano et al., 2023, p. 656).

Monotropism helps us understand, as Autistics, what it means to ‘be in a state of flow’ and just how important that can be for our mental and emotional health.

To tease out mental health issues from Autistic issues, as well as recognize how to build mental fitness as Autistic people is hard. Overshadowing is a constant issue as autism rarely travels alone and has many cousins. Yet, it’s so important to interlace a neurodiversity affirming lens with DSM-5™ diagnostic criteria. The interplay between poor mental health and autism and the impact of unmet needs for Autistic individuals is far too often overlooked. It’s also important to note how autism presents differently in different genders. If these things are not considered and accommodated, then the impact of a misdiagnosis in adulthood can lead to self-injury, employment and relationship problems, mental health issues including suicidality. As physical and/or mental health professionals, we need to understand this, be equipped to ‘action’ on these points that need out attention, and not back down from supporting our Autistic clients/patients.

This chapter explores monotropism from a strength-based perspective. If an adult has lived most of their life ‘struggling’ to fit in with the Allistic community (where they do not fit) and have adaptively morphed (Lawson, 2020) into social society to keep safe, find friendship, employment, and so on, then separating autism from the ‘mask’ adapted for that purpose, can be hard to do. If we are not careful, we will miss such individuals and this only adds to their woes, unemployment numbers, poor mental health, suicidality, and relationship break-ups.

All too often we do not see the Autistic individual, but the mask they might have become. Constantly adapting to cover or hide who you are is exhausting and very damaging (Hayworth, personal communication, 2023).

When autism stops being a pathology and individuals are recognized for all they can be in the right environment, they will blossom. As an analogy, we see the healthy, colourful, and beautiful gardens when those gardens are tended to, with the appropriate soils and conditions that they need to help them grow into the best they can be. Orchids need a different environment to primulas and roses thrive in an environment quite different to cacti.

So, exploring key characteristics of autism from a strength-based foundation will consider the levels of anxiety that are experienced during social contact as well as concepts central to capabilities. Adaptive morphing in social situations, meaning individuals live their whole lives masking who they are to fit in, find a partner or a job, can have disastrous consequences upon mental health. The concept of object permanence (OP) and how it interjects with monotropism from a psycho-social model, not the medical model, is paramount to this understanding.

By the time a neurodivergent individual presents for counselling, especially if they are Autistic or/and ADHD, they may have lived their lives in a fog of confusion, disconnection, and a sense they don’t really know who they are. This is especially

the case for an individual who is late in identifying their neurodivergence. Neurodiversity includes every human on the planet, yet the desire to ‘find our mob’ rules. Therefore, identifying as ‘neurodivergent’ (those of us who ‘diverge from the majority’) beneath the neurodiversity umbrella can be key to gaining a sense of belonging.

Until we comprehend the reasons for the issues presented to us or capture the essence of the person before us, we will not be able to offer them insights that support them in the work they need to do to move forward in their lives.

I love the film series based upon being ‘divergent’ or having divergent skills, not just in one area.

Divergent Me

It’s scary to be funnelled into ‘one’ way of being!
 As humans we are too often sent
 Into places of another’s seeing.
 I may not fit the ‘standard’,
 Whatever that may be.
 But, to diverge from the ‘typical’
 I now know that is me.

Is being neurodivergent in a world of neurodiversity a contradiction? Some might argue it is. When Singer’s (1998) thesis explored the idea of ‘neurological diversity’ or neurodiversity, Judy meant it to be used as a political term for acceptance and support for difference, especially in the autism/Autistic community. The bulk of her thesis was focused upon disability and the differences between the medical and social models of disability (Singer, 1998), as Judy wanted to highlight the disparities between these. For some, however, this raised concerns that those Autistic individuals with complex needs may actually ‘miss out’ on support if we adopted the term ‘neurodiversity’ as a way of promoting inclusion and human rights (Jaarsma & Welin, 2012). While others (Legault et al., 2021) view neurodiversity as all-inclusive and neurodivergence as encouraging marginalization.

So, you can see the topic is still being debated. However, as the years moved on, neurodiversity, as a term, seemed to move on in public understanding too. The term was either adopted to mean ‘all brain variations’ or, only those with maladaptive brains, such as a brain ‘with autism’. For example, ‘... I have a neurodiverse child in my class...’ one teacher said to me. At other times, I have had someone ask me ‘... how do I help a child with neurodiversity?’ Hearing such phrases tells me that even though the term has been around a very long time, society still separates its members into a variety of states of being. For a fuller explanation and debate on neurodiversity, see Milton (2019, 2020). There is also an enlightening video by Jac den Houting: <https://www.youtube.com/watch?v=A1AUdaH-EPM>

While neurodiversity indeed does mean ‘All Brains’, in many minds it still tends to refer to a paradigm in which ‘all brains’ proposes a ‘standard’ that we associate with usual. So, when a brain or neuro-system doesn’t act as we propose it should, we could describe it as part of a ‘neurodiverse minority’ or we could call this being neurodivergent that which diverges from the majority. Maybe we could imagine neurodiversity as the overall umbrella term for all of humanity, but beneath that heading we also have ‘neurodivergent’ that helps us understand a major difference which impacts communication. Damian Milton addresses this beautifully in his work on ‘the double empathy problem’ (Milton, 2012).

So, yes, neurodiversity is to the brain and nervous system what biodiversity is to the plants and animal life on earth. Such biodiversity is not disputed, and we commonly appreciate the vast differences that exist between animal and plant life. Neurodiversity implies there is a huge difference in how our human brains appear. For example, gender appears to play a role in how our brains encode memories, detect emotions, recognize faces, problem solve, and make decisions. Cognition and behaviour are controlled by the brain, and various functional differences may be associated with the gender-specific structure of the brain (Cosgrove et al., 2007). Cosgrove et al. (2007) concluded: ‘...the gender-related differences exist in the whole-brain’. These gender-related brain structural differences might be related to gender differences in cognition, emotional control as well as neurological differences. Then again, Pang et al. (2023) found that brain ‘shape’ or geometry dictates behaviour, rather than individual aspects of brain connectivity:

Using mathematical models, we confirmed theoretical predictions that the close link between geometry and function is driven by wave-like activity propagating throughout the brain, just as the shape of a pond influences the wave ripples that are formed by a falling pebble

(Professor Fornito, 2023): [Landmark study finds that the shape of the brain influences the way it works - Turner Institute for Brain and Mental Health \(monash.edu\)](#)

This means all brains are different and such differences may only enrich human life, not detract from it. However, some neurodiverse individuals may find it perplexing to identify emotions of self or others and hard to generalize concepts or interpret context, as well as use attention differently to the majority, or may have learning differences and/or cognitive challenges. As one of those individuals myself, I am very happy to be neurodivergent; I know my cognitive style ‘diverges’ from the majority.

What do all the above have to do with empathy, Theory of Mind (ToM) neurodiversity or time travel? How do these play into a strength-based perception of monotropism?

In daily life concepts and context (e.g. up, over, yours, mine, ours, never, now, and so on) divided attention enables connection from self to ‘other’ for social interaction and communication between self and others to happen well. Difficulties in these areas can be misinterpreted as poor theory of mind (ToM) (Lawson & Dombroski, 2017), when they may be more connected to OP. When an individual doesn’t respond cognitively, as the majority might, we try to find a reason. We might conclude they lack empathy or are not able to ‘put themselves into the shoes of

another'. Milton's argument concerning 'The Double Empathy Problem' (Milton, 2012) demonstrates clearly however that Allistic individuals might 'get' other Allistic individuals and Autistic individuals might 'get' other Autistic individuals, but it's when the two groups try to communicate across the divide that issues occur (see: Crompton et al., 2021; DeBrabander et al., 2019). Milton argues:

The basis of the theory is that a mismatch between two people can lead to faulty communication. This disconnect can occur at many levels, from conversation styles to how people see the world. The greater the disconnect, the more difficulty the two people will have interacting

See: [Double empathy, explained | Spectrum | Autism Research News spectrumnews.org](https://spectrumnews.org/news/double-empathy-explained/)

Also, because we are all neurodiverse, understanding ourselves and each other is a challenge. Scripture tells us that 'like calls to like'. Bahns et al. (2017) found in their study rather than 'opposites attract', it's those who are 'likeminded' that are more likely to relate well together. In fact, the study suggests we are 'hard wired' for this. As complex humans we need to comprehend 'other' as being 'different' and also 'like us'. This needs a connected system that operates from a mature sense of OP. But both underdeveloped object permanence and disconnected interoception play a role in the social difficulties Autistic individual's experience. For example: '...some of us may not notice we are tired, hungry, thirsty, anxious and so on (Goodall & Brownlow, 2022) especially when our attention is elsewhere focused'. But, when we are tuned in and our attention is in a state of flow, we connect more broadly (Rapaport et al., 2024).

The young woman who lives not far from me completely 'loses it' and has a tantrum. When asked what the problem is she replies '...with my shoes on I can't "see" my painted toenails. How do I know they are still there?' When given her open-toed shoes (sandals) to wear, she was happy because she was satisfied with the coloured nail polish she had painted onto her toenails was still there. It's not a ToM issue, she isn't being thoughtless about the noise and disruption her 'tantrum' is causing, she is extremely anxious and fearful. All her hard work and time given to painting her nails was experienced as 'lost' during the time she couldn't see her toes. But when her Autistic disposition is accounted for, she is able to relax and enjoy her usual activities (dancing on this occasion).

OP enables forward-thinking (time travel), the sun is behind the clouds, I just can't see it at this moment... Forward-thinking and moving through time to understand consequences may be harder for some people. I know this is so for me, as a neurodivergent person.

Our interoceptive connection and awareness of inner bodily functions (heart rate, anxiety, pain and discomfort, thirst, desire, etc.) might mean regulation of emotions and the ability to respond 'appropriately' in a number of situations is negatively impacted for some neurodivergent people. But, when the interoceptive self is attended to, however, this awareness and emotive regulation can successfully occur. As an Autistic ADHD person who is dyslexic, dyspraxic, living with auditory processing disorder, and synaesthesia, I experience a heap of disconnection in a variety of ways. However, being neurodivergent for me doesn't make me less of a

human being, but it does make me different to most of my peers. So, during times of conversation, social activity and other types of human interaction, I might be disadvantaged, if my type of neurodivergence isn't recognized. However, when it is recognized, I can thrive.

To help myself, I use several tools and try to let others know what I need from them to enable successful interaction to occur (e.g. noise cancelling headphones, time to process an action, conversation or expectation, a quiet place for conversation to take place that is not a noisy café or shopping hub, a meeting place free from strong fragrances, and so on). I also use the medium of a smart watch and my mobile phone to keep track of time and expectations. Sometimes, due to my executive functioning challenges, I feel like a fraud. Experiencing 'Imposter Syndrome' can rob me of those things that I do well, See: [Mike Cannon-Brookes: How you can use imposter syndrome to your benefit | TED Talk](#) and: [\(9\) Neurodivergent Imposter Syndrome \(Autism & ADHD\) - YouTube](#)

Research is currently helping us to understand how much more 'connected' and able an Autistic person is when they can navigate their daily lives and relationships via their interests (Gunn & Delafield-Butt, 2016; Lawson, 2011; Rapaport et al., 2024). Although it's true we all do better when we are interested, Autistic people can only 'connect' when their attention is captured via their interest. This was clearly demonstrated in the GAMMA study by Kiser (2013) and Lawson (2011). It's also heralded in the work by Cherewick and Matergia (2024); Koegel et al. (2012); Lee et al. (2020); O'Connor and Kirk (2008), and in many Autistic people's stories of how they developed social understanding and gained access to education and employment (Elder-Robison, 2008; Grandin, 2006; Lawson, 1998; Lawson, 2011; Wood, 2019; Williams, 1992, 2006).

Instead of trying to 'fit' an autistic person into a societal lens that says 'you must be this shape and conform to this social concept to be "normal" and acceptable', we should be understanding of how monotropism dictates Autistic cognitive perception, processing, and behaviour. This means working together to find a middle road that accepts difference makes needed adaptations to the social, physical, and emotive environments allowing for each of us to relate to the other via a common interest, with sensory supports as necessary and a commitment to our common good.

Conclusion

Fitting all the pieces together completes the puzzle and gives us access to the picture as it was meant to be takes time and energy. Some of us are better at it than others! If we take the time to see and hear the person in front of us we could be saving lives. As an Autistic person who didn't receive their autism diagnosis until the age of 42 years, I know what life can be like living in a fog of non-connection. For me this meant frequent visits to our General practitioner (GP) to try to find out what was 'wrong' with me? I was to be misdiagnosed with a few 'mental health conditions' before my autism was recognized. These required me to take medications that were wrong for

me and served only to push me deeper into depression and despair. Once my autism was recognized though, along with my learning differences, I was able to steadily and over time put the pieces of the puzzle of my life in the right places. Looking back over those pieces can still cause me confusion, but now autism gives me the answers I so very much needed back then. It also helped me come to terms with the ‘cousins’ that accompany my autism and join with a community of others that share in a common goal. We want others to understand us, accept and support us as well as allow us to share our lives with you.

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Chapter 8

Autism, Gender Differences, and Monotropism



Introduction

For decades, autism was thought to be predominantly associated with being male, more than female (Asperger, 1943; Duvekot et al., 2017; Wing, 1981a, 1981b). However, our awareness of autism and its impact upon gender is changing. Not only do we recognize women and other genders (non-binary, trans) are just as likely to be Autistic as men, but we also understand that autism, which is a spectrum, impacts the gender spectrum in various ways too (Boorse et al., 2019).

This chapter briefly explores autism via the lens of gender and monotropism. It's commonly understood that Autism, initially gaining access into the medical history books in the early 1940s, via the writings and research conducted by Leo Kanner and Hans Asperger, was potentially only found in males. This misconception coloured how autism was seen and what autistic behaviour looked like (very male). So, when females began to show up in the statistics questions were asked as to why they had been missed in earlier years.

Gender Differences in How Autism Presents

Although DSM-5 (APA, 2013) does not state females (Assigned Female At Birth (AFSB)) on the spectrum present any differently to males (AMAB), there is an increasing body of research suggesting they do (e.g. Attwood, 2007; Cruz et al., 2024; Duvekot et al., 2017; Gould & Ashton-Smith, 2011; Hendrickx, 2015). See also: <https://www.healthline.com/health/autism-in-women#symptoms-in-women>.

So, even though the core 'difficulties' remain similar in either male or females on the spectrum, for females, AFAB, presentation is impacted and moderated by double X chromosome. Of course, not all AFAB have double X chromosomes, but the

majority will. This implies females without cognitive impairment (e.g. of average or above, IQ) often ‘fly beneath the radar’ and are overlooked. This may be happening for several reasons:

Due to being naturally more socially aware (having XX chromosome: See: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4785899/>) social observational behaviour may be more available. This allows females to learn early on how to act ‘as if they have understood’ a social convention or expectation, even when they haven’t. Therefore, their actions act as a type of mask (Allely, 2019; Lawson, 2020). This is not a mask of their choosing, but one as seen by others, allowing an individual to cope outwardly, while desperately paddling below the surface to stay afloat (Allely, 2019; Lawson, 2020).

This external coat offering a ‘masking’ or camouflage effect is not so much a chosen state, but rather the result of enacted behaviour to avoid discomfort (e.g. from not fitting in; from being punished and so on) (Allely, 2019; Cruz et al., 2024; Lawson, 2020).

What Do We See

- (a) Often specific passionate interests seen in females are more commonly accepted as ‘typical’ in females (AFAB) (e.g. books, animals, people, fashion, caring profession, music, art, theatre) and are not recognized as being of a single focused or more rigid nature and may pass unnoticed.
- (b) Autistic females (AFAB) are prone to internalizing their feelings, including anxious emotions, depressive states and self-directed behaviours, such as self-blame, eating disorders, and low self-worth. So, they often exhibit less externalized behaviours, when compared to males (AMAB).
- (c) This means difficulties may not be noticed by themselves, teachers, parents, families, or others, due to not being externalized or seen, or they may be misunderstood and labelled as naughty, lazy, and/or disobedient.
- (d) Mental health is often impacted negatively so females are more often diagnosed with other issues (e.g. personality disorder).

Autistic females come in all shapes and sizes and accommodating the broader understanding of what this means for these individuals will require a fresh look at autism and being female. For example, being female ranges across a host of characteristics inclusive of the ‘girly girls’ through to ‘Tom boy’ girls.

Female variability can pose problems for clinicians if they do not take this into consideration. For example, although we appreciate the female gender presents in a variety of ways we can be ‘thrown’ off track if the individual before us presents as more masculine, non-binary, or ‘over the top’ as their favourite cartoon person such as Elsa from ‘Frozen’.

Some females demonstrate characteristics more commonly associated with being male; exhibiting poorer intellectual functioning and more externalizing behaviour

difficulties, such as more irritability, aggression, or self-injurious behaviours (Frazier et al., 2014). It could mean certain presentations can cause us to ‘miss’ exploring autism as foundational to the individual’s challenges because they don’t seem ‘typically autistic’.

Although there will also be some males and other genders who demonstrate more traditional ‘feminine’ or ‘masculine’ traits and, at times, it will be difficult to tease these differences out, but if we have this understanding, we are more equipped to look deeper and beyond the possibly superficial presentation by an individual.

Gender can be a very fluid state and some females, although happy in their gender identity, present more traditionally female as any male might present more traditionally too. However, some women say they feel more at home among ‘the guys’ preferring male company and ‘masculine’ activities (Grandin, 2006; Hendrickx, 2015; Holliday Willey, 2015), while others would feel awkward and lost. In other words, it’s not one size fits all!

Although often female interests mirror that of their peers (dolls, animals, books, fashion, music, people, writing) (Grandin, 2006; Prince-Hughes, 2005), the difference is in intensity and how the interest dominates above all others. While, for males, special or particular interests, are often quite unique in character (e.g. collecting signs, fans, motor parts), for females their interests may seem less odd, and the passionate intensity of that interest often goes unnoticed.

Gould and Ashton-Smith (2011) suggest female diagnosis is being seen through the male lens, and this is one reason females are not recognized or are being misdiagnosed, many receiving mental health diagnoses like: ‘social phobia; schizophrenia; personality disorders, various eating disorders’; and so on. Of course, autism and a mental health diagnosis may co-occur. Rivet and Matson (2011) found females who were later diagnosed as being on the Autistic spectrum had previously been diagnosed with other condition such as learning difficulties, personality disorders, obsessive compulsive disorder, or eating disorders.

Statistically, it’s accepted that fewer females than males meet criteria for an autism spectrum diagnosis. Therefore, another reason assumptions have been made that autism is predominantly a male condition.

I think as a culture we just kind of got caught in the idea that being autistic is a male thing, when really, it’s just another way of being human

(Brooks, cited in Robertson, 2016: <https://www.pri.org/stories/2016-03-15/understanding-gender-bias-autism-research>).

Brooks also states:

... What I’m worried about is that because scientists and researchers and doctors and medical professionals are trying so hard to figure out what autism looks like in girls and women, they’re going to accidentally end up coming up with a gender stereotyped idea of ‘girl autism’ versus ‘boy autism...’ If people look at autism through a medical lens as something that needs to be fixed and treated, that’s a problem in and of itself, instead of seeing it as a different way of being and figuring out how you can support people whose minds are different with being successful in the world, *she says*.

Many females, influenced by their double X chromosome, are more socially inclined than males, and they may be practiced at ‘pretending to be normal’ (Holliday Willey, 2015) so they fit into societal expectations. However, this pretense takes much energy and is exhausting way beyond usual limits seen in the typical population (Heyworth, personal communication, 2023).

Clinicians seeing females present with issues relating to ‘unknown causes’ of tiredness (e.g. low iron, other physical or health reasons having been ruled out) often diagnose depression. Depression may be reasonable, but there may be root causes that are not obvious. Further investigation is warranted if the individual demonstrates some of the following: seems socially awkward, has unusual eye contact, complains of feeling isolated, disconnected from life, of never fitting in, shows evidence of ‘obsessive behavior’, indicates they have difficulty keeping a job or finding/maintaining friendships, is slow to process information/slow to respond and/or may have skin and other sensory sensitivities.

Some young girls (and older females) may display signs of demand avoiding behaviour (Christie et al., 2011), where it can appear they are being lazy, defiant, and difficult. Rather than assuming the former clinicians can check into the individual’s history further to see if avoiding tasks (even small ones such as getting dressed, eating, personal hygiene, responding in conversation, even avoiding things they would love to do, and so on) is an established pattern. The demand avoidance may be due to not having understood the nature of a task required, not felt able to manage the task or not felt connected to an outcome through lack of interest, and this makes the task inaccessible.

Hyper-sensitive individuals may fear being laughed at, being embarrassed, being exposed as not being able to meet performance expectations or being criticized and judged. These commonly lead to avoidant behaviour. Attwood (2012) reminds us to look at the DSM criteria of social difficulties in autism but add the female perspective. So, some females will be very sociable and outgoing, but fail to maintain friendships, while for others sustaining imaginary friends longer than is typically age appropriate might occur, and still for others social interaction will happen only by way of peer pressure or peer mediation, or masking.

Difficulties for Clinicians

There are a number of difficulties facing clinicians exploring the issues females present with, who tell their doctor they lack motivation, find it difficult to face the outside world and/or are not succeeding at school, work or in their social lives. Typically, clinicians jump to mental health reasons, such as depression, social phobia, or life demands such as grief and so on. However, the clinician may need to schedule a longer appointment, take more time over family history and attempt to ‘read between the lines’.

It's also difficult if the person wants some 'quick fix' and isn't willing to take the time needed to build a bigger picture. Without looking at the wider picture, assumptions such as those above will continue to be made.

Eating Disorders

Eating disorders (e.g. Anorexia Nervosa, AN) are now known to have a genetic link to autism (Nielson et al., 2015) and outcomes for Autistic individuals and AN are not as favourable as those with AN alone. Difficulties in separating these are well known. Physicians need to understand how similar they are and how commonly linked.

So, although autism and AN can appear to be completely different, they share many traits. For example, both show the features below:

- A fascination for detail
- A tendency to focus on oneself
- Inflexible behaviours
- Rigid attitudes

In AN, typical individuals may suffer due to feelings of being fat and wanting to 'fit' into societal pressures of what's considered 'beautiful'. In autism, it might be a means to subdue overwhelming sensory issues or avoid growing up if being an adult seems too demanding. It's difficult for clinicians to see these differences unless they go below the surface of presenting issues.

If a clinician doesn't feel confident or has their doubts, referring to a psychiatrist might be necessary. However, it's difficult for a physician to know or have local access to an 'autism aware/autism friendly' psychiatrist. The Royal College of Psychiatrists acknowledges that many working in general psychiatry have very little training in autism. Those working in intellectual disability and psychiatry services have more awareness of autism due to the comorbidity of autism and ID. This means that intellectually able females can be missed, or their behaviour misinterpreted, and the wrong diagnosis given. The lack of resources can present issues and may tempt the physician to treat the presenting condition without looking any further to check what might be behind it.

Challenges for Females Seeking a Diagnostic Assessment for Autism

The challenges for those seeking a diagnosis are many. These range from the lack of knowledge on autism among professionals in autism, social care, and health to lacks in general education and in public awareness.

There is less work on developing awareness of autism in females in general and less understanding of the impact of autism on women and girls.

Many females have been mis-diagnosed with a variety of mental health issues (social phobia; depression, personality disorder, and so on) so they may not realise their issues could be autism related. If this is the case unfortunately, they won't seek further help, or the support they receive will not be appropriate and may not help.

Things clinicians need to know about the current diagnostic tools and process, and how to change it to better suit women and girls on the autism spectrum

Currently the diagnostic process involves a child/individual being brought to the attention of a medical professional. This is usually due to some challenging behavior noted at school, a breakdown at work, a possible relationship issue, physical health and/or mental health queries causing distress for the individual, family or organization. However, as noted above, the reasons girls and women present to their GP are often different to those seen in males.

Women often present with various areas of difficulty, and at a much later stage in their life. The clinician may not link the areas of difficulty or associate these with autism if they don't add the 'how social and communication difficulties may present in females' tag.

Questions or areas of concern for clinicians to check re: daily living

In Autistic women it is not uncommon for them to have financial debts. (e.g. through misuse of credit cards, buying clothes on a whim, going out to eat etc.). Many find it difficult to navigate domestic duties, organize paperwork and, in general, stay on top of daily living requirements. This is due to difficulties in executive functioning and rigidity which affect the ability to plan, prioritize and/or multitask. Being monotropic accounts for this, but all too often the person is labelled as lazy, messy, not organized and so on.

The AQ 50 (<https://psychology-tools.com/autism-spectrum-quotient/>) a tool often used to check for autism houses questions that many females would 'pass' due to the 'male orientation' of question content. For example: 'I usually notice car number plates... ..OR, I am very good at playing games that involve pretending... .' In females the issue of pretense games is more about who dictates, leads, controls the game and at keeping the game rules set, rather than upon issues of pretense per say.

Being put on the spot and questioned can be very uncomfortable for potential Autistic females. Many are not good with abstract concepts or open-ended questions, but Autistic individuals are honest and will tell you the truth as they experience it. So, when asking questions around the areas set out below, a clinician would need to be aware of this.

Social: Do you have time for friends? Do you feel understood by your friends? Are you happy with the friends you have? Do you go out often? Do you enjoy parties? If not, how do you socialize?

Emotional: Do you find it easy to relate to the emotional states of others? If a family member died what might be your first response?

Financial: Do you find it easy to budget? Are you good with money?

Relational: Do you feel you are successful in your relationships? Do you long for a partner or are you happy choosing a single life? Do you share your home with others? If so, how does that work out for you, if not, why not?

A great autism diagnostic tool for adults which is proving popular and has gained validity and reliability is the Autism Clinical Interview for Adults (ACIA). This tool is comprehensive, does not depend upon the individual having parents and/or family who can support their assessment (although there is a place for this if possible) which is useful and practical when exploring adult autism. See link for further details: <https://research.ncl.ac.uk/neurodisability/ourstudies/autismclinicalinterviewadultstrainingacia/>

Conclusion

The above chapter paints a picture of some of the typical differences that ‘gender’ may present. Traditionally autism was considered a male disposition, and females were overlooked. However, the more we have come to understand about autism, gender presentations, adaptive morphing, reasons why males are more often noticed and capture attention of the medical community than females, internalizing and externalizing of behavior’s and the much broader autism spectrum than in previous times, the more we recognise the reality that too many females have either been mis-diagnosed with some mental health condition or just dismissed for various reasons. This means there are many females in our society living as Autistic people who do not know they are Autistic. Closing the gap for these individuals will not be easy to do. Many have given up seeking out reasons for their challenges; many are ‘self-diagnosing’ due to the long waiting lists and the costs involved with an autism or ADHD assessment, and many will continue to live their lives prone to mental health conditions, suicidality, employment and relationship frictions and many other states that induce trauma, loneliness and isolation. As practitioners we can and must do better!

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Chapter 9

LGBTQIA+ and Monotropism



Lesbian, Gay, Bi-sexual, Trans, Queer, Intersex, A-sexual, and others (LGBTQIA+) are all forms of either gender or sexuality identity. Intersex, though, which is a biological physiological disposition, implies in one individual lies some attributes of both male and female and their gender and sexuality can be anywhere along the gender and sexuality continuum. So, a person is born with both male and female biological traits, but these may not be initially recognized. This is not, of course, of their choosing and although it may be evident at birth, during adolescence or adulthood, individuals may choose to live as intersex individuals, or they may opt towards conforming their body to fit their most desired gender.

Over many decades, however, intersex individuals were often given surgical intervention (as infants) to change their genitalia to best fit the social or governing opinion of the times. So, if they appeared more male or more female, decisions were made by parents and doctors to surgically conform their genitalia to look more in line with conventional male or female genitalia. Over time human rights groups and intersex advocates protested at the potential devastating effects such surgeries had on human lives. Many hospitals have stopped the practice of such surgeries, but some continue this barbaric practice (Kingsbury & Hegarty, 2021).

The 2005 Chicago consensus statement was the principle medical response to these controversies in recent times. It acknowledged harms of some surgeries, significant gaps in knowledge, emphasised that the overall aim of clinical medicine was societal adjustment, aimed for a new classification system grounded in genetics, and introduced the neologism disorders of sex development, or dsd, as a new umbrella term in this area

(Lee et al., 2006)

Understanding intersex individuals who happen to be Autistic would be a particular bonus. For reasons related to the writing of this book I was not able to find any ‘research’ concerning ‘autism & intersex’ but, in general the feedback from the community is:

Contemporary clinicians should seek and prioritize their patient’s own language and goals around their intersex variations and bodily difference (rather than imposing DSD terminology), should acknowledge that young patient’s goals may ultimately differ from their

family's goals long term through putting off any unnecessary interventions until the patient is old enough to seek them out and give informed consent, and should source patient support information from large-scale internationally recognized intersex organizations (including those for all intersex people such as OII, and local chapters for patients' specific variations such as TS and AIS societies ...

(Jones, 2018, p. 111).

The + symbol at the end of LGBTQIA denotes other forms of sexual and gender identity that are not spelt out but are just as valid. Traditionally, 'heterosexuality' is viewed as the most common form of sexuality and gender has been traditionally thought of as either male or female. It's of interest that homosexuality compared to heterosexuality appears to be more common among the Autistic population when compared to the Allistic population. There is much speculation as to why this is so (Graham et al., 2022). Dewinter et al. (2017, p. 2929) state:

A substantial proportion of autistic adolescents and adults are LGBTQIA+. Autistic people are more likely to be transgender or gender nonconforming compared with non-autistic people, and findings from a recent autism registry study suggest that among autistic people able to self-report on a survey, up to 18% of men and 43% of women may be sexual minorities.

This is supported by research from George and Stokes (2018b), who found similar outcomes from their work.

Gender identity, based within the neurological system (Swaab, 2007; Swaab & Garcia-Falgueras, 2009; Swaab et al., 2021) focuses attention upon one's inner and expressed sense of self. Sexuality is physiological and includes biological functions of the reproductive system as well as sexual behaviour (Schwarz, 2024).

Gender and sexuality may be experienced differently by the Autistic population compared to the allistic population. The Autistic monotropic brain possibly leads to being more in-tune with individual foci rather than societal pressures. Generally, the Allistic population is more socially adept, but only within the allistic world (Milton, 2012). Their default setting is to shift attention with ease, allowing for a wider social engagement. This has implications for gender identity, sexual development, comprehension, and experience. For example, as demonstrated in earlier chapters of this book, the external sensory world (seven senses) and internal sensory world (eighth sense) connect 'differently' in the monotropic brain than in the polytropic brain (e.g. mono: sensory sensation can be heightened or diminished (DSM-5-TR, 2022), but, for Allistics sensory information is diffused by nature of it being shared across many senses. In autism, if an individual is focused and interacts with elements of that focus, they can experience a flow state. But, shifting from a flow state to an interrupted, discordant state can be detrimental. This is because it can be confusing, uncomfortable through to painful and even lead to inertia (Rapaport et al., 2024).

Accordingly, Autistic individuals may experience higher levels of gender incongruence, non-binary gender, and a variety of sexuality domains, but be less connected to the wider social underpinnings (e.g. easier to believe 'porn' is real rather than scripted; harder to appreciate, recognize or use flirtatious behaviour; may 'feel' like 'just me' rather than specifically 'male' or 'female' and may conform less to traditional social gender or sexuality 'norms' (Purkis & Lawson, 2021).

Monotropism leading to passionate interests that capture our attention completely, leaving no spare attention for other things, may lead clinicians to associate our ‘perceived’ obsessive behaviour around gender or/and sexuality, as being ‘an autism thing’ that shouldn’t be taken too seriously’. For some this understanding may lead them to wonder if the Autistic person they see before them, who may be gender questioning or even insistent their gender needs to change, is actually experiencing ‘sensory or even body dysphoria’, rather than gender dysphoria. It’s certainly very important that the differences be explored, and the issues separated. However an individual presents to us, we must respect their experience, listen to their story, and support them to explore where their experience may be leading them. Please do not dismiss us!

Obviously, we know gender and sexuality are different (gender refers to our inner identity and how we perceive ourselves and is based within the neurological system, whereas sexuality is physiological and includes biological functions of the reproductive system as well as sexual behaviour) (Lawson & Purkis, 2021). Both will include our sensory systems (external and internal) and, as suggested in earlier chapters, these systems will be experienced differently for Autistic individuals from those within the Allistic population.

Because in autism our brain leads us to be single focused and not traditionally bound to being socialized in ways the Allistic population experiences, this will dictate to all areas of our lives. This obviously has implications for gender and sexuality development, comprehension, and experience. Some studies have shown that sexuality and gender are much more diverse in Autistic individuals (Graham et al., 2020; Sala et al., 2020; Sarris, 2020; Strang et al. 2018a, 2018b).

The idea that males and females experience sexuality and gender differently in autism, compared to the Allistic population, is not new but, what will this mean for diagnosis, individual development of sexual and gender identities as well as how to support this diverse population? These questions will face clinicians and professionals supporting Autistic people. With the emphasis on ‘Nothing about us without us’, we must walk beside Autistic people, listening to us and supporting us with respect and understanding.

One Example Knowing that ‘gender’ is how we feel about and relate to society’s experiences of being male or female will mean different things to different people. But, in autism this may not be so connected to a feeling state so much as it is to ‘a state of being’. As an Autistic person, I might have a ‘knowing’ within me that I am of the gender that leads me to being more ‘at home’ but I might not always have a word for this. I might not be comfortable with typical gender roles that are assigned to me. This is especially true if there is an expectation, for example, being born ‘male’ but feeling more at home with traditional female roles, such as cooking, sewing, domestic chores, or organizing. I might not ‘feel’ feminine but I might know I’m not male. Some Autistic people say things like ‘I’m not male or female, I’m just me’ and the term ‘non-binary’ gender might fit them best. One could argue that gender roles are socially constructed while gender per se is an inner knowing (Lawson & Purkis, 2021).

A Bit of Wenn's Story As a trans masculine Autistic individual, I transitioned over a period of years (2013–2021), before I could say my 'trans puberty' was complete. Initially, it took a long time for me to recognize my male gender (I was 61 when I began to recognize I wasn't female but male in my gender identity). Many of my challenges with being Autistic and trans are written about in our book, Lawson and Lawson (2017). But, here in this book I am only going to share in relation to my own experience and tell you about a few of the things that happened to me.

All through growing up and well into my adult years, although I knew I wasn't female, (in the traditional sense of the term) I didn't have the words for describing how I felt or what my gender might be. Instead, I told myself and others 'I must be middle sex/gender'. I loved a woman so I thought I must also be a lesbian although I never frequented the 'gay' scene and didn't know others who were like me. I felt very much alone and was totally unaware of the unique population of Autistic and LGBTQIA+ that existed in the wider community. These days, social media has made accessing our community a lot easier.

During the process of transitioning and still so today, I have challenges that relate to my being Autistic and to being trans. Some of these are connected to my sensory system which is mostly heightened. This means I need to wear tinted lenses, wear a baseball cap to help minimize the lighting impact upon me, and sometimes I use ear plugs or ear defenders when noise is too much for me. I also avoid crowded spaces, smelly restaurants, itchy clothing, being required to multi-task and be involved with socials I've not got a say in. Many of these issues are due to my autism but they impact being trans because the non-Autistic trans community often expect trans folk to be sociable, enjoy parties, and be quite extravert. Of course, this doesn't apply to everyone, but mainstream trans folks might not understand those of us who are Autistic.

My 'unique' challenges may not be yours, but it is important to identify these for yourself. Although I think the sensory things can be unique to autism, they may be highlighted for/in, Autistic trans people because being trans might present some novel situations we may not be prepared for. For example, I wasn't prepared for the smell that hit my senses when I first stood to pee. Somehow sitting down to pee, I hadn't noticed the strong smell of urine. There are two ways to deal with this, if this is one of your challenges. Initially, we can drink more fluids (especially water) as this will dilute the urine and remove the unwanted smell. Secondly, we can choose to sit to pee, so we are one more step removed from the fragrance of urine. Of course, this means using a stall in public conveniences and not the urinal. If you are transfeminine, you might welcome this change.

Another challenge I faced during transition after a week on testosterone was having painful nipples. I have a high pain threshold in general, but this one really was something that caught me off guard. I took one paracetamol to help and avoided loose, flappy clothing that moved lots and irritated my chest. I am used to wearing loose fitting clothing because it helped to disguise my shape so moving towards more fitted clothing was a challenge.

I was unsure how testosterone might affect me... I guess it might be the same for those taking female hormones. It is important to check blood levels pretty often, especially at the beginning of transition. I appreciate not everyone chooses to transition medically, but if you do, it's a challenge to trust others to advise you. As an Autistic person I'm not keen on doctors and medicos, let alone attempting to explain my concerns to them. But it's important we each take hold of our own fate and let others know what we need from them. For example, if you can't cope with being in a waiting room, ask the nurse to text/call you when it's your turn to see the doctor, so you can wait in your car or close by, but not in the waiting room.

If you have any allergies (latex; Elastoplast), make sure you let the medicos know because they might use such as a dressing after a shot or after surgery. I come up in welts if these are used on my skin.

As Autistic people, we may be very sensitive in a number of ways. Negotiating these is also a challenge! Separating any issues that are trans related from those that are autism related is a challenge in and of itself. As Autistic people, we are very literal and single-minded. This might mean we have particular beliefs about our gender and how we 'should' look, or what we 'should' feel.

Quite often you might 'feel' you are 'not trans enough'. Being trans and moving/moved away from the gender you were assigned at birth is a process of change and not necessarily a feeling state. I have a good friend who was trying to be helpful and make me feel good about myself. He said: 'you pass well as a man Wenn, no-one who meets you and didn't know you were trans would guess you were'. His words are actually 'genderist' (like racist but with gender rather than race). He didn't mean it, but being the gender you are at home in has nothing to do with whether you 'pass' as that gender. The literal take on this is offensive and not helpful. If this happens to you, don't take it personally because the individual doesn't know their words are hurtful. Rather, you can point out that gender is a spectrum and no one gender is like another. Meaning, you are not attempting 'to pass' as much as you are working towards being your authentic self.

There are a lot more challenges with being Autistic and trans, I have only mentioned a few that were part of my experience. It might be helpful to join a Facebook page or a trans/Autistic group, where you can share the challenges you have with others going through similar stuff. Don't struggle alone!

Gender Dysphoria Historically

Historically, explanations for gender dysphoria were known as mental illness possibly due to trauma, sexual abuse, and/or a lack of role models. But today, we know that biology and hormones contribute to and influence gender development (Korpaisarn & Safe, 2019; Ettner, 2020). We know that our physiology as a system is laid down in utero during week seven but gender identity, which is formed in the neurological system isn't laid down until week 14. There may be some discrepancy, therefore, between hormone washes at these stages and they are not always in sync, see: <https://www.dailymotion.com/video/x38hzm7>.

Gender Discomfort or Gender Dysphoria?

Traditionally, gender and sexuality have tended to be seen as ‘binary concepts’ which means they have been ‘set in stone’ and understood as fixed and not changeable. However, this idea has radically changed over time as we have listened to the experience of individuals along with the science of gender and sexuality. We now know that gender and sexual identities are not set and may change over a lifetime, more than once (e.g. Kaestle, 2019). We may be familiar with the idea of gender dysphoria, but in autism some may experience this on a spectrum of mild, to moderate and severe gender discomfort.

It’s important to understand that ‘gender’ discomfort might not be the same as ‘gender dysphoria’. Many individuals feel ‘uncomfortable’ with gender role expectations placed upon them but are content to be the gender they have been known by since birth. Others have an inherent discomfort with their assigned gender, but it’s not so dramatic that it could be called ‘dysphoria’. Yet, for many of us, our ‘discomfort’ might mean we need to consider if our known gender is the right one for us (Lawson & Lawson, 2017).

In the past, ‘gender dysphoria’ conjured up images of mental torment that haunted individuals who were often suicidal and desperately unhappy. This will still be the case for many, but for some of us, we were not terribly unhappy, so much as just not at home. I know many of my friends’ responses to the news of my being ‘transgender’ were: ‘but you weren’t unhappy with your gender’ or ‘but you never seemed unhappy, so why change your gender?’ For me, not being ‘at home’ but not having words to express this, meant I took much longer to join the dots and to realize I was actually living in the wrong gender (Lawson & Lawson, 2017).

DSM-5 and Gender Dysphoria

The DSM 5 (APA, 2013, p. 22) defines Gender Dysphoria (GD) as a ‘marked incongruence between an individual’s experienced or expressed gender and the one they were assigned at birth’. Specifically, GD describes the emotional turmoil and distress experienced due to such an incongruence.

Living with gender discomfort is not a mental health issue, but it does need recognition; otherwise, it could contribute to mental distress and mental ill health (Dhejne et al., 2016; Cleveland Clinic, 2021; Pinto et al., 2022). More and more people are presenting with either gender discomfort or gender dysphoria, than ever before (Judge et al., 2014; Kaltiala et al., 2020; Wiepjes et al., 2018; Wattel et al., 2024).

Why is this so? Maybe due to wider societal conversations about gender identity, maybe due to wider acceptance of gender differences, it’s hard to tell. But certainly, gender roles and gender expectations have changed over time and this impacts Autistic people too.

Again, I emphasize, gender roles and expectations are not the same thing as one's internal sense of gender identity. I am very happy, as a man, doing the dishes, cooking the dinner, arranging flowers for the table and creating gift cards for my friends. Traditionally, these may seem more like 'feminine' things to do rather than masculine. But these are more to do with personality, likes and dislikes, skills and abilities and general 'feeling comfortable' with, rather than one's inner sense of gender identity.

Changing Gender Roles

In 2023, especially in the Western world, there is an expectation that men will be more caring and more involved with family life, than they have in the past. This is especially so since the 'extended family', where aunts, uncles, and grandparents were close by to help shoulder the family load, may have moved away. But, as the nuclear family is now the more dominant (mum, dad, and their offspring), the shared responsibility of parenting for their children no longer is spread around among the extended family, so fathers and mothers carry the load alone. In some countries, government assistance aids the family income, but this is not the case in all areas of the world. This does mean gender roles are not so explicit, and some may feel there is confusion around this idea. However, sharing the load in this way is not a bad thing, it's just different.

Recognizing Gender Discomfort or Dysphoria

How do we recognize the differences between gender discomfort, gender roles discomfort, and gender dysphoria? It's uncommon that adults show a strong desire to live as a different gender to that assigned them at birth. But we will come across individuals wanting to rid themselves of all connections to the gender they were assigned at birth, which might mean surgical intervention and transition to enable them to have the anatomy of the gender they feel most at home in. They will also have feelings and emotive states associated with the gender they are inclined towards, rather than those associated with their assigned gender.

For children there will be a strong preference for clothes of another gender, cross-gender roles in make-believe play, a disdain for and refusal to accept the physical attributes of their body (i.e. ignoring menstruation and hygiene), toys and games associated with another gender, playmates of another gender, and a strong desire for the physical sex characteristics to match their experienced gender (Boston Children's Hospital, [n.d.](#); Frank, 2020).

As humans developing in utero, the distribution of hormones will vary across gestation of our physiology and neurology (oestrogen and androgen). There will be receptor activation variations that impact hormone distribution and uptake during brain development in utero (Fernández et al., 2022). This should encourage us to be less judgmental and more aware about the issues of gender dysphoria.

As stated above, we know gonadal differentiation occurs around week seven or eight of pregnancy, while differentiation of the brain and nervous system occurs after hormonal wash activates the pathways around 14 weeks in utero. Due to the different timing, sexual distinctions in the gonads and brain may take different directions. The resulting asynchronization can leave a developing anatomical female with a masculinized brain and vice versa (BU Medical Center, 2015; Swaab & Garcia-Falgueras, 2009; Fernández et al., 2022). Therefore, predicting gender based on external anatomy is not entirely accurate because there is no way to assess underlying neurobiology and development explicitly.

As far as autism is concerned, the literature on the intersection of autism and gender dysphoria has increased in the last decade. Early studies showed that Autistic individuals reported higher rates of gender diversity than the Allistic population (Glidden et al., 2016; Van der Miesen et al., 2016; George & Stokes, 2018). Therefore, it's not unusual that Autistic individuals will experience lower identification with specific gender(s) than Allistic people. This is especially true for those assigned female at birth (Cooper et al., 2018). Research also suggests that this relationship is bidirectional, where the rate of autism in transgender individuals is between 6% and 26% compared to a 1.85% prevalence in the general population (Thrower et al., 2020; Maenner, 2020; Walsh et al., 2018).

A 2022 systematic review of current theories on a proposed link between autism and gender dysphoria found 15 published hypotheses, all lacking substantial empirical support (Wattel et al., 2024). Similarly, only two longitudinal studies on the intersection of autism and gender dysphoria have been published (Nobili et al., 2020; Russell et al., 2021). As the needs and experiences of trans-Autistic individuals are heard, treatments and research will become more accessible to and inclusive of our population (Wattel et al., 2024; Zupanič et al., 2021). Much distress is experienced by Autistic adults with gender dysphoria that includes social stigma about gender as well as being neurodivergent (the term neurodivergent implies individuals who 'diverge' from the majority). If gender transition is sought, it will impact an individual's known world and such individuals need respectful support that is 'autism informed'. During gender transition, the Autistic individual's need for structure and routine will be interrupted, and even though we want the gender changes to take place, it will be hard for us to imagine these changes. We may have a picture of the finished product, but not the process to get there (Cooper et al., 2022).

Some agencies and organizations are hesitant towards us and find our concerns about gender, as Autistic people, worrying. They may not take us seriously, often dismissing us as people with little awareness of our inner selves and even suggest our concerns are possibly based in sensory issues, specific interests or are born from what we may have seen on social media. However, Autistic youth and adults are clear that autism does not impair their understanding of gender (See: [Trans 101: Neurodiversity \(youtube.com\)](#)). In fact, many individuals feel that being Autistic facilitates their understanding of gender identity and self (Cooper et al., 2022; Cooper et al., 2022b).

Sexual Harms and Risks

We know Autistic children and youth are more at risk of sexual victimization than the Allistic population. This is especially true for gender-diverse Autistic individuals (Pecora et al., 2020; Ohlsson Gotby et al., 2018; Gibbs et al., 2023). The statistics for the intersection of autism, mental health, and gender discomfort tell an important story. A 2022 review revealed that individuals with gender dysphoria showed elevated rates of depressive symptoms (64%), suicidality (42.9%), substance use disorders (40.2%), anxiety (25.9%), and general distress (33.8%) compared to their non-Autistic peers.

The review also highlighted individuals with GD encounter more social stressors and face higher levels of discrimination, contributing to worsening psychiatric conditions (Pinto et al.). Non-white individuals were particularly impacted (Mitchell et al., 2022). These socioeconomic disparities mirror those found in autism care, highlighting the underlying inequalities that shape the identification and management of autism and gender dysphoria (Aylward et al., 2021; Nevison & Parker, 2020; McDonnell et al., 2019).

Autistic individuals also experience a notably higher prevalence of mental health conditions than Allistic individuals. Moreover, research has shown that Autistic LGBTQIA+ individuals experience significantly elevated rates of mental illness, physical health challenges, unmet healthcare needs, limited insurance coverage, and refusal of services by medical providers compared to cisgender Autistic people (Hall et al., 2020). Given the high co-occurrence of gender discomfort and gender dysphoria with autism, comprehensive mental health support is paramount for this population (George & Stokes, 2018a, 2018b). Other drivers of poor mental health and suicidality for Autistic LGBTQIA+ people include gender-based victimization, bullying, violence, harassment, and rejection from family, friends, and community (Virupaksha et al., 2016; Hall et al., 2020).

Monotropism and Technology

As an aspect of monotropism technology use in autism, which is often our route to connection, may lead to concerns about porography. If we are not finding support from family, friends, and those in our daily lives, we will turn to social media and other ‘online’ mediums to search for answers. What does being gay look like in autism? What and how do I go about trying to understand sexuality and relationships? ‘Years ago, if you had looked up “sex” in a dictionary, which was a book, you could read about sex without it connecting you to any other topic. These days if you type sex into a search engine you will find porn’ (Maree Crabbe, 2024, personal communication).

Porn for monotropic individuals can lead us down rabbit holes that take us deeper and deeper into a realm of sexual experience we may not want to go. But, if watching porn is paired with the amazing feelings of orgasm, we are only reinforced to seek it out more. Being sexual beings is healthy and normal, just as is our sexual curiosity and our need for sexual exploration. But we can get ourselves into ‘a deep pit’ of all kinds of risks and harms, including legal ones, if we are not aware of the pitfalls. A useful programme that aids the support of young Autistic people with this domain can be found at: <http://www.notthenorm.com.au>

The good news is that despite the daily challenges faced by the Autistic transgender community, our resilience is notable. Research indicates that transgender individuals with high self-esteem, assertiveness, and perceived social support from loved ones demonstrate greater resilience to psychiatric conditions than their counterparts (Hall et al., 2020). Further, a growing body of evidence reveals that trans-Autistic individuals experience significant reductions in poor mental health and suicidality when provided with gender-affirming care (Cooper et al., 2023a, 2023b; Dhejne et al., 2016; Lawson & Lawson, 2017; Virupaksha et al., 2016). These improvements can be attributed to greater comfort within one’s body and the validation and hope that accompany acceptance and treatment (Dhejne et al., 2016; Lawson & Lawson, 2017).

Diagnosing gender discomfort and gender dysphoria in Autistic individuals, formal or otherwise, should be done carefully, taking into account the signs and outcomes related to autism and GD. Autism traits like rigid thinking, sensory sensitivities, resistance to change, and social differences may compound the issues, making the diagnosis and treatment process more difficult. Special interests, however, are not seen as contributing to gender discomfort or dysphoria although a passion for needing to be of a different gender to the one assigned, will take over all of one’s attention, due to being monotropic. Therefore, practitioners must be educated on the intersection of these experiences (Cooper et al., 2022b).

For example, Autistic children diagnosed with gender dysphoria often exhibit elevated passionate interest around gender themes (Zucker et al., 2017; Vanderlaan et al., 2015). Without proper knowledge of both gender dysphoria and autism, fixations on gender identity or disdain for one’s physical appearance may be considered as special interests or sensory processing issue in Autistic individuals (Paradiso et al., 2022). The intensity and persistence of attention on gender identity can vary among people and lead to different outcomes. For some, it may indeed be a temporary intense focus, while others may fully experience gender dysphoria. Given the irreversible nature of many gender-affirming treatments, distinguishing between issues caused by being Autistic and actual gender dysphoria is crucial (Valdés et al., 2022).

To ensure a comprehensive assessment, researchers recommend using a team of practitioners, including professionals knowledgeable in both autism and gender dysphoria (Hall et al., 2020; Mitchell et al., 2022; Pinto et al., 2022). Based on the high association between gender dysphoria and autism, researchers also recommend routine assessment of autism in individuals who seek treatment for gender dysphoria (Shumer et al., 2016). For Autistic individuals seeking and receiving gender-

affirming care, treatment can improve mental health issues. It is, therefore, imperative that supports focus on the lived experiences of the individual instead of the potential drivers for the co-occurrence of autism and gender dysphoria (Nobili et al., 2020).

Several case reports indicate positive outcomes from gender-affirming care for trans-Autistic people (Lawson & Lawson, 2017; Van der Miesen et al., 2016; Zupanič et al., 2021). Gender-affirming treatments include anything from changes in gender expression and role to hormone therapy or surgery. For Autistic individuals with gender dysphoria who decide to transition, the process often takes longer than for Allistic trans people. This is due to autistic traits associated with resistance to change, rigidity, and sensory perception. However, over time and with continual support, Autistic individuals who transition experience less anxiety, depression, and suicidal tendencies and higher self-esteem and overall quality of life (Lawson & Lawson, 2017; Zupanič et al., 2021). Recent findings underscore the benefits of providing appropriate support and treatment tailored to the unique needs of Autistic individuals with gender dysphoria.

Conclusion

Autism and the Future of Gender

Most Autistic people being single-minded, focus intensely on one or two aspects of interest that capture attention, but Allistics tend to focus less intensely but on a broader range of topics. As previously mentioned, Allistic focus is compared to a wide beam of light (e.g. a lighthouse) and Autistic focus as a large torch operating on the narrow beam. With a wider view (light beam), one can create contextual understandings of broader circumstances, unlike a narrow or single-minded view which may not see the full picture in context (Murray, 2018; Lawson, 2019). Therefore, because concepts of gender are very cultural, they can be difficult for Autistic people to access, and many of us never assume the gender roles that society would have us accept. On a fundamental level, most Autistic individuals experience and understand gender differently from, and often in contrast to, the wider population (Lawson, 2019).

Autistic individuals offer an intersectional perspective of humanity, encouraging a more nuanced and inclusive view of gender. Researchers, practitioners, and communities must continue to shed prejudices and work to love and accept our Autistic and gender-diverse population. One day, I imagine that gender will be understood as part of humanity that is changing and adapting to an ever-changing world. It will not be seen as ‘either or’ but as individual as one’s love of coffee; it comes in all shades and tones, with darkest to lighter and lightest, but perfectly usual!

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Chapter 10

Autism and Cousins: Living Life Well



Introduction

Throughout this book, we have explored various aspects of what it means to be Autistic. Monotropism has been defined and explained as ‘autism’s dominant foundational cognitive style’ along with its impacts on various intersections, such as OP, interoception, and our external sensory sensing system. Of course, being monotropic impacts all of who we are so must be the lens through which we view all of the experiences life takes us through. For example, we might have been told our thinking is black and white, that we are literal and not great at reading the ‘hidden curriculum’.

But, interpreting this area of our processing style could mean we ruminate in our thinking and home in on the non-black and white aspects of life, which seems like a contradiction. But, actually being monotropic might mean for some of us it’s not so much about being ‘black and white’ in the traditional sense of the term, as it is about almost ‘obsessing over’ the bit our attention latches onto, which might be the grey areas, the bit between the black and white, but we do so in a ‘black and white’ manner which is rule based, factual, and taken literally. Rumination is a ‘hard fact’ of Autistic life that is often not understood. Comprehending this though can be a real ‘game changer’ for both client and clinician.

Learning more about autism and these areas of the human domain and how they apply to Autistic people will aid us all in our communications. Once an understanding is reached this learning can translate into appropriate support, care and interaction with Autistic people, reduce stress for all concerned, and enable Autistic individuals to live a fuller life.

Autism isn’t heterogeneous and this means each of us is so very different from the other (like most humans). But however we present with our various personalities, gender, socioeconomic status, religious beliefs, education, cultural norms, ethnicity and employment status, monotropism will be at the core of our autism.

It's important to recognize the lens through which we encounter other individuals, so we don't get sidetracked by either their achievements or their challenges, their education or employment status. It's tempting to see 'autism' only in the superficial way that is depicted by the DSM-5 (2013, p. 22) (e.g. difficulties with social understanding, communication, rigid interests, repetitive behaviours, and sensory challenges) but if we don't 'see' the individual and their individuality we might miss how they are impacted by these.

For example, 'difficulties in the Allistic arena' for Autistics might not show if the individual is operating within areas of passion or/and interest. But when the same individual is in an environment outside of their 'interest' we see a marked difference in their social encounters. They may avoid eye contact, not be comfortable with 'small talk', mask to enable them to 'fit in' and/or not be able to stay very long in such a social situation, needing to leave early or having to retire due to headache and so on. This isn't a situation where there is an occasional headache, but one that is persistent in certain circumstances. Often the masking an individual feels compelled to 'wear' is exhausting and cannot be managed for a long time without physical affects.

Autistic adults have a right to flourish. It must be 'nothing about me without me':

'A Capabilities Approach' to understanding and supporting autistic adulthood', was a research project undertaken by a team of autistic and allistic researchers. The study found little research concerning "thriving" autistic adulthood'. Autism was largely conceived as a condition of childhood, governed by an orthodox scientific approach that conceptualizes autistic experience as a series of biologically derived functional deficits. The capabilities approach focuses on ten core elements of a thriving human life identifying areas autistic adults currently thrive and/or struggle with; thus, effectively signposting directions for services and support (Pellicano et al., 2022, p. 624).

So far in our exploration of Autism, we have come to understand Autism as a wide spectrum with various conditions often that co-occur. Cognitively, the literature mentions learning disability, intellectual disability, and various others in between. We note individuals can have average to very high intelligence, good verbal skills, rich vocabulary, an ability to think in visual images and identify patterns, a propensity to think outside the box and generate novel solutions to problems, an ability to absorb and retain large amounts of information, especially about topics of particular interest, and are detail oriented, upon areas of interest. We have an ability to focus for long periods on areas of interest, being non-judgmental, have an ability to perform repetitive tasks where accuracy, rules, and routine are important; we are reliable punctual, honest, loyal, fair, and usually have a propensity towards social justice.

Along with this very positive list of assets (many not so likely in the Allistic population) we have a range of challenges. These are usually related to a tendency to take words, people, situations, and emotive states literally; a disposition towards undeveloped OP which can make it hard to forward-think and/or appreciate consequences; we have an internal sensing system that has to compete for our attention making it harder to recognize hunger, thirst, temperature, pain, sexual desire, anxiety, and so on, due to all of our attention being elsewhere focused; an external

sensing system which can mean our senses can be over or underwhelmed from things we see, hear, taste, hear, feel, and so on, for the same reasons as our interoception which may be 'off line' and we have the added challenge of trauma. Trauma comes from just being a square peg in a round hole. It comes from a lack of place in a shared community (family, school, college, workforce, romantic relationships, friendships, colleagues, and neighbourhood). Sometimes it presents like any other trauma might, but often it goes undefined as it might be masked by our need to 'be like most others'. Then there are the cousins, cognitive, mental, and physiological in make-up.

Autism Can Sometimes be Accompanied by a Cousin Called Dyslexia

Being Autistic and dyslexic might mean having great visual processing and pattern recognition with good spatial knowledge allowing access to big picture thinking, when connected, as well as having an ability to see things holistically. But others of us who are Autistic and dyslexic may not be pattern or visual thinkers, but we may have conceptual ideas that are really practical and/or imaginative. It's not one size fits all so it's important to get to know us and appreciate our strengths not only our challenges. We may or may not have good mechanical aptitude, sharper peripheral vision, and be highly creative with entrepreneurial proclivities. It's of interest that 50% of NASA employees are reportedly dyslexic (<http://www.dyslexia-aware.com/dawn/dyslexic-thinking#:~:text=The%20dyslexic's%20abilities%20have%20been,excellent%20D%20and%20spatial%20awareness>).

Autistics and Their Cousin: Attention With or Without Hyperactivity Disorder (ADHD)

An ADHD individual may be hyper-focused, have higher levels of creativity and curiosity, with innovation and inventiveness leadership abilities, plus high energy, spontaneity, and a disposition towards productivity. But they may also battle some of the challenges that come with being monotropic (Dwyer et al., 2024) such as needing to focus on one thing at any one time but their ADHD causing them to lose concentration quickly, become quickly bored and driven by a need to move from activity to activity, skim over things, and not take in the wider picture. Impulsivity can rule and in social situations there may be a lack of 'filtering' between appropriate and not being appropriate in behaviours, thinking processes, actions, and even belief systems.

For Example When viewed via the monotropic lens, an Autistic individual may know how to wash the dishes in theory (you do all glass ware first, then crockery, cutlery, and then pots and pans)... but I will look at these and my comments may be: 'I can't wash up, these dishes are all too dirty and everything is too messy, not organized right... It's too overwhelming'. Quite often my wife will rinse off the dishes and organize them so I can wash them. At other times, if I'm already out of spoons (used up all my energy), she will do them. When it comes to what a clinician may experience from us? We may have difficulty 'keeping appointments' so need a text or some type of agreed upon reminder; we may have difficulty acting on any prescribed actions from you, so might benefit from smart phone reminders, visual walk-through exercises as we watch these on a YouTube video, and other creative ways we may work on together.

Autism and Its Cousin, Dyspraxia

Dyspraxia may mean creative and original thinking, having good strategic thinking and problem-solving, great determination and being hard-working, being highly motivated and able to develop our own strategies to overcome difficulties on the one hand, but with monotropism in the mix can mean what you see in us, looks very different. We may stumble over our words, be physically clumsy, seem 'disconnected' from various expectations upon us and very unorganized.

There are three ways that dyspraxia (Developmental Co-ordination Disorder—DCD) presents in us, as Autistic individuals, this may be verbally, orally, or in our motor (muscle and movement system) connections. Characteristics can include struggling to produce clear, fluent speech or to say particular words or phrases, even when you feel like you know them, they seem out of reach in our minds. We may also have difficulty with fine motor skills like handwriting or tying shoelaces. Some of us struggle to kick a ball, climb stairs, ride a bike, and walk in step or dance.

For example: I found it particularly difficult to follow a physiotherapist as she tried to demonstrate a routine she wanted me to do with particular exercises, I just couldn't do it! I also struggle to shave while looking in a mirror, figuring out which way to glide my razor, and I can't copy dance steps, struggle to walk up a flight of stairs correctly or walk in step.

When you look at this through a monotropic lens, you might appreciate why an Autistic individual might be echolalic (repeat and echo certain phrases, words, advertising jingles) and, or be passionate about escalators, find it difficult to follow your spoken instruction, process language, and/or note how to process and apply what we are being told or what we, ourselves are hearing, feeling, or noticing.

Autism and Its Cousin, Dyscalculia

Being ‘mathalexia’ (as I think of it) or living with dyscalculia means lots of difficulties around numbers, but still often being creative, with strategic and intuitive thinking, practical ability, problem-solving aptitudes, and a love of words.

For example: Numbers make my head sore. If a person is speaking out loud as they work out a math problem, I need to ask them to stop doing so, out loud. I can feel physical aggression towards them! I’m not good with money, bills, copying numbers from one space to another, remembering dates, working out sums or comprehending time. I struggle to remember how to write numbers in decimal format (e.g. 10,000,00 or 100,00,00) and to appreciate a lower sum compared to a higher sum. This has nothing to do with IQ, but everything to do with how my brain interprets numbers and plays it back to me.

Autism and Its Cousin, Dysgraphia

Although an individual may have good verbal reasoning, moving that reasoning from the spoken format to getting it down on paper, is a whole different task. For me, living with dysgraphia means there is this space between thinking the words, speaking the words, and getting them down on paper. The space between them is one I struggle to jump over. Some of us will manage to write but run all our letters together. Some of us will mix up capitals and lower-case letters in the wrong order. Judging distance and how to navigate a space without walking into things is another challenge!

Autism and Its Cousin, Aphantasia

If you are living with Aphantasia visual imagery will be allusive. For example: I ask individuals to close their eyes and picture an orange on their kitchen bench. Keeping their eyes closed I ask them to focus upon that orange. About 30 s later, I ask them to open their eyes and describe to me, what they saw. Some will tell me they saw a round fruit with rough skin that was an orange colour. Others will tell me they saw lots of round orange fruits sitting next to their juicer waiting to have their sweet juice squeezed from them. But another will tell me they saw ‘nothing, just a blackness’. They know what an orange looks like and can describe it fine, just not ‘see it’ in their mind’s eye. I’ve asked some of my Autistic friends to imagine a sunset, most can see a picture of the sun setting over the ocean nearby. They ‘see’ this within their mind’s eye, as a visual mental image. People with *aphantasia* lack this ability (no clear ‘mental picture’ in the mind) even though they can describe what a sunset looks like (Zeman et al., [2015](#)).

This lack of visual imagining is compensated for as our other senses work together to ‘give’ us a sense of an orange on our kitchen bench or the sun setting over the ocean. Most people who lack visual imagery are unaware of this, until they are asked to close their eyes and ‘picture’ a scene. It can be quite a surprise for them, even a shock, when they can’t ‘see’ the image requested. Some will say that they think it’s: ‘silly... of course you cannot see something if your eyes are shut’. When I tell them that most people can, they find it hard to believe. I also let them know that their other senses compensate and they are not disadvantaged in any big way, if they can’t see within their ‘mind’s eye’; they are the same person after doing this small task as they were before they did it.

Research does demonstrate however, that if you are living with Aphantasia it does have an impact upon emotional relating (Dance et al., 2021). Visual imagination is also a spectrum and people live anywhere from being ‘hyperphantasics’ with an incredibly vivid imagination; so strong it feels as real as if they were seeing it in the flesh. I’m reminded of a film I saw where all the characters constantly existed in all aspects of the writer’s daily life, so much so they had conversations together as the writer would ask them each for their opinion. Then there are those who live somewhere in between, hyper and hypo, with phantasia and hypophantasia, experiencing varying degrees of imagery vividness.

For Autistic individuals living with aphantasia, there will be unique challenges not experienced by Allistic individuals living with aphantasia. Many of our differences, as Autistic individuals with this intersection, are compounded by Aphantasia. How we understand social communication will have an extra degree of difficulty placed over all our social interactions. It’s another lens that needs placing over our decoding of social expectations. For example: If you can’t ‘imagine’ or ‘see again’ an experience of a social interaction, it might be harder to retain a connection to it. So, if you experience a particular romantic gesture from a partner, which includes the romance amplification due to scenery, environment, people present, certain tastes and sounds, and so on, it will impact upon our processing and recall of information, our understanding of abstract concepts, our ability to plan for the future, and our navigation of social interactions. Of course, the other side of this coin means there will be strengths that can come with this intersection. Many individuals with aphantasia and autism have exceptional abilities in logic, detail-oriented thinking, and focusing on the present moment.

Autism and Its Cousin, Central Auditory Processing Disorder (CAPD)

CAPD is a neurological disorder that impacts how the brain receives and interprets spoken sounds. The intention of a partner to engage in conversation with us, with the words chosen, their intonation, meaning and intentions behind the spoken words, all supported by body language and facial expressions, allow a two-way conversation,

or more, between people to take place. If the individual however is monotropic and living with CAPD, understanding the conversation will be compacted by (a) the breakdown of ‘hearing & translating’ of speech and (b) not accessing the wider picture intended but only the narrow one gleaned from the monotropic lens. Without the truer picture of intended meaning being gleaned from the conversation the hearer will ‘gestalt’ the missing links to a broader understanding and be left with a disjointed, far from true, picture. This is innately problematic and will heavily impact the autistic CAPD individual in negative ways. If you further add (c) to the mix, possibly of an individual growing up in a restrictive, paranoid, hypersensitive family, then this lens will also impact the communication received.

Autism and Its Cousin, Alexithymia

Alexithymia is the term given to a condition when an individual has difficulty identifying emotion in themselves and in others. Often there is difficulty in experiencing, identifying, and expressing emotions. People with alexithymia may have problems maintaining relationships and taking part in social situations. They may have a co-occurring mental health condition, such as depression, or no diagnosable mental health conditions. Alexithymia also has links with autism. See: <https://www.medicalnewstoday.com/articles/326451>

But not having the words to name an emotion is not the same as not ‘feeling’ emotions. As Autistic people, we have lots of emotions turning and tossing around inside of us. But, being monotropic makes it hard to (a) notice what we are feeling (our attention may be focused elsewhere), (b) when we do notice we may have missed the context, so it’s difficult to identify what the feeling is connected to, and (c) having missed the reason and situation for the emotion we may find it difficult to intuitively act or respond genuinely because the main source and connecting material is out of focus. By the time our ‘discomfort’ or whatever registers as an emotion the moment may have passed. This means it can appear that we lack empathy, are cold, calculating, and not caring!

Understanding the reasons for our Alexithymia should guide a medical professional in how to respond to us. For example, it may not be helpful to say to an autistic individual ‘where do you feel the pain?’ or, ‘...how does that make you feel?’ My usual response would be ‘I don’t know’. Because, at the time of asking I’m not connected enough to the source, which then means I’m not able to identify it. I often know if I’m uncomfortable or comfortable, but to lift off the layers and dig deeper can take some extra time and creative resources. For example: ‘...once, when I was visiting a friend, her two year old daughter fell over and was crying. She kept calling ‘Mummy, Mummy’. My friend came running to the scene and I was nearby reading a book (my attention was occupied). My friend said: ‘Wenn, Jesse fell over and she is crying, why didn’t you see if she was alright?’ I replied, ‘I was reading so I didn’t hear Jess at first, then, when I did hear her she was calling “Mummy”. She didn’t want me, she wanted you’. I was being monotropic, autistically literal, not able at

that time to forward-think (as in, Jesse has fallen over, she might be hurt, I know she is calling for Mummy, but I am a responsible adult and I can check on her while her Mummy comes) and I was not immediately connected to the situation. I was not being, cold, unthinking, and uncaring! On the occasion described my friend said to me, ‘...in the future, if you notice Jesse being upset you have my permission to check in on her...’. This was an appropriate way to respond to me. I was being given a type of map that told me what I could do in the future.

Autism may have other cousins not written about in this book. These could include intellectual disability, epilepsy, some physical conditions such as Ehlers-Danlos Syndrome (EDS), stomach and intestinal conditions such as Coeliac disease, Crohn’s Disease, and sensitive ‘tummies’ that might lead to constipation and/or diarrhoea, anxiety, and various allergies. Understanding the condition and then adding on the monotropic lens to explore the condition through is vital for Autistics in understanding how best to support us. Regular treatments may not work out as well for us. At times the therapeutic amount of a ‘drug’ may be too high for us, and we do better on smaller amounts. Temple Grandin, an American Autistic Older says:

Now, I’ve been on low doses of antidepressants for over 30 years—it saved me. What happens is that they put kids on a drug like Prozac and then they experience excitation as a side effect. Excitation keeps coming up in the literature that I’ve been reading. I’ll tell you what’s wrong—they are giving too high a dose. Forget the label, sometimes you need to go lower than the starter dose. I’ve seen this over and over again. Every time they do a research study, they just use a set dose. If it’s too high, it’s a mess when dealing with the anxiety. For years I have been preaching the use of a lower dose—if they experience excitation, lower the dose. I’ve seen that work over and over again

See: <https://autismspectrumnews.org/an-autism-spectrum-news-interview-with-temple-grandin/>

To live life well as an Autistic person requires that, as Autistic people, we welcome our autism and our cousins. Acceptance of difference and commitment to well-being requires us to listen, learn, and act. Our actions can be in any area of our being: our thinking, emotive states, and our physical responses. Comprehending intersectional borders, such as being Autistic, black and gender diverse, female, male or ‘other’, having learning differences, being from differing social economies and so on, require our creative adaptation to Autistic needs. Putting Autistic people before curriculum, before procedures and processes, takes knowledge, courage, and willingness to go the extra mile. When clinicians and professionals walk in our shoes and understand why we do what we do, this will greatly enhance the relationship and the joint experiences of both practitioner and patient/client. Taking time to get to know us, see what’s ‘beneath’ the DSM-5 criteria for autism, work with our strengths and support us through our challenges. This is the recipe for a successful outcome.

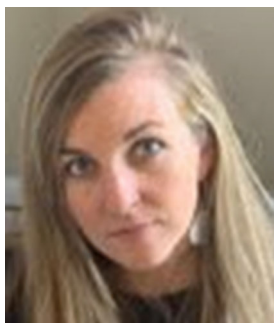
For more on resources visit: www.wennlawson.com and on YouTube: https://www.youtube.com/channel/UCD8L-Ht_jSQxBmWe6MzZvJA

And: www.notthenorm.com.au

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Appendix A



Megan Anna Neff

The following sensory ‘check list’ is used with permission from Dr. Neff.

The original website is: <https://neurodivergentinsights.com/blog/sensory-preferences-the-complete-checklist>

Sensory Preference Checklist: Lighting

LIGHTING



- Natural lighting/outdoors
- Dim lights
- Colorful lights (LED)

☐ **Natural lighting:** The gentle, warm glow of natural sunlight can have a calming and uplifting effect. Being in spaces with ample natural light may enhance your mood and overall sensory experience.

☐ **Dim lighting:** Soft, low-level lighting can create a cosy and relaxing atmosphere. It can help reduce sensory overload and create a sense of calm.

☐ **Bright lighting:** Bright, well-lit environments can promote alertness and focus. Some individuals may find bright lighting energizing and beneficial for certain activities.

☐ **Artificial light:** Standard indoor lighting, such as fluorescent or incandescent bulbs, provides consistent and reliable illumination. It is commonly found in most indoor environments.

☐ **Colourful lighting (LED):** LED lights come in various colours and can create a vibrant and visually stimulating atmosphere. Colourful lighting options allow you to customize the ambiance to suit your sensory preferences.

☐ **Moderate lighting:** Moderate lighting strikes a balance between bright and dim, providing sufficient illumination without being too overwhelming. It offers a comfortable environment for various activities.

☐ **Yellow hues (warm lighting):** Warm, yellow-toned lighting can create a cosy and inviting atmosphere. It can contribute to a sense of relaxation and comfort.

☐ **Cool spectrum light (white):** Cool spectrum or white light tends to have a brighter and more energizing effect. It can mimic natural daylight and may promote focus and productivity.

☐ **Other _____:** Identify any additional visual preferences or experiences that resonate with you personally.

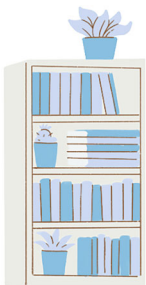
When considering your sensory preferences for lighting, take note of how different lighting conditions make you feel.

- Do you prefer the warm and soothing glow of natural sunlight or the cosy ambiance of dim lighting?
- Are you drawn to colourful lighting that sparks joy and visual interest?

Reflect on your responses and identify the lighting options that align with your sensory preferences. Also, note how your preferences may change depending on your activity (if you are studying vs. socializing vs. resting, for example). By understanding and incorporating your sensory preferences for lighting, you can create visually appealing, calming spaces conducive to your overall sensory preferences.

Sensory Preference Checklist: Visual

VISUAL



- Visually sparse & organized
- Neutral colors
- Patterns
- Bright decor/colors

- ☐ **Visually sparse and organized:** Some individuals find comfort and clarity in a visually minimalist and organized environment with minimal distractions.
- ☐ **Neutral colours:** A preference for neutral colour palettes, such as shades of white, beige, or grey, can create a calming and soothing visual atmosphere.
- ☐ **Vibrant colours:** Others may enjoy the energy and stimulation provided by vibrant and bold colours in their surroundings.
- ☐ **Patterns and bright decor:** Patterns, intricate designs, and visually stimulating decorations can be appealing to individuals who enjoy visual variety and stimulation.
- ☐ **Solid colours:** A preference for solid colours, without complex patterns or designs, can create a visually calming and harmonious environment.
- ☐ **Minimalism:** Embracing a minimalist visual aesthetic with clean lines and uncluttered spaces can provide a sense of serenity and simplicity.
- ☐ **Everything spread out/visible:** Having a preference for visual accessibility and being able to see everything in one's environment can contribute to a sense of control and reduced anxiety.

☐ **No preference:** Some individuals may not have a strong preference for specific visual elements and feel comfortable in a variety of visual environments.

☐ **Other _____:** Identify any additional visual preferences or experiences that resonate with you personally.

Consider the visual elements that bring you a sense of comfort, focus, or inspiration. Reflect on the colours, patterns, organization, or openness that enhance your visual experience. Tailor your surroundings to align with your preferences and create a visually supportive environment that promotes your well-being.

Sensory Preference Checklist: Tactile

TACTILE



- Deep pressure touch
- No touch
- Fidget toy
- Petting a dog or cat

☐ **Light touch:** Some individuals may enjoy gentle, feather-like touches on their skin. Light touch can provide a soothing and comforting sensation.

☐ **Deep pressure touch:** Firm and deep pressure can be grounding and calming for many individuals. Activities such as hugging, squeezing, or using weighted blankets can provide deep pressure stimulation.

☐ **No touch:** Some individuals may prefer minimal tactile stimulation and feel more comfortable without physical contact.

☐ **Twirling hair:** Twirling or playing with hair can be a self-soothing behaviour for many individuals. The repetitive motion can provide sensory comfort.

☐ **Fidget toy:** Fidget toys, such as stress balls or textured objects, can offer tactile stimulation and help individuals regulate their sensory experiences.

☐ **Receive a massage:** Massage therapy can provide a combination of deep pressure touch and relaxation, promoting sensory regulation and stress reduction.

☐ **Pet a dog or cat:** Interacting with pets, such as stroking or petting a dog or cat, can offer tactile comfort and a sense of connection.

- ☐ **Warm bath or shower:** The sensation of warm water can be soothing and relaxing. Taking a warm bath or shower can provide tactile and sensory enjoyment.
- ☐ **Rub or pick at clothes:** Some individuals may engage in rubbing or picking at their clothes as a way to self-soothe or regulate sensory input.
- ☐ **Soft textures:** Exploring fabrics or materials with a soft and plush texture can be comforting and enjoyable.
- ☐ **Smooth textures:** Smooth surfaces or objects can offer a pleasing tactile experience for those who appreciate sensory input.
- ☐ **Preferences for specific fabrics:** Some individuals may have preferences for certain types of fabrics, such as silky, fuzzy, or textured materials.
- ☐ **Other _____:** Identify any additional visual preferences or experiences that resonate with you personally.

Consider the tactile sensations that bring you comfort and joy. Reflect on the activities or textures that make you feel calm, grounded, or stimulated in a positive way. Everyone's tactile preferences are unique, so take the time to explore and discover what works best for you. Incorporate these preferences into your sensory experiences and create a sensory-friendly environment that supports your well-being.

Sensory Preference Checklist: Sound

SOUND



- Quiet
- Background noise
- Stimming music (song on repeat)
- White/brown noise

- ☐ **Quiet:** Some individuals prefer a quiet environment, free from excessive noise or distractions. They find peace and focus in a calm and serene auditory atmosphere.
- ☐ **White Noise/Brown Noise:** Background noise, such as white noise or brown noise, can be soothing and help individuals concentrate or relax. It masks other sounds and provides a constant and consistent auditory input.

- ☐ **Loud and lively:** Certain individuals thrive in an environment with energetic and vibrant sounds. They enjoy the buzz and excitement of a lively atmosphere.
- ☐ **Music/Background noise:** Having music or background noise, such as ambient sounds or soft music, can be pleasing and provide a comforting backdrop to their activities.
- ☐ **Television on in the background:** Some individuals find comfort in having the television on as background noise, creating a sense of familiarity and cosiness.
- ☐ **No preference:** Some individuals may not have specific preferences for auditory stimuli and feel comfortable in various sound environments.
- ☐ **Other _____:** Identify any additional auditory preferences or experiences that resonate with you personally.

Consider your auditory preferences and how different sound environments impact your well-being. Whether it's finding solace in silence, embracing the presence of background noise, or seeking out specific types of sounds for relaxation or stimulation, honour and create an auditory environment that aligns with your preferences.

Sensory Preference Checklist: Taste

GUSTATORY (ORAL)



- Chewing gum
- Chewing/sucking on ice cubes
- Eating a crunchy snack

- ☐ **Chewing gum:** Some individuals find chewing gum to be satisfying and helpful for sensory regulation.
- ☐ **Crunchy snacks:** Enjoying crunchy snacks provides a sensory experience and can contribute to a sense of satisfaction and stimulation.
- ☐ **Chewing on straws:** Chewing on straws or similar objects may provide oral sensory input and a soothing effect for some individuals.

- ☐ **Suck, lip, or bite lips or cheeks:** Some individuals engage in oral habits like sucking, biting, or pressing lips or cheeks for sensory stimulation and comfort.
- ☐ **Chewing on ice:** Chewing on ice can provide a cooling, refreshing, and grounding sensory experience for some people.
- ☐ **Spicy food:** The sensation of spicy flavours can be stimulating and enjoyable for those who have a preference for spicy food or are sensory seeking.
- ☐ **Drink a warm beverage:** Drinking a warm beverage, such as tea or coffee, can provide a soothing and calming sensation.
- ☐ **Drink a carbonated drink:** The effervescence and bubbles of carbonated drinks can provide a unique sensory experience and is great for sensory seekers.
- ☐ **Other _____:** Identify any additional gustatory preferences or experiences that resonate with you personally.

Be mindful of your gustatory preferences and explore different tastes and textures that bring you comfort and satisfaction. Consider incorporating sensory-friendly foods and beverages into your daily routines to enhance your sensory well-being.

Sensory Preference Checklist: Smell

SMELL



- Mild, subtle scents
- Aromatherapy
- Fresh or clean scents
- Bakery or food-related scents

- ☐ **Mild or subtle scents:** Preference for gentle or delicate fragrances that are not overpowering.
- ☐ **Fresh or clean smells:** Enjoyment of scents associated with cleanliness, such as fresh laundry or a clean breeze.
- ☐ **Citrus or fruity scents:** Pleasure derived from the aroma of citrus fruits or other fruity fragrances.
- ☐ **Bakery or food-related scents:** Appreciation for scents reminiscent of baked goods, cooking, or specific food items.

- ☐ **Herbal or earthy scents:** Fondness for scents derived from herbs, plants, or natural earthy aromas.
- ☐ **Aromatherapy scents (e.g. lavender, eucalyptus):** Preference for scents commonly used in aromatherapy for relaxation or invigoration.
- ☐ **Natural scents (e.g. flowers, trees):** Enjoyment of the fragrances found in nature, such as floral scents or the smell of trees.
- ☐ **Specific scents that bring joyful memories:** Identification of particular scents that evoke positive memories or feelings of joy.
- ☐ **Other _____:** Recognition of any additional olfactory preferences or experiences that resonate with you personally.

Consider the scents that bring you comfort, relaxation, or a sense of joy. Explore the variety of fragrances available and identify those that align with your olfactory (smell) preferences. Incorporate scents into your environment that promote a sensory experience that supports your well-being.

Sensory Preference Checklist: Proprioception

- ☐ **Lift weights:** Engaging in weightlifting or resistance training exercises can provide deep pressure and sensory input, promoting a sense of body awareness and regulation.
- ☐ **Pressure:** Seeking out activities or items that apply pressure to the body, such as firm hugs, compression garments, or weighted items, can provide a grounding and calming effect.
- ☐ **Weight:** Appreciating the sensation of additional weight or pressure on the body, whether through weighted clothing, weighted blankets, or carrying heavy objects, can offer a sense of stability and comfort.
- ☐ **Pillow/weighted blanket:** Finding comfort and relaxation by using pillows or weighted blankets, which provide a gentle, even pressure sensation, promoting a sense of security and calm.
- ☐ **No pressure (avoidance):** Avoiding or feeling discomfort with activities or items that apply pressure or weight to the body, preferring lighter touch or gentle movements instead.
- ☐ **Other _____:** Identify any additional proprioceptive preferences or experiences that resonate with you personally.

Understanding your proprioceptive preferences can help you create an environment that supports your sensory well-being. Whether it's engaging in weightlifting exercises, seeking pressure or weight sensations, or opting for a lighter touch, embrace the activities and items that bring you comfort and regulate your proprioceptive system.

Sensory Preference Checklist: Movement

MOVEMENT



- Jumping or bouncing
- Stimming
- Gentle rocking or swaying
- Swinging or being in motion

- ☐ **Rocking:** Finding comfort and sensory regulation through rhythmic rocking movements, which can provide a soothing and grounding effect.
- ☐ **Rock in a rocking chair:** Seeking the repetitive back-and-forth motion of a rocking chair, which can promote relaxation and sensory regulation.
- ☐ **Swaying:** Engaging in gentle swaying movements, whether self-initiated or through the use of a swing or hammock, to promote a sense of calm and sensory integration.
- ☐ **Ride bike:** Finding enjoyment and sensory regulation through cycling or bike riding, which can provide a combination of physical activity and rhythmic movement.
- ☐ **Being still:** Preferring minimal movement and maintaining a relatively static posture, finding sensory comfort in stillness.
- ☐ **Other:** _____ (Identify any additional movement preferences or considerations that resonate with you personally.)

Recognizing and incorporating movement preferences can play a significant role in supporting sensory regulation and overall well-being. Explore different movement options and activities that align with your sensory needs, providing comfort, relaxation, and sensory integration opportunities.

Sensory Preference Checklist: Environment

ENVIRONMENT



- Outdoors
- Indoors
- Nooks and crannies
- Flexible seating (e.g., wobble stool, foam mat, rocking chair)

☐ **Outdoors:** Being in an outdoor environment, surrounded by nature and fresh air, can be invigorating and calming for some individuals.

☐ **Indoors:** Finding comfort and security in indoor environments, where one can control the sensory elements, can provide a sense of stability and relaxation.

☐ **Outside and covered:** Enjoying the outdoors while having the option of finding shelter or shade can provide a balanced sensory experience for those who prefer a mix of indoor and outdoor elements.

☐ **Nooks and crannies:** Seeking out cosy and secluded spaces, such as small corners or tucked-away areas, can offer a sense of security and a calming environment.

☐ **Open space:** Feeling at ease in open and expansive environments, where there is ample room to move and breathe, can provide a sense of freedom and sensory satisfaction.

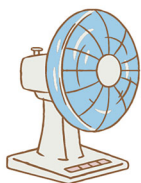
☐ **No preference:** Having no specific preference for a particular environment, being adaptable to different surroundings, or finding comfort in a variety of settings.

☐ **Other** _____: Identify any additional environmental preferences or experiences that resonate with you personally.

Pay attention to your surroundings and create an environment that aligns with your sensory preferences. Whether it is finding solace in the tranquillity of nature or seeking comfort in a cosy nook, your environment can play a significant role in your sensory well-being.

Sensory Preference Checklist: Temperature

TEMPERATURE



- Cool or warm
- Breezy, flowing air (fan)
- Heated blankets
- Cooling blankets or mats

☐ **Cool:** Finding comfort and sensory regulation in cooler temperatures or environments, which can help in reducing sensory overwhelm and promoting a sense of calmness.

☐ **Warm:** Seeking warmth and cosiness, finding sensory comfort in warmer temperatures or using warm materials such as blankets or heated pads for relaxation.

☐ **Breeze/airflow:** Enjoy gentle air movement or breezes, whether through open windows, fans, or outdoor environments, to promote a sense of freshness and sensory soothing.

☐ **Neutral Temperature:** Preferring a temperature that is neither too hot nor too cold, finding sensory comfort in a balanced and moderate environment.

☐ **No preference:** Not having a specific temperature preference, feeling comfortable in various temperature ranges.

☐ **Other:** _____ (Specify any additional temperature preferences or considerations that resonate with you personally.)

Understanding and attending to temperature preferences can contribute to creating a more comfortable and supportive sensory environment. Adjusting the temperature settings, using appropriate clothing, and considering personal temperature preferences can promote optimal sensory well-being and regulation.

Sensory Preference Checklist: Clothing

CLOTHING



- Tight, compressive clothing
- Loose-fitting clothing
- No tags or seams
- Breathable material

- ☐ **Constriction clothing (tight):** Finding comfort and sensory regulation through wearing clothing that provides a snug or tight fit, creating a sense of grounding and security.
- ☐ **Loose-fitting clothing:** Preferring clothing that allows for freedom of movement and minimal restriction, providing a sense of ease and comfort.
- ☐ **Cotton fabric:** Enjoying the softness and breathability of cotton fabric, which can be gentle on the skin and help regulate body temperature.
- ☐ **Tagless clothing:** Seeking clothing without tags or labels that can cause discomfort or irritation against the skin.
- ☐ **Seamless garments:** Choosing clothing items that are seamless, reducing the potential for sensory irritation and providing a smooth, comfortable experience.
- ☐ **Compression garments:** Finding relief and sensory regulation through the use of compression clothing, which applies gentle pressure to the body.
- ☐ **Breathable materials** (e.g. bamboo, linen): Opting for fabrics that promote airflow and ventilation, enhancing comfort and preventing overheating.
- ☐ **Soft and gentle textures** (e.g. velour, fleece): Preferring clothing with soft and cosy textures that provide a soothing sensory experience.
- ☐ **Non-irritating fabrics** (e.g. organic cotton, modal): Selecting fabrics that are hypoallergenic and less likely to cause skin irritation or discomfort.
- ☐ **Adjustable closures** (e.g. Velcro, snaps): Choosing clothing with adjustable closures to customize the fit and avoid sensory discomfort from tight or restrictive closures.
- ☐ **Stretchy and flexible clothing:** Embracing clothing that offers stretch and flexibility, allowing for ease of movement and accommodating sensory sensitivities.

- ☐ **Weighted or pressure-based clothing:** Seeking the comforting sensation of weighted or pressure-based clothing items that provide gentle, constant pressure on the body.
- ☐ **Moisture-wicking fabrics:** Opting for fabrics that draw moisture away from the skin, keeping you dry and comfortable, especially during physical activities.
- ☐ **Non-restrictive waistbands (e.g. elastic-free):** Preferring waistbands that are not constricting or elastic, providing a more comfortable and sensory-friendly fit.
- ☐ **Layering options for temperature regulation:** Choosing clothing that allows for layering, providing flexibility to adjust to different temperatures and sensory needs.
- ☐ **Odour-resistant or hypoallergenic materials:** Selecting clothing made from materials that resist odour or are hypoallergenic, reducing sensory discomfort and irritation.
- ☐ **Other:** _____ (Identify any additional clothing preferences or considerations that resonate with you personally.)

Understanding your clothing preferences can help create a sensory-friendly wardrobe that supports your comfort and well-being. Whether it's the choice of fabric, fit, or closure, prioritize clothing options that align with your sensory needs and enhance your overall sensory experience.

Sensory Preference Checklist: People

PEOPLE



- Lots of people, big crowds
- Only a few people
- Solitude/time alone

- ☐ **Lots of people:** Some individuals thrive in social environments with large crowds and enjoy the energy and buzz of being around many people.

- ☐ **People who you don't know:** Interacting with unfamiliar people or strangers may be exciting and stimulating for individuals who enjoy meeting new individuals and exploring new social connections.
- ☐ **Only a few people:** Others prefer smaller social gatherings or interactions with a limited number of individuals, finding comfort and ease in more intimate settings.
- ☐ **A small group of people:** Some individuals feel most comfortable and engaged in small group settings where they can actively participate and connect with others.
- ☐ **Alone:** Enjoying solitude and time spent alone is a preference for individuals who find solace, recharge, and rejuvenation in their own company.
- ☐ **Other _____:** Identify any additional preferences or experiences related to being around people that resonate with you personally.

Consider your social preferences and the types of social interactions that bring you joy, comfort, and fulfilment. Whether it's thriving in a crowd, cherishing one-on-one connections, finding solace in solitude, or embracing the excitement of meeting new people, honour and create opportunities for social experiences that align with your preferences.

1. **Reflect and self-observe:** Take time to reflect on your sensory experiences throughout the day. Pay attention to how different sensory inputs affect your mood, energy levels, and overall well-being. Notice what feels calming, overwhelming, or stimulating to your senses.
2. **Experiment with different environments:** Create opportunities to experience different sensory environments. Visit different places, such as parks, libraries, or coffee shops, and observe how the sensory elements in each setting impact your comfort level and sensory experience.
3. **Engage in sensory activities:** Try out various sensory activities to discover what resonates with you. Experiment with different textures, sounds, tastes, and movements. Engaging in activities like listening to soothing music, engaging in art or crafts, or spending time in nature can help you identify your sensory preferences.
4. **Seek input from others:** Engage in conversations with trusted friends, family members, or therapists about your sensory experiences. Their observations and insights can provide valuable perspectives and help you gain a deeper understanding of your sensory preferences.

The goal is to develop a better understanding of your own sensory landscape and create a sensory-friendly environment that supports your well-being. By embracing the journey of self-discovery, you will continue to explore and refine your understanding of your sensory profile over time.

Conclusion

Cultivating a sensory lens and identifying your sensory preferences is a powerful journey of self-discovery and self-care. By understanding how your sensory experiences shape your perception and impact your daily life, you gain valuable insights into your unique sensory profile. Through sensory checklists, sensory diets, reflection, experimentation, and input from others, you can begin to uncover the sensory inputs that bring you comfort, joy, and regulation, as well as those that may overwhelm or dysregulate you.

Armed with this knowledge, you can create a sensory-friendly environment that supports your well-being and allows you to thrive. Your sensory preferences may evolve and change over time, so continue to explore, adapt, and prioritize your sensory well-being. Embrace the power of your sensory lens and embark on a journey of self-discovery, acceptance, and self-care. Your sensory well-being matters, and by honouring it, you are nurturing your overall health and well-being.

Appendix B: Sensory Information Checklist and How to Help

Sensory differences—A guide for all audiences. National Autistic Society UK. Available from: <https://www.autism.org.uk/advice-and-guidance/topics/sensory-differences/sensory-differences/all-audiences>

Processing everyday sensory information can be difficult for autistic people. Any of their senses may be over- or under-sensitive, or both, at different times. These sensory differences can affect how they feel and act and can have a profound effect on a person's life. Here we help you to understand autism, the person and how to help.

You can also find out about synaesthesia, therapies, and equipment.

Too Much Information

Sometimes an autistic person may behave in a way that you wouldn't immediately link to sensory differences. A person who finds it difficult to process everyday sensory information can experience sensory overload, or information overload. Too much information can cause stress, anxiety, and possibly physical pain. This can result in withdrawal, distressed behaviour, or meltdowns.

If I get sensory overload then I just shut down; you get what's known as fragmentation...it's weird, like being tuned into 40 TV channels.
—an autistic person

If someone is having a meltdown, or not responding, don't judge them. There are things that you can do to help. This can make a world of difference to autistic people and their families. Often, small changes to the environment can make a difference. Creating a sensory profile may help you to work out what changes are needed.

Three Points to Remember are:

- **Be aware.** Look at the environment to see if it is creating difficulties. Can you change anything?
- **Be creative.** Think of some positive sensory experiences.
- **Be prepared.** Tell the person about possible sensory stimuli they may experience in different environments.

Sensory Differences

Here we look at some of the effects of hypersensitivity or hyposensitivity to sights, sounds, smells, tastes, touch, balance and body awareness, and ways you could help.

Sight

Under-Sensitive

- Objects appear quite dark, or lose some of their features.
- Central vision is blurred but peripheral vision quite sharp.
- A central object is magnified but things on the periphery are blurred.
- Poor depth perception, difficulties with throwing and catching, clumsiness.

Over-Sensitive

- Distorted vision—Objects and bright lights can appear to jump around images may fragment.
- Easier and more pleasurable to focus on a detail rather than the whole object.
- Has difficulty getting to sleep as sensitive to the light.

You could make changes to the environment, such as reducing fluorescent lighting, providing sunglasses, using blackout curtains and/or creating a workstation—a space or desk with high walls or divides on both sides to block out visual distractions.

Sound

Under-Sensitive

- May only hear sounds in one ear, the other ear having only partial hearing or none at all.

- May not acknowledge particular sounds.
- Might enjoy crowded, noisy places or bang doors and objects.

You could help by using visual supports to back up verbal information and ensuring that other people are aware of the under-sensitivity so that they can communicate effectively. To meet the person's individual sensory need, include experiences they enjoy in their daily timetable or medical appointments.

Over-Sensitive

- Noise can be magnified and sounds become distorted and muddled.
- May be able to hear conversations in the distance.
- Inability to cut out sounds—notably background noise—leading to difficulties concentrating.

Do you hear noise in your head? It pounds and screeches. Like a train rumbling through your ears.

(Powell, J., in Gillingham (1995), p. 41)

You Could Help by:

- Shutting doors and windows to reduce external sounds.
- Preparing the person before going to noisy or crowded places.
- Providing ear plugs and music to listen to.
- Creating a screened workstation in the office, positioning the person away from doors and windows.

Smell

Under-Sensitive

- Some people have no sense of smell and fail to notice extreme odours (this can include their own body odour).
- Some people may lick things to get a better sense of what they are.

You could help by creating a routine around regular washing and using strong-smelling products to distract people from inappropriate strong-smelling stimuli (like faeces).

Over-Sensitive

- Smells can be intense and overpowering. This can cause toileting problems.
- Dislikes people with distinctive perfumes, shampoos, etc.

Smells like dogs, cats, deodorant and aftershave lotion are so strong to me I can't stand it, and perfume drives me nuts.

(Gillingham, 1995, p. 60)

You could help by using unscented detergents or shampoos, avoiding wearing perfume, and making the environment as fragrance-free as possible.

Taste

Under-Sensitive

- Likes very spicy foods.
- Eats or mouths non-edible items such as stones, dirt, soil, grass, metal, and faeces. This is known as pica.

Over-Sensitive

- Finds some flavours and foods too strong and overpowering because of very sensitive taste buds. Has a restricted diet.
- Certain textures cause discomfort—may only eat smooth foods like mashed potatoes or ice cream.

Some autistic people may limit themselves to bland foods or crave very strong-tasting food. As long as someone has some dietary variety, this isn't necessarily a problem. Find out more about over-eating and restricted diets.

Touch

Under-Sensitive

- Holds others tightly—needs to do so before there is a sensation of having applied any pressure.
- Has a high pain threshold.
- May be unable to feel food in the mouth.
- May self-harm.
- Enjoys heavy objects (e.g. weighted blankets) on top of them.
- Smears faeces as enjoys the texture.
- Chews on everything, including clothing and inedible objects.

You Could Help by:

- For smearing, offering alternatives to handle with similar textures, such as jelly, or cornflour and water.
- For chewing, offering latex-free tubes, straws, or hard sweets (chill in the fridge).

Over-Sensitive

- Touch can be painful and uncomfortable—people may not like to be touched and this can affect their relationships with others.
- Dislikes having anything on hands or feet.
- Difficulties brushing and washing hair because head is sensitive.
- May find many food textures uncomfortable.
- Only tolerates certain types of clothing or textures.

Every time I am touched it hurts; it feels like fire running through my body.
(Gillingham, 1995, p. 3)

You Could Help by:

- Warning the person if you are about to touch them—always approach them from the front.
- Remembering that a hug may be painful rather than comforting.
- Changing the texture of food (e.g. purée it).
- Slowly introducing different textures around the person's mouth, such as a flannel, a toothbrush, and some different foods.
- Gradually introducing different textures to touch (e.g. have a box of materials available).
- Allowing a person to complete activities themselves (e.g. hair brushing and washing) so that they can do what is comfortable for them.
- Turning clothes inside out so there is no seam, removing any tags or labels.
- Allowing the person to wear clothes they're comfortable in.

Balance (Vestibular)**Under-Sensitive**

A need to rock, swing, or spin to get some sensory input.

You could encourage activities that help to develop the vestibular system. This could include using rocking horses, swings, roundabouts, seesaws, catching a ball, or practising walking smoothly up steps or curbs.

Over-Sensitive

- Difficulties with activities like sport, where we need to control our movements.
- Difficulties stopping quickly or during an activity.
- Car sickness
- Difficulties with activities where the head is not upright or feet are off the ground.

You could help by breaking down activities into small, more easily manageable steps and using visual cues such as a finish line.

Body Awareness (Proprioception)

Our body awareness system tells us where our bodies are in space, and how different body parts are moving.

Under-Sensitive

- Stands too close to others because they cannot measure their proximity to other people and judge personal space.
- Finds it hard to navigate rooms and avoid obstructions.
- May bump into people.

You Could Help by:

- Positioning furniture around the edge of a room to make navigation easier.
- Using weighted blankets to provide deep pressure.
- Putting coloured tape on the floor to indicate boundaries.
- Using the ‘arm’s-length rule’ to judge personal space—this means standing an arm’s length away from other people.

Over-Sensitive

- Difficulties with fine motor skills, for example, manipulating small objects like buttons or shoe laces.
- Moves whole body to look at something.

You could help by offering ‘fine motor’ activities like lacing boards.

Synaesthesia

Synaesthesia is a rare condition experienced by some autistic people. An experience goes in through one sensory system and out through another. So a person might hear a sound but experience it as a colour. In other words, they will ‘hear’ the colour blue.

Therapies and Equipment

We can’t make recommendations as to the effectiveness of individual therapies and interventions or equipment.

Music therapists use instruments and sounds to develop people’s sensory systems, usually their auditory (hearing) systems.

Occupational therapists design programmes and often make changes to the environment so that people with sensory differences can live as independently as possible.

Speech and language therapists often use sensory stimuli to encourage and support the development of language and interaction.

Some people say they find coloured filters helpful although there is only very limited research evidence. Find out more from [UK Irlen Centres](#).

Sensory integrative therapy and Sensory Integration Network.

The [Brain in Hand](#) digital self-management support system.

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Further Reading and Resources

Network Autism links:

[Sensory challenges for autistic pupils—by Phoebe Caldwell](#)

[Top 5 tips for professionals—by Dr Olga Bogdashina](#)

Books

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Resources and More Information

Autism and mental health. Free online course from Curtin University & edX. <https://www.edx.org/course/autism-and-mental-health>

Porn is Not the Norm (PINN). *Supporting autistic young people and their communities to safely navigate pornography's influence*. www.notthenormorg.au
<https://www.youtube.com/watch?v=0gCXwBh2saQ>
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